

# RELATIVE PERMEABILITY CALCULATIONS FROM PORE SIZE DISTRIBUTION DATA

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## ABSTRACT

Formulas for calculating relative permeability from pore size distribution data are derived from basic laws of fluid flow in porous media. The tortuosity factors that appear in the equations are described in terms of the physical properties of the medium and the saturations of the contained fluids. Procedures for calculating and methods for measuring the tortuosity factors are discussed. Values of relative permeability calculated by the formulas are compared with results obtained by experimental measurements.

## INTRODUCTION

Recently, several investigators<sup>1,2,3,4</sup> have derived relative permeability equations from Darcy's and Poiseuille's Laws in which some physical factor or factors describing the porous system appear. The factor that has caused the greatest amount of comment, and which is probably least understood, is the tortuosity of the fluid path in the porous sample. Values of the tortuosity factors appearing in the equations range from a constant for all conditions of the fluid flow system to an inverse function of the pore radii.

It is the purpose of this paper to compare some experimentally determined relative permeability curves with those calculated by a method which utilized capillary pressure information. The capillary pressure data were obtained preferably from pore size distribution information as determined by mercury injection but, in some cases, in the absence of suitable mercury injection information, data were taken from measurements made by the porous semi-permeable diaphragm method. The physical significance of certain tortuosity concepts is inferred and the determinations of the tortuosity factors from pore size distribution data and by electrical measurements are discussed.

## THEORETICAL BACKGROUND

In previously reported studies on pore size distribution in petroleum reservoir rocks, by the author and co-workers,<sup>5</sup> an equation was derived relating specific permeability,  $k$ , to pore entry radii,  $R_i$ , porosity,  $\phi$ , tortuosity factor,  $X_i$ , and incremental pore volume,  $V_i$ . The equation is

$$k = 0.126 \phi \sum_{i=0}^n \frac{V_i \bar{R}_i^4}{X_i^2 \bar{R}_i^2} \dots \dots (1)$$

<sup>1</sup>References given at end of paper.

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where  $X_i = L_i/L$ ,  $L_i$  being the "effective" length of the flow path in the porous sample, and  $L$  the length of the porous sample. The constant, 0.126, is a units conversion factor. (This definition of tortuosity factor  $X_i = L_i/L$  does not conform to all such definitions found in the literature.<sup>6</sup>)

By similarity, in a two-phase system, the effective permeability of the wetting-phase,  $k_{ew}$ , is

$$k_{ew} = 0.126 S_w \phi \sum_{i=0}^n \frac{V_{iw} \bar{R}_i^4}{X_{iw}^2 \bar{R}_i^2} \dots \dots (2)$$

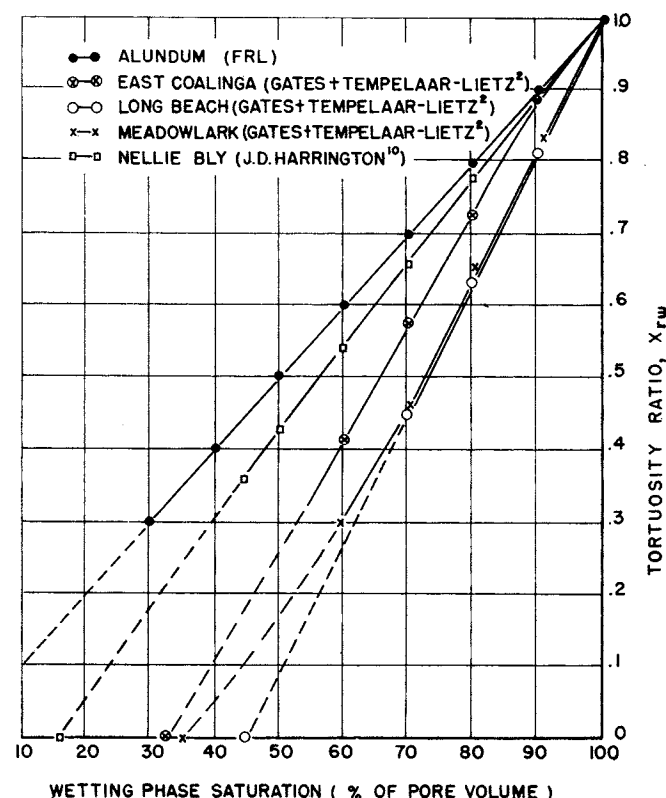


FIG. 1 — CALCULATED TORTUOSITY RATIO CURVES.

where  $V_{iw}$  is the incremental pore volume of the wetting-phase, and  $X_{iw} = \frac{L_{iw}}{L}$ ,  $X_{iw}$  being the tortuosity of the wetting phase and  $L_{iw}$  being the "effective" length of the flow path for the wetting-phase in the two-phase system. Let the tortuosity ratio

$$X_{rwi} = \frac{X_i}{X_{iw}} = \frac{L_i}{L_{iw}} \quad \dots \quad (3)$$

or

$$X_{iw} = \frac{X_i}{X_{rwi}} \quad \dots \quad (4)$$

and, if one assumes that

$$V_{iw} = V_i/S_w \quad \dots \quad (5)$$

then, substituting Equations (4) and (5) in Equation (2) gives

$$K_{ew} = 0.126 \phi \sum_{i=0}^n \frac{X_{rwi}^2 V_i R_i^{-4}}{X_i^2 R_i^2} \quad \dots \quad (6)$$

The equation for the wetting-phase relative permeability therefore becomes

$$k_{rw} = \frac{k_{ew}}{k} = \frac{0.126 \phi}{k} \sum_{i=0}^n \frac{X_{rwi}^2 V_i R_i^{-4}}{X_i^2 R_i^2} \quad \dots \quad (7)$$

The ratio of tortuosities,  $X_{rwi}$ , is the only unknown factor in Equation (7). This equation would otherwise permit the calculation of the wetting-phase relative permeability from pore size distribution or other capillary pressure data. It is assumed

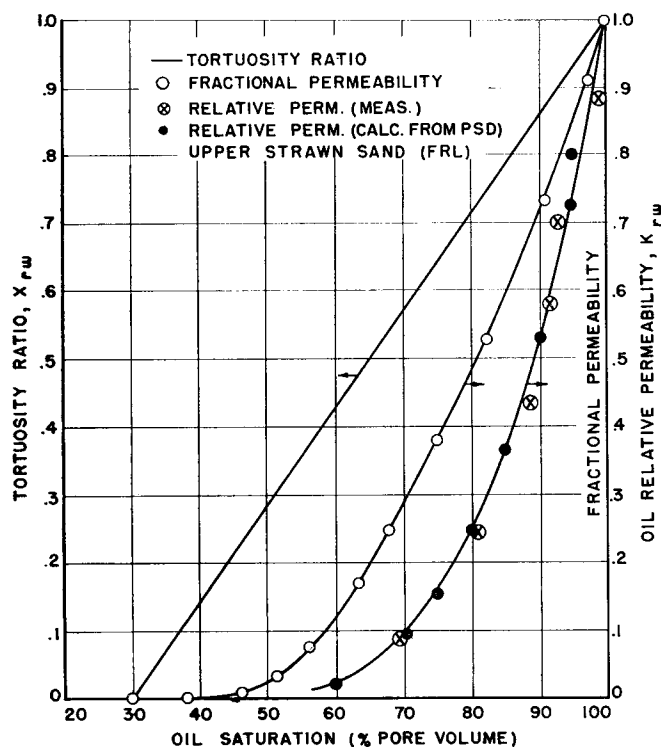


FIG. 2 — RELATIVE PERMEABILITY CURVES. SAMPLE: UPPER STRAWN SAND. GAS PERMEABILITY: 227 MD. EFFECTIVE POROSITY: 21.2 PER CENT. DATA: MEASUREMENTS BY FIELD RESEARCH LABORATORIES CALCULATIONS FROM MERCURY INJECTION DATA.

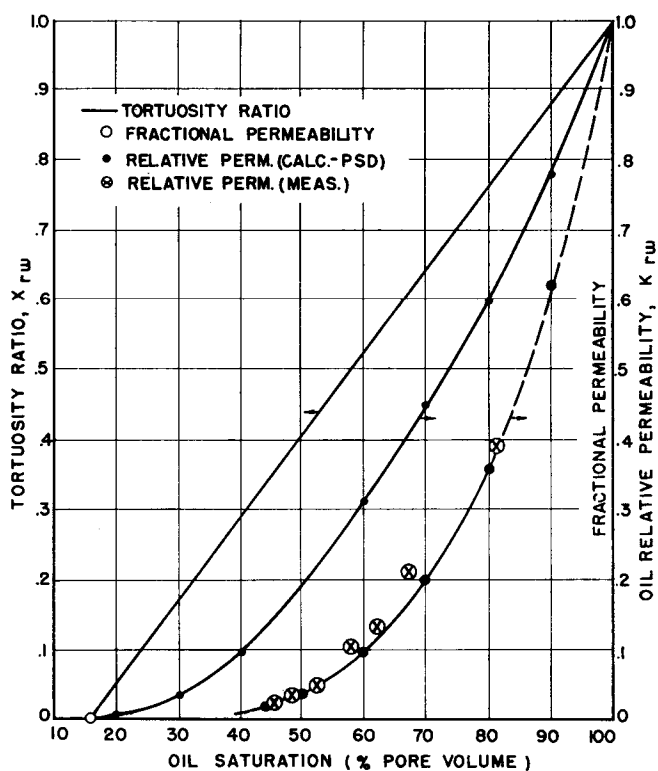


FIG. 3 — RELATIVE PERMEABILITY CURVES. SAMPLE: NELLIE BLY SANDSTONE. GAS PERMEABILITY: 1,080 MD. EFFECTIVE POROSITY: 27.3 PER CENT. FLUIDS: GAS AND OIL. DATA: MEASUREMENTS BY J. D. HARRINGTON.<sup>10</sup> CALCULATIONS FROM MERCURY INJECTION DATA.

that  $X_i$  is known from single-phase permeability and pore size distribution measurements.

By similar reasoning the non-wetting phase effective permeability,  $k_{en}$ , in a two-phase system can be shown to be given by the equation,

$$k_{en} = 0.126 (1 - S_w) \phi \sum_{i=0}^n \frac{V_{in} R_i^{-4}}{X_{in}^2 R_i^2} \quad \dots \quad (8)$$

where  $V_{in}$  and  $X_{in}$  are respectively the incremental pore volume and the tortuosity for the non-wetting phase. Let

$$X_{rni} = \frac{X_i}{X_{in}} = \frac{L_i}{L_{in}} \quad \dots \quad (9)$$

where  $X_{rni}$  is the tortuosity ratio and  $L_{in}$  is the "effective" length of the flow path for the non wetting phase.

If one assumes that

$$V_{in} = V_i / (1 - S_w) \quad \dots \quad (10)$$

and if Equations (9) and (10) are substituted in Equation (8),

$$k_{en} = 0.126 \phi \sum_{i=0}^n \frac{X_{rni}^2 V_i R_i^{-4}}{X_i^2 R_i^2} \quad \dots \quad (11)$$

The non-wetting phase relative permeability,  $k_{rn}$ , becomes

$$k_{rn} = \frac{k_{en}}{k} = 0.126 \frac{\phi}{k} \sum_{i=0}^n \frac{X_{rni}^2 V_i R_i^{-4}}{X_i^2 R_i^2} \quad \dots \quad (12)$$

Equation (12) permits the calculation of non-wetting phase relative permeability from pore size distribution data provided the ratio of tortuosities can be determined.

### TORTUOSITY RATIO

Several investigators<sup>1,3,4,6,7,8,9</sup> have studied the tortuosity ratio for multiphase flow in porous media. The large number of variables, resulting from various combinations of fluids and porous media, makes exact solutions to the tortuosity problem highly improbable, but does not eliminate approximate solutions which are within engineering accuracy.

Let Equation (7)

$$K_{rw} = \frac{0.126 \phi \sum_{i=0}^n \frac{X_{rw}^2 V_i \bar{R}_i^4}{X_i^2 \bar{R}_i^2}}{K} = \frac{0.126 \phi X_{rw}^2 \sum_{i=0}^n \frac{V_i \bar{R}_i^4}{X_i^2 \bar{R}_i^2}}{K} \quad (13)$$

where  $X_{rw}$  is an "effective" wetting phase tortuosity factor. When

$$S_w = 1$$

then, by definition

$$X_{rw} = 1$$

and when

$$S_w = S'_w$$

then it is assumed that

$$X_{rw} = 0$$

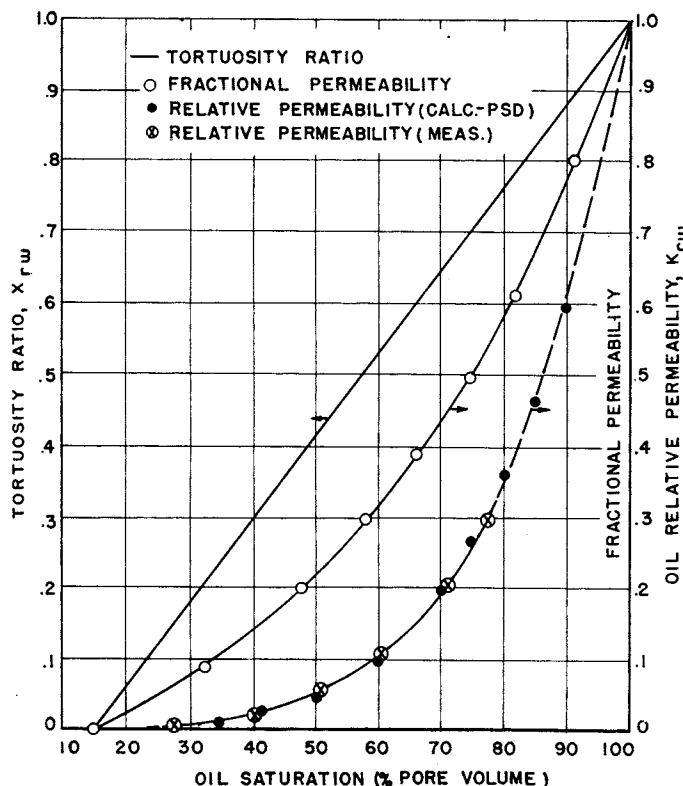


FIG. 4—RELATIVE PERMEABILITY CURVES. SAMPLE: BASAL BROMIDE SANDSTONE. GAS PERMEABILITY 450 MD. EFFECTIVE POROSITY: 16 PER CENT. FLUIDS: GAS AND OIL. DATA: MEASUREMENTS BY J. D. HARRINGTON.<sup>10</sup> CALCULATIONS FROM MERCURY INJECTION DATA.

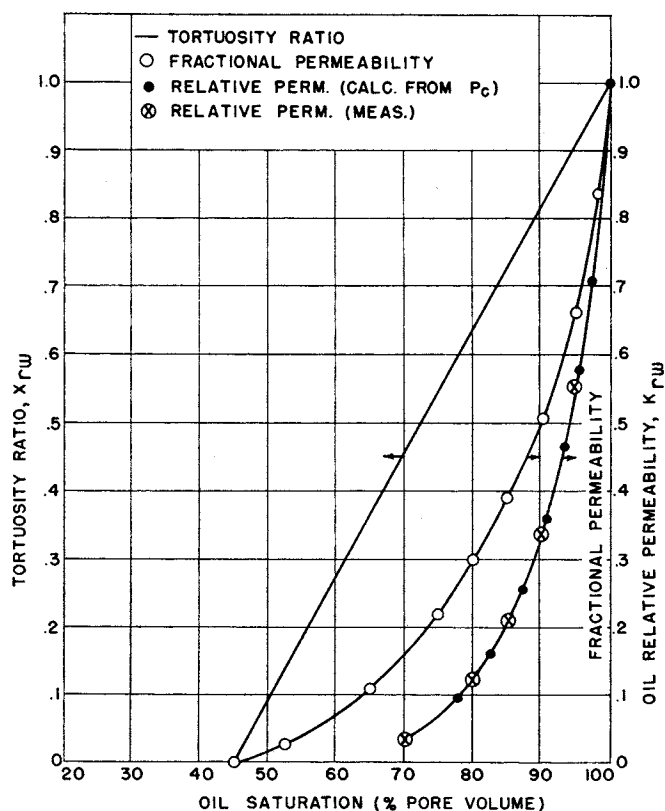


FIG. 5—RELATIVE PERMEABILITY CURVES. SAMPLE: LONG BEACH FIELD. GAS PERMEABILITY: 55 MD. EFFECTIVE POROSITY: 18.2 PER CENT. FLUIDS: AIR AND KEROSENE. DATA: MEASUREMENTS BY GATES AND TEMPELAAR-LIETZ.<sup>2</sup> CALCULATIONS FROM POROUS DIAPHRAGM CAPILLARY PRESSURE DATA.

where  $S'_w$  is the "irreducible" minimum saturation of the wetting phase. Since theoretical values of  $X_{rw}$  are unknown for values of saturation intermediate between the irreducible minimum and one, values of the wetting-phase tortuosity ratio have been calculated at a number of saturations from experimental data obtained in these laboratories and from measurements published by other experimenters.<sup>2,10</sup> Representative results are plotted in Fig. 1. The deviation of these calculated values from a straight line is small, and apparently the deviation is in no preferred direction. As a first approximation, it therefore is assumed throughout the remainder of this paper that  $X_{rw}$  is some linear function of the saturation,  $S_w$ , such as

$$X_{rw} = A S_w + B \quad (14)$$

where  $A$  and  $B$  are undetermined constants. By use of the boundary values referred to above, Equation (14) becomes

$$X_{rw} = \frac{1}{1-S'_w} (S_w - S'_w) \quad (15)$$

### APPLICATION OF FORMULAS

In application of Equation (15) in conjunction with Equation (13) for calculating wetting-phase relative permeability from pore size distribution data, the phases (wetting and non-wetting) are assumed to be distributed in the porous matrix in a particular way. The irreducible minimum saturation of the

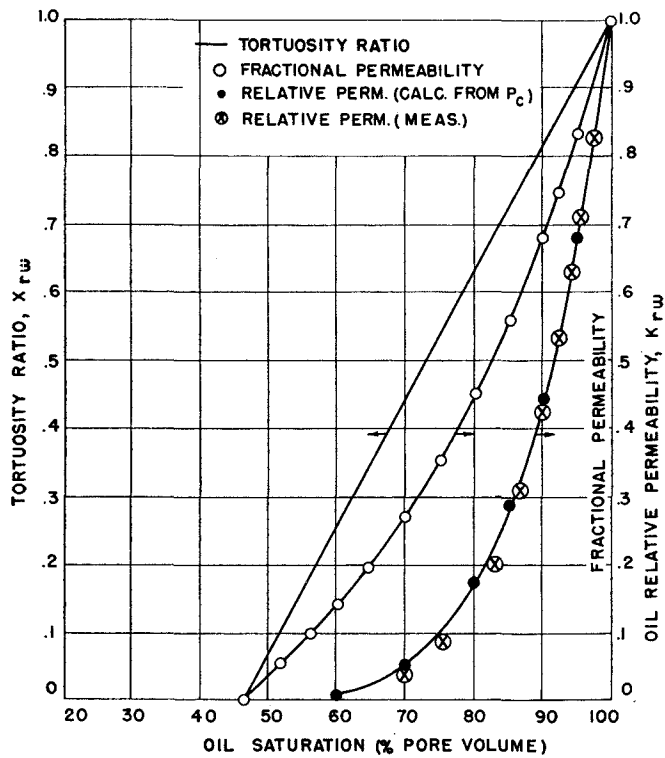


FIG. 6—RELATIVE PERMEABILITY CURVES. SAMPLE: DOMINGUEZ FIELD. GAS PERMEABILITY: 36 MD. EFFECTIVE POROSITY: 26.7 PER CENT. FLUIDS: AIR AND KEROSENE. DATA: MEASUREMENTS BY GATES AND TEMPELAAR-LIETZ.<sup>2</sup> CALCULATIONS FROM POROUS DIAPHRAGM CAPILLARY PRESSURE DATA.

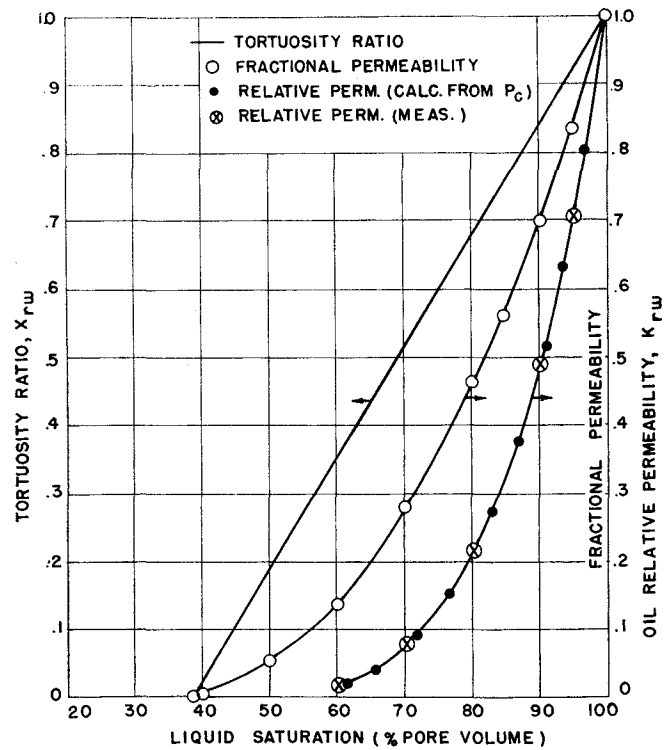


FIG. 7—RELATIVE PERMEABILITY CURVES. SAMPLE: BEREA CANYON FIELD. GAS PERMEABILITY: 27 MD. EFFECTIVE POROSITY: 15.8 PER CENT. FLUIDS: AIR AND KEROSENE. DATA: MEASUREMENTS BY GATES AND TEMPELAAR-LIETZ.<sup>2</sup> CALCULATIONS FROM POROUS DIAPHRAGM CAPILLARY PRESSURE DATA.

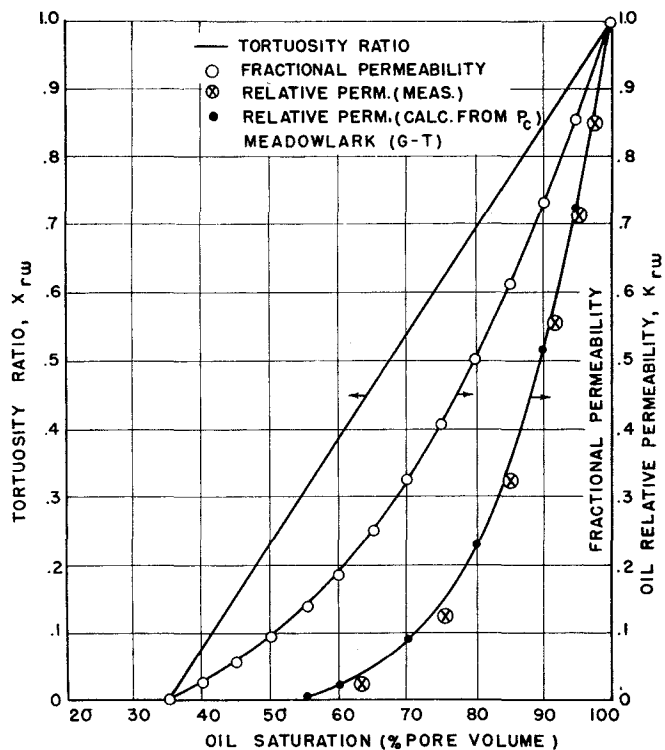


FIG. 8—RELATIVE PERMEABILITY CURVES. SAMPLE: MEADOWLARK FIELD. GAS PERMEABILITY: 370 MD. EFFECTIVE POROSITY: 24.5 PER CENT. FLUIDS: AIR AND KEROSENE. DATA: MEASUREMENTS BY GATES AND TEMPELAAR-LIETZ.<sup>2</sup> CALCULATIONS FROM POROUS DIAPHRAGM CAPILLARY PRESSURE DATA.

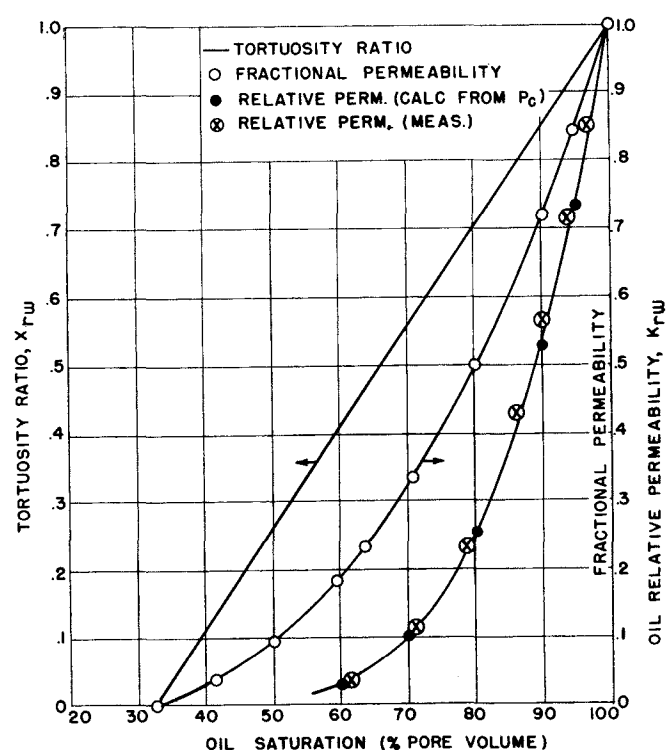


FIG. 9—RELATIVE PERMEABILITY CURVES. SAMPLE: EAST COALINGA FIELD. GAS PERMEABILITY: 318 MD. EFFECTIVE POROSITY: 27.4 PER CENT. FLUIDS: AIR AND KEROSENE. DATA: MEASUREMENTS BY GATES AND TEMPELAAR-LIETZ.<sup>2</sup> CALCULATIONS FROM POROUS DIAPHRAGM CAPILLARY PRESSURE DATA.

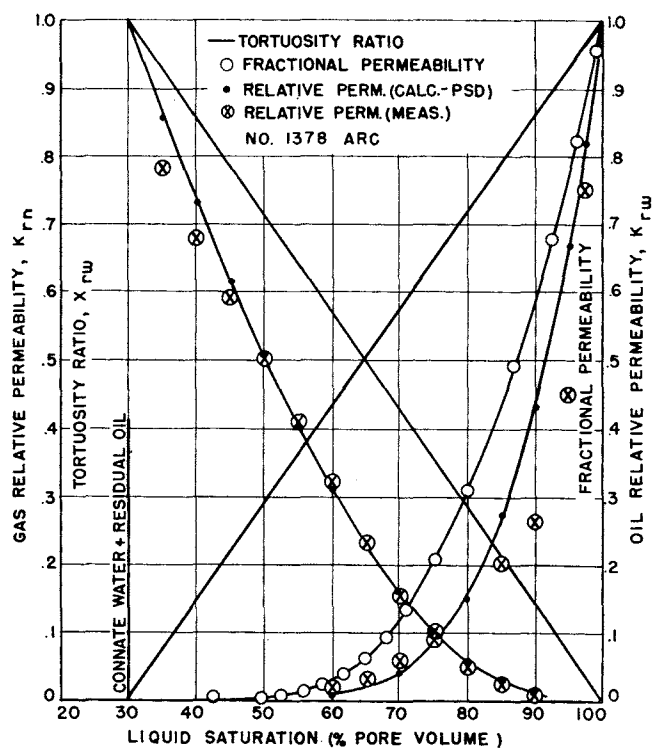


FIG. 10 — RELATIVE PERMEABILITY CURVES. SAMPLE NO. 1378 ATLANTIC REFINING CO. GAS PERMEABILITY: 95 MD. EFFECTIVE POROSITY: 23.8 PER CENT. FLUIDS: GAS, OIL, AND CONNATE WATER. DATA: MEASUREMENTS BY ATLANTIC REFINING CO. CALCULATIONS BY MERCURY INJECTION DATA.

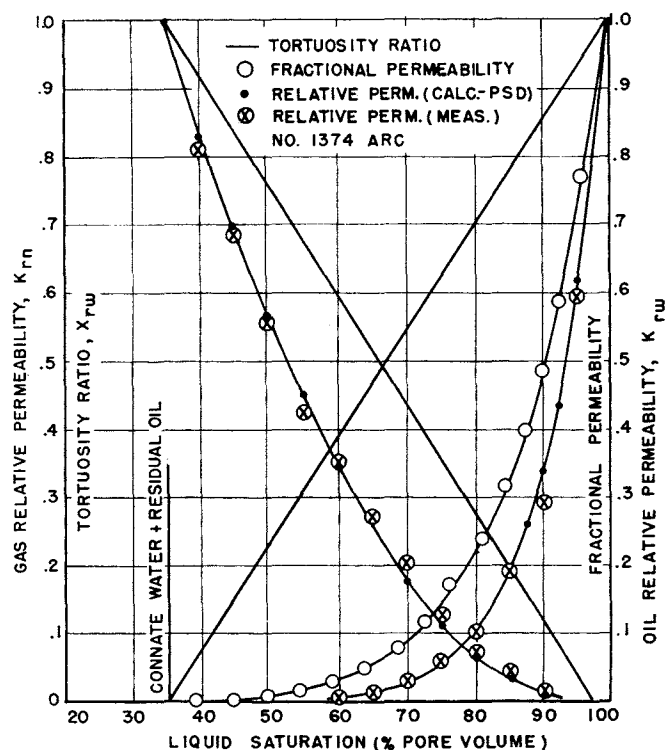


FIG. 11 — RELATIVE PERMEABILITY CURVES. SAMPLE NO. 1374 ATLANTIC REFINING CO. GAS PERMEABILITY: 72 MD. EFFECTIVE POROSITY: 21.9 PER CENT. FLUIDS: GAS, OIL, AND CONNATE WATER. DATA: MEASUREMENTS BY ATLANTIC REFINING CO. CALCULATIONS FROM MERCURY INJECTION DATA.

wetting-phase is assumed to be contained in the smaller pores and the residual non-wetting phase occupies the larger pores according to the mercury injection measurements. For intermediate saturations the permeability to the phase under consideration is adversely affected by the presence of the other phase. This is consistent with the assumption that the wetting-phase surrounds the non-wetting phase which, in turn, occupies the center of the pore space, particularly for media composed of irregular particles, and having crevices, inter-connecting smaller pores, and other irregularities.

The calculation from pore size distribution data of wetting-phase relative permeability in a two-phase system involves the following procedure. The irreducible wetting-phase saturation,  $S'_{w,}$  as determined by the semi-permeable porous diaphragm method or by other capillary pressure methods is substituted in Equation (15). The resulting equation provides a simple linear relation between the tortuosity ratio and the saturation, from which values for the tortuosity ratio may be determined either graphically or by numerical substitution. The single-phase fractional permeability for chosen saturations is then calculated from pore size distribution data. The single-phase fractional permeability is the ratio of the calculated permeability of the pores with radii less than a specified radius to the total permeability of the sample, where the specified radius is determined from the pore size distribution curve for the particular value of fractional volume under consideration. The value of the fractional permeability obtained is multiplied by the square of the tortuosity factor ( $X_{rw}^2$ ) to give the wetting phase relative permeability. The procedure is repeated for a number of other wetting phase saturations ranging from the irreducible minimum to complete saturation. In case the mer-

cury injection experiment has not been performed on the sample under study, approximate pore size distribution data may be estimated from the capillary pressure curve obtained by the porous semi-permeable diaphragm method. This procedure was followed for some of the examples cited below.

Typical results, calculated and measured, are shown in Figs. 2 to 12 for data taken in Magnolia's Field Research Laboratories, Atlantic Refining Co.'s Laboratory, and for other data reported in the literature.<sup>2,10</sup> The linear tortuosity ratio *vs* saturation relationship and the calculated fractional permeability *vs* the saturation are also shown on all figures. In general, the calculated relative permeability values agree very closely with the experimental values for those core samples and fluids studied to date. It must be emphasized that the amount of data now available for correlation purposes is as yet insufficient to arrive at any definite conclusions as to the general validity of the procedure.

Although the major emphasis in this paper is devoted to procedures for calculating wetting phase relative permeability from pore size distribution data, three illustrative examples of calculated non-wetting phase relative permeability are given in Figs. 10, 11, and 12. The procedure for calculating non-wetting phase relative permeability from pore size distribution data is similar to that used for calculating wetting phase relative permeability. The tortuosity ratio for the non-wetting phase (gas) is assumed to have the value of one at the irreducible minimum saturation of the wetting phases consisting of the connate water and the residual oil, and a value of zero at the minimum saturation of the non-wetting phase; this tortuosity ratio is assumed to vary in a linear manner with liquid saturation between these two values corresponding to the limits of

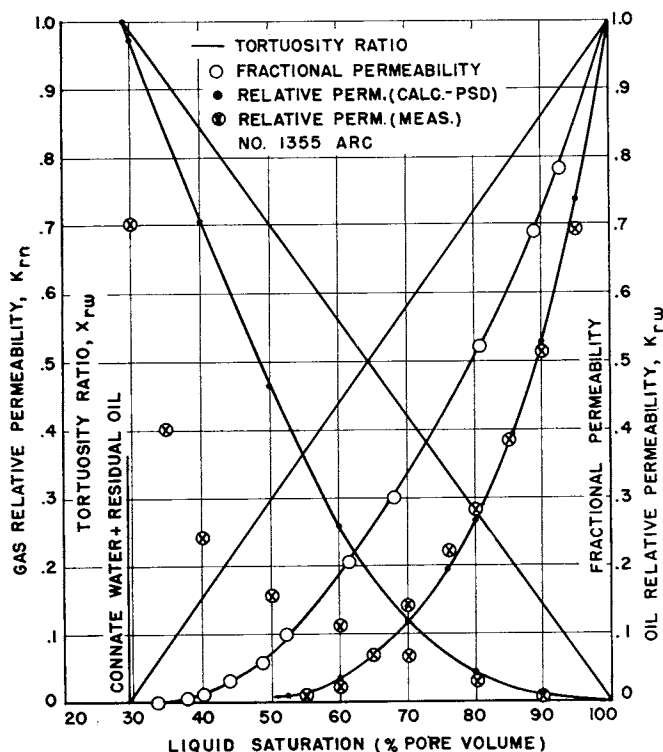


FIG. 12 — RELATIVE PERMEABILITY CURVES. SAMPLE NO. 1355 ATLANTIC REFINING CO. GAS PERMEABILITY: 163 MD. EFFECTIVE POROSITY: 21.3 PER CENT. FLUIDS: GAS, OIL, AND CONNATE WATER. DATA: MEASUREMENTS BY ATLANTIC REFINING CO. CALCULATIONS FROM MERCURY INJECTION DATA.

non-wetting phase saturation. In general, the calculated results are consistent with the measured non-wetting phase relative permeability values.

### DETERMINATION OF TORTUOSITY RATIO BY RESISTIVITY MEASUREMENTS

Thornton<sup>6</sup> has assumed that the average lengths of path for fluid flow and electric current flow in a porous medium are equal. If this assumption is valid, the magnitude of the wetting phase tortuosity ratio,  $X_{rw}$ , could be determined directly by measurement of electrical resistivity of a brine-saturated sample at various saturations.

According to Thornton<sup>6</sup>

$$I = \frac{1}{S_w X_{rw}} \quad (16)$$

where  $I$  is the resistivity index. Therefore,

$$X_{rw} = \frac{1}{I S_w} \quad (17)$$

Equation (16) expresses the tortuosity factor as an inverse function of resistivity index and saturation. If Equations (17) and (13) are combined and rearranged with the tortuosity ratio outside of the summation sign, an equation results which is similar to one derived by Wyllie and Spangler:<sup>11</sup>

$$k_{rw} = \frac{0.126 \phi}{F^2 S_w^2 k} \sum_{i=0}^n \frac{V_i R_i^4}{X_i^2 R_i^2} \quad (18)$$

Generally, for partially saturated samples, the experimental results obtained by the author's co-workers indicate that the resistivity index calculated from Equation (18) using measured relative permeability data, is larger than the resistivity index measured by standard methods. One typical example is illustrated in Fig. 13. This could mean that the electrical conductivity of the matrix is greater than zero and, hence, the "effective" ion path is less than the fluid flow path. Further investigations of this phenomenon are in progress.

### CONCLUSIONS

A procedure has been outlined and formulas have been developed for calculating wetting and non-wetting phase relative permeability in two phase systems from pore size distribution data. The tortuosity factors which appear in the equations are assumed to be linear functions of saturation. Some consideration has been given to the measurement of the tortuosity ratio by the electrical resistivity index analogy as proposed by other investigators.<sup>6,11</sup> So far, this method of measuring the tortuosity ratios has been unsuccessful.

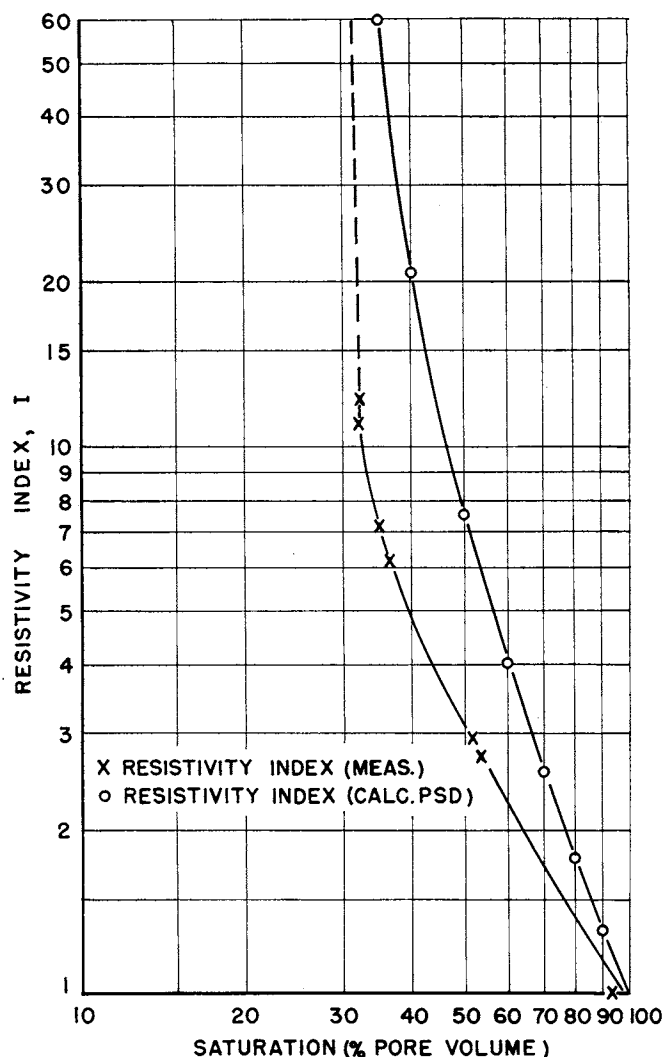


FIG. 13 — RESISTIVITY INDEX CURVES. SAMPLE: MIOCENE SAND. GAS PERMEABILITY: 256 MD. EFFECTIVE POROSITY: 25 PER CENT.

For the samples studied and the fluids used, the calculated relative permeability data agree favorably in most cases with the values measured by standard methods. The results indicate that the calculated data are more consistent and probably contain less maximum error than the measured data. (The error in measurement is unknown.) Of major importance and utility to fluid flow studies in porous media are the speed and ease with which the calculated data may be obtained for either reservoir determinations or for laboratory standards.

Methods and procedures for calculating relative permeability data are being extended and experimental measurements of relative permeability on a variety of samples are being made for correlation purposes. It is emphasized that the results obtained to date are not sufficient for complete evaluation of the method.

### ACKNOWLEDGMENTS

Acknowledgment is made to the management of the Field Research Laboratories, Magnolia Petroleum Co., for permission to publish the results of this investigation. The author wishes to express his appreciation to J. D. Harrington for the release of the experimental data and the use of the core samples on which the data were taken at the University of Oklahoma. Acknowledgment is also due the Atlantic Refining Co. for supplying both experimental relative permeability data and the core samples on which the data were taken. Appreciation is also due L. S. Gournay, L. J. Hirth and W. R. Wallace for their assistance in obtaining the experimental data.

### NOMENCLATURE

|           |                                                                                            |
|-----------|--------------------------------------------------------------------------------------------|
| $A, B$    | — Constants                                                                                |
| $I$       | — Resistivity index                                                                        |
| $k$       | — Permeability (darcys)                                                                    |
| $k_e$     | — Permeability, effective (darcys)                                                         |
| $k_{en}$  | — Permeability, effective non-wetting phase (darcys)                                       |
| $k_{ew}$  | — Permeability, effective wetting phase (darcys)                                           |
| $k_{rn}$  | — Permeability, non-wetting phase relative                                                 |
| $k_{rw}$  | — Permeability, wetting phase relative                                                     |
| $L$       | — Length, sample (cm)                                                                      |
| $L_i$     | — Length, "effective" capillary (cm)                                                       |
| $L_{in}$  | — Length, non-wetting phase "effective" capillary (cm)                                     |
| $L_{iw}$  | — Length, wetting phase "effective" capillary (cm)                                         |
| $R$       | — Radius, pore (cm)                                                                        |
| $R_i$     | — Radius, incremental pore entry, $i = 1, 2, 3, \dots$ (cm)                                |
| $S_w$     | — saturation, wetting phase (fractional)                                                   |
| $S'_w$    | — Saturation, irreducible minimum (fractional)                                             |
| $V_i$     | — Volume, incremental pore, $i = 1, 2, 3, \dots$ (fraction of total pore volume)           |
| $V_{in}$  | — Volume, non-wetting phase incremental pore (fractional)                                  |
| $V_{iw}$  | — Volume, wetting phase incremental pore (fraction of wetting-phase-saturated pore volume) |
| $X_i$     | — Tortuosity factor ( $L_i/L$ )                                                            |
| $X_{in}$  | — Tortuosity factor, non-wetting phase ( $L_{in}/L$ )                                      |
| $X_{iw}$  | — Tortuosity factor, wetting phase ( $L_{iw}/L$ )                                          |
| $X_{rni}$ | — Tortuosity ratio, non-wetting phase ( $L_i/L_{in}$ )                                     |
| $X_{rwi}$ | — Tortuosity ratio, wetting phase ( $L_i/L_{iw}$ )                                         |
| $\phi$    | — Porosity (fractional)                                                                    |
| $X_{rw}$  | — Tortuosity factor, "effective"                                                           |

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### DISCUSSION

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Burdine has presented an interesting modification of one of the equations which has been proposed for calculating relative permeabilities from pore size distribution or capillary pressure data. The equation Burdine uses for calculating relative permeabilities is derived from a very simple model of a porous medium. The model, a bundle of parallel tubes, is so very much over-simplified that calculated and measured relative permeability often do not agree. To bring these into better agreement, the concept of tortuosity has been introduced. Tortuosity, being difficult to evaluate experimentally, is empirically adjusted to improve the fit of calculated data to measured data. An improvement of fit by such empirical adjustment cannot be accepted as making the parallel tube model more valid nor the tortuosity concept more sound.

Burdine's treatment of the tortuosity terms,  $X_i$  and  $X_{rw}$ , seems to be questionable. The difficulty goes back to his earlier paper<sup>1</sup> where the tortuosity  $X_i$  was defined as the tortuosity of tubes of radius  $R_i$ . If this definition is adhered to, then  $X_i$  cannot be taken out of the summation (in Equation 17 of his earlier paper) when  $X_i$  varies with  $R_i$ . In extending his equation to the calculation of relative permeability, Burdine has postulated that tortuosity varies with radius and therefore saturation. However, in establishing the empirical tortuosity corrections, he has taken the tortuosity term out of the summation. The tortuosity term is then a complex average for the pores which are conducting fluid. This inconsistency makes Burdine's method of improving the fit of calculated-to-measured relative permeability strictly an empirical one. The relation between Burdine's tortuosity correction term and the concept of varying tortuosity with varying pore radius is only indirect.

Although the agreement between calculated and measured wetting phase relative permeability indicated in this paper seems adequate for engineering purposes, the same cannot be said for the agreement between calculated and measured non-wetting phase relative permeability. Burdine's method does not include the concept of a critical non-wetting phase saturation.

tion, although this quantity is of great engineering importance. In Figs. 10 and 12, the critical non-wetting phase saturation is taken as zero. In Fig. 11, a finite critical saturation is shown, but Burdine does not indicate how this could be obtained from capillary pressure data. Engineering calculations require very accurate non-wetting phase permeabilities in the region of the critical saturation; the method of calculating these permeabilities, as proposed by Burdine, does not appear adequate. The reader could better evaluate the method if the non-wetting phase relative permeabilities were plotted as semilogarithmic graphs.

Burdine assumes that mercury injection data and water-oil or water-air capillary pressure data are equally useful for calculating relative permeabilities. If the mercury injection curve differs from the water-oil or water-air capillary pressure curve by only a pressure multiplying factor, Burdine's assumption is correct. There are, however, some data which indicate that for many reservoir samples there is no simple relationship between the mercury injection curve and the water-oil or water-air capillary pressure curve.

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1. Burdine, N. T., Gournay, L. S. Reichertz, P. P.: "Pore Size Distribution of Petroleum Reservoir Rocks," *Trans. AIME*, (1950) 189, 195.

#### AUTHOR'S REPLY TO MR. FATT

Fatt states that a very simple model has been used to represent a porous system and that, for this reason, agreement between calculated and experimental relative permeability values is often poor. We agree as to the simplicity of the model and to the frequent disagreement observed but point to the significant number of cases in which really remarkable agreement is observed. This very agreement, coupled with the desire, as in all scientific effort, to retain the maximum of simplicity led to the consideration of possible modifications in the use of the capillary tube model which might result in more frequent and superior agreement. That such modifications as have been used required the use of empirical procedures was not unexpected. In the absence of more fundamental approaches, the use of empirical methods seems entirely valid. Whether such empirical procedures are able to make "the parallel tube model more valid" or "the tortuosity concept more sound" depends solely on the predictive capacity of the resulting relationships and we consider this to be a matter as yet undetermined.

Referring to the tortuosity terms,  $X_i$  and  $X_{rw}$ , Fatt states correctly that if the original definition of  $X_i$  in Equation (17) of an earlier paper is adhered to, then  $X_i$  cannot, in general, be removed from the summation sign. If, however, the "effective" capillary lengths of all the pores are known or assumed to be equal, the factors  $X_i^2$  in Equation (17) of reference 1, may be removed from the summation and replaced by a "dividing factor,"  $f$ , where

$$f = \frac{\sum_{i=0}^n X_i^2}{X_i^2}$$

Hence Equation (17) may be rewritten as

$$K = \frac{100 \phi}{8Cf} \sum_{i=0}^n \frac{V_i \bar{R}_i^4}{\bar{R}_i^2}$$

An analysis of pore size distribution data for petroleum reservoir rock samples indicated that the tortuosity factor,  $X_i$ , must be considered a function of the pore radius. This resulted in the formulation of the empirical formula, Equation (18) of reference (1),

$$X_i = 5e^{-0.4 R_i} + 1.83$$

Since  $X_i$  is an inverse function of the pore radius,  $R_i$ , it has been retained inside the summation sign for all single phase and multiphase permeability calculations.

Referring to Equations (7) and (12) of the present paper, the tortuosity ratios are assumed to be functions of the pore radii and, as such, the tortuosity ratio symbols should be written  $X_{rw_i}$  and  $X_{rn_i}$ , for the wetting and non-wetting phases, respectively. Since no methods, either experimental or theoretical, have been developed to evaluate the tortuosity ratios,  $X_{rw_i}$  and  $X_{rn_i}$ , an approximate procedure has been employed with Equations (7) and (12) in an attempt to circumvent this difficulty. On this basis Equation (7) has been rewritten in the following manner:

$$K_{rw} = \frac{0.126 \phi}{K} \sum_{i=0}^n \frac{X_{rw_i}^2 V_i \bar{R}_i^4}{X_i^2 \bar{R}_i^2} = \frac{0.126 \phi X_{rw}^4}{K} \sum_{i=0}^n \frac{V_i \bar{R}_i^4}{\bar{R}_i^2}$$

where  $X_{rw}$  may be considered an "effective" wetting phase tortuosity factor that is a function of the saturation only. It is the empirical evaluation of the factor  $X_{rw}$  as a function of saturation which is the main subject of the paper.

As pointed out by Fatt, the calculated non-wetting phase relative permeability data presented in the original paper do not show the same order of agreement with experiment as observed in the case of wetting phase data on which was placed, to quote the paper, "the major emphasis." In fact, in the paper no strong claim was made for the use of the method for the calculation of non-wetting phase relative permeability values. It was stated only that results "consistent" with measured data were obtained. Further, contrary to the statement made by Fatt, the method of calculation of non-wetting phase relative permeability does include the concept of critical saturation. It is explicitly stated in the paper that the tortuosity ratio for the non-wetting phase has "a value of zero at the minimum saturation of the non-wetting phase," i.e., at the "critical non-wetting phase saturation" referred to by Fatt. The value of saturation must be determined, of course, by independent means and, insofar as the author is aware, cannot be obtained from capillary pressure data. Such independent data also are required in the case of calculation of either non-aqueous wetting phase relative permeability or non-wetting phase relative permeability in the presence of interstitial water, in which case the minimum non-aqueous wetting phase saturation and the minimum interstitial water content are also required for the calculations (see Figs. 10, 11 and 12 of original paper).

Fatt states that the author has assumed that mercury injection data and water-oil or water-air capillary pressure data are equally useful for calculating relative permeabilities. Actually the paper, by stating that "approximate pore size distribution data may be estimated from the capillary pressure curve obtained by the porous semi-permeable diaphragm method" implies that the mercury injection technique is preferred. The author has not investigated in any detail the use of data obtained by the porous diaphragm method and is unable to generalize on the relative usefulness of the two types of capillary pressure data. It is to be presumed, however, that the data from the porous diaphragm method will prove superior to the data from the mercury method for samples where interaction between fluids and the porous matrix occurs during the course of a relative permeability determination. No experimental verification of this presumption exists, to the best of the author's knowledge.

★ ★ ★

# Multiphase flow predictions from carbonate pore space images using extracted network models

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[1] A methodology to extract networks from pore space images is used to make predictions of multiphase transport properties for subsurface carbonate samples. The extraction of the network model is based on the computation of the location and sizes of pores and throats to create a topological representation of the void space of three-dimensional (3-D) rock images, using the concept of maximal balls. In this work, we follow a multistaged workflow. We start with a 2-D thin-section image; convert it statistically into a 3-D representation of the pore space; extract a network model from this image; and finally, simulate primary drainage, waterflooding, and secondary drainage flow processes using a pore-scale simulator. We test this workflow for a reservoir carbonate rock. The network-predicted absolute permeability is similar to the core plug measured value and the value computed on the 3-D void space image using the lattice Boltzmann method. The predicted capillary pressure during primary drainage agrees well with a mercury-air experiment on a core sample, indicating that we have an adequate representation of the rock's pore structure. We adjust the contact angles in the network to match the measured waterflood and secondary drainage capillary pressures. We infer a significant degree of contact angle hysteresis. We then predict relative permeabilities for primary drainage, waterflooding, and secondary drainage that agree well with laboratory measured values. This approach can be used to predict multiphase transport properties when wettability and pore structure vary in a reservoir, where experimental data is scant or missing. There are shortfalls to this approach, however. We compare results from three networks, one of which was derived from a section of the rock containing vugs. Our method fails to predict properties reliably when an unrepresentative image is processed to construct the 3-D network model. This occurs when the image volume is not sufficient to represent the geological variations observed in a core plug sample.

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## 1. Introduction

[2] In the Middle East, more than 60% of the hydrocarbon reserves are contained in carbonate reservoirs [Konyuhov and Maleki, 2006]. The design of improved oil recovery in these fields requires better reservoir characterization. At the fundamental scale, pore scale, carbonate rocks can be highly irregular because of their complex geological history, including diagenesis by animal organisms and chemical reaction. An understanding of flow and transport processes in carbonates remains a challenge.

[3] Pore network models provide a tool to understand and predict single and multiphase flow and transport in porous media [Bryant and Blunt, 1992; Blunt et al., 2002; Valvatne and Blunt, 2004; Piri and Blunt, 2005a, 2005b; Øren et al., 1998; Al-Futaisi and Patzek, 2003]. To build predictive models, topologically representative pore throat networks

need to be used. The starting point for generating these networks is a three-dimensional (3-D) image of the pore space.

### 1.1. Three-Dimensional Rock Images for Carbonates

[4] Micro-CT scanning can image the pore space of rocks with a resolution of around a few microns [Dunsmuir et al., 1991]. However, most carbonate pore systems, with submicron features, lie below the resolution of CT scanning tools. It is possible to image the pore space in two dimensions using thin sections and electron microscopy. From this a statistically representative 3-D image may be generated. In this paper we will use multiple-point statistics that attempt to reproduce typical patterns in the pore space and the long-range connectivity of the rock [Okabe and Blunt, 2004, 2005; Al-Kharusi and Blunt, 2007].

### 1.2. Network Extraction

[5] Various techniques have been proposed to construct topologically representative networks representing sandstones from 3-D images [Lindquist and Lee, 1996; Lindquist and Venkatarangan, 1999; Lindquist, 2002; Vogel and Roth, 2001; Silin et al., 2003; Silin and Patzek, 2006; Arns et al., 2003, 2004a, 2004b; Al-Raoush et al., 2003; Al-Raoush and

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Willson, 2005a, 2005b; Delerue and Perrier, 2002; Delerue et al., 1999]. To date, however, the only networks that have then been used for successful predictions of multiphase flow properties are those generated from representations of the pore space derived from simulating the packing and cementation of grains, the process-based method [Øren et al., 1998; Øren and Bakke, 2003]. The networks generated by Øren and Bakke used grain size distribution information and other petrographical data obtained from 2-D sandstone thin sections. Then, knowing the grain centers, pores and throats in an equivalent network model could be identified. However, for carbonates, with a complex diagenetic history, it is not easily possible to generate a representative grain packing.

[6] Sheppard et al. [2005] used a medial axis transform pioneered by Lindquist and coworkers [Lindquist and Lee, 1996; Lindquist and Venkatarangan, 1999; Lindquist, 2002] with morphological measures to extract networks from large micro-CT images. Arns et al. [2005] imaged a range of carbonate core and outcrop material using their in-house micro-CT imaging system. The porosity and permeability computed on these 3-D images compared well with experimental data. They confirmed the large difference between the topological properties of carbonates and sandstones.

[7] In this work, we use the maximal ball concept of Silin et al. [2003] and Silin and Patzek [2006] to extract a network. This approach identifies pores by growing spheres in the void space; chains of spheres of decreasing size represent throats. We have shown that networks for both Berea and Fontainebleau sandstones have similar topological properties to those extracted from the same images using a process-based approach by Øren and Bakke [Al-Kharusi and Blunt, 2007]. We further constructed network models from carbonate subsurface samples and compared absolute permeability values with laboratory measurement and lattice Boltzmann calculations. In this paper we will construct additional carbonate networks for which we will predict multiphase flow properties and compare to experimental measurements.

### 1.3. Multiphase Fluid Flow Predictions

[8] Lattice Boltzmann techniques can be used to compute multiphase flow properties, but because of the high computational expense to simulate quasi-static displacement, their use has been limited [Gunstensen and Rothman, 1993; Chen and Doolen, 1998; Hazlett et al., 1998]. Semiempirical methods for predicting multiphase flow properties have been suggested, such as combining the *Mualem* [1976] relative permeability model with the *van Genuchten* [1980] effective capillary pressure model [Dullien, 1992]. The capillary pressure is first matched and then relative permeability is predicted. However, the method assumes a strongly water-wet medium and uses parameters that may not properly encapsulate the displacement processes and pore space geometry.

[9] Pore network models provide an elegant method to compute two and three-phase flow properties; semianalytic expressions for quasi-static flow and displacement in individual pores and throats are combined to make predictions for small samples, 10–100 pores across (a few mm) [Øren et al., 1998; Bakke and Øren, 1997; Øren and Bakke, 2003; Blunt et al., 2002; Valvatne and Blunt, 2004; Al-Futaisi and Patzek, 2003; Lerdahl et al., 2000; Piri and Blunt, 2005b;

Svirsky et al., 2007; Suicmez et al., 2007]. These models can simulate any sequence of oil, water and gas injection in elements (pores or throats) of arbitrary wettability (contact angle). Each element can have an angular cross section. This allows wetting fluids to reside, connected, in the corners while the nonwetting phase occupies the centers.

[10] One key issue is the assignment of contact angles in the model. While the network captures the geometric structure of the rock, it does not provide information about its wettability that has a significant impact on relative permeability and capillary pressure [Valvatne and Blunt, 2004]. Most, if not all, reservoir rocks are not strongly water wet. Current models account for wettability alteration due to contact of crude oil with the rock surface after primary drainage and allow the contact angles to be different for waterflooding and oil reinvasion [Øren and Bakke, 2003; Valvatne and Blunt, 2004]. In carbonates this is of particular significance since most reservoir carbonates display oil-wet characteristics [Dullien, 1992].

[11] Valvatne and Blunt [2004] predicted relative permeability for a carbonate sample using a quasi-static network model. They used a network based on Berea sandstone and modified the pore and throat size distributions to match the primary drainage capillary pressure. They then assigned contact angles to match the measured residual oil saturation. In this paper instead we use a network derived directly from a carbonate sample and so we can predict the primary drainage capillary pressure without tuning the network properties. However, we do still require a method to assign contact angles.

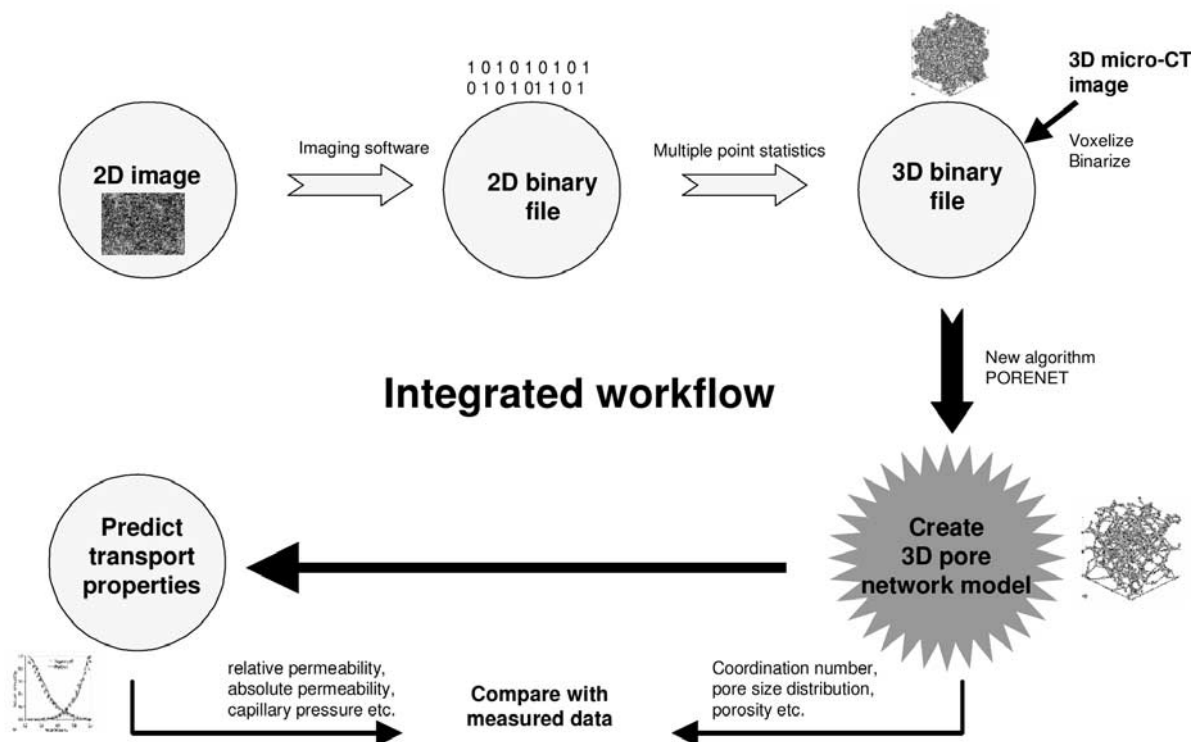
## 2. Carbonate Network Extraction and Comparison With Laboratory Data

### 2.1. Workflow for Carbonates

[12] The network extraction algorithm uses the maximal ball concept originally described by Silin et al. [2003]. A complete description of the algorithm is provided in Al-Kharusi and Blunt [2007] but the essential features are repeated here for completeness.

[13] Maximal balls are first computed: they are the largest spheres centered on any voxel in the image that just fit in the pore space. The largest maximal balls, that have no overlapping neighbors that are larger, identify pores. Throats are found by following chains of balls between these pores. Pores can be constituted from either a single maximal ball or a group of balls adjacent to or overlapping each other. Throats are made of balls decreasing in diameter, connecting two maximal balls, or a group of neighboring balls equal in size [Al-Kharusi and Blunt, 2007].

[14] The pore body and pore throat volumes are computed by referring each pore or throat to the original image voxels comprising the spheres involved. Once the volumes are determined then an equivalent effective pore body radius is calculated by equating the pore body volume obtained to the volume of an equivalent sphere. In the case of a throat, an equivalent effective throat radius is calculated by equating the throat volume obtained to an equivalent cylindrical volume. The length of the cylinder is the maximum distance computed between any two spheres comprising the throat entity. Note that these shapes are only used to find effective



**Figure 1.** The integrated multistaged workflow approach used in this study.

radii but the actual shapes assumed for flow computations are governed by the shape factors described below.

[15] Shape factors, the ratio of cross-sectional area of an element to perimeter squared, are assumed to have a truncated Weibull distribution ranging from a sharply edged triangular shape (having a shape factor value of 0.01) to an equilateral triangular shape (shape factor of 0.048) with a standard deviation of 0.008. In this study, investigation of shape factors of half that of an equilateral triangle (0.024) has also been looked at, refer to Section 2.5. A proportion of the pore space to represent clays that always remain full of water, is assigned to the model; it is given a fixed, small value of 0.5% to match the measured irreducible water saturation in the experiments.

[16] The extracted network is input into a two-phase pore-scale simulator, originally developed by *Valvatne and Blunt* [2004]. Absolute permeability and formation factors are calculated when the network is fully saturated with a single phase. For comparison purposes, absolute permeability was also directly computed on the 3-D image using the lattice Boltzmann method [*Al-Kharusi and Blunt*, 2007; *Okabe and Blunt*, 2004].

[17] In this study, a multistaged workflow was implemented where we started from easy-to-obtain data (a 2-D image) and finished by predicting difficult-to-measure data such as relative permeability. This is achieved in five different stages (Figure 1). The first stage is to acquire a representative 2-D image of the void space. This is done by imaging a thin section of the rock using electron microscopy at a resolution of less than  $1\ \mu\text{m}$ . The second stage is to threshold and then to digitize the image. The third stage is to convert the image into a 3-D representation of the pore space. The fourth stage is to extract a topologically representative network model of pores and throats from the 3-D

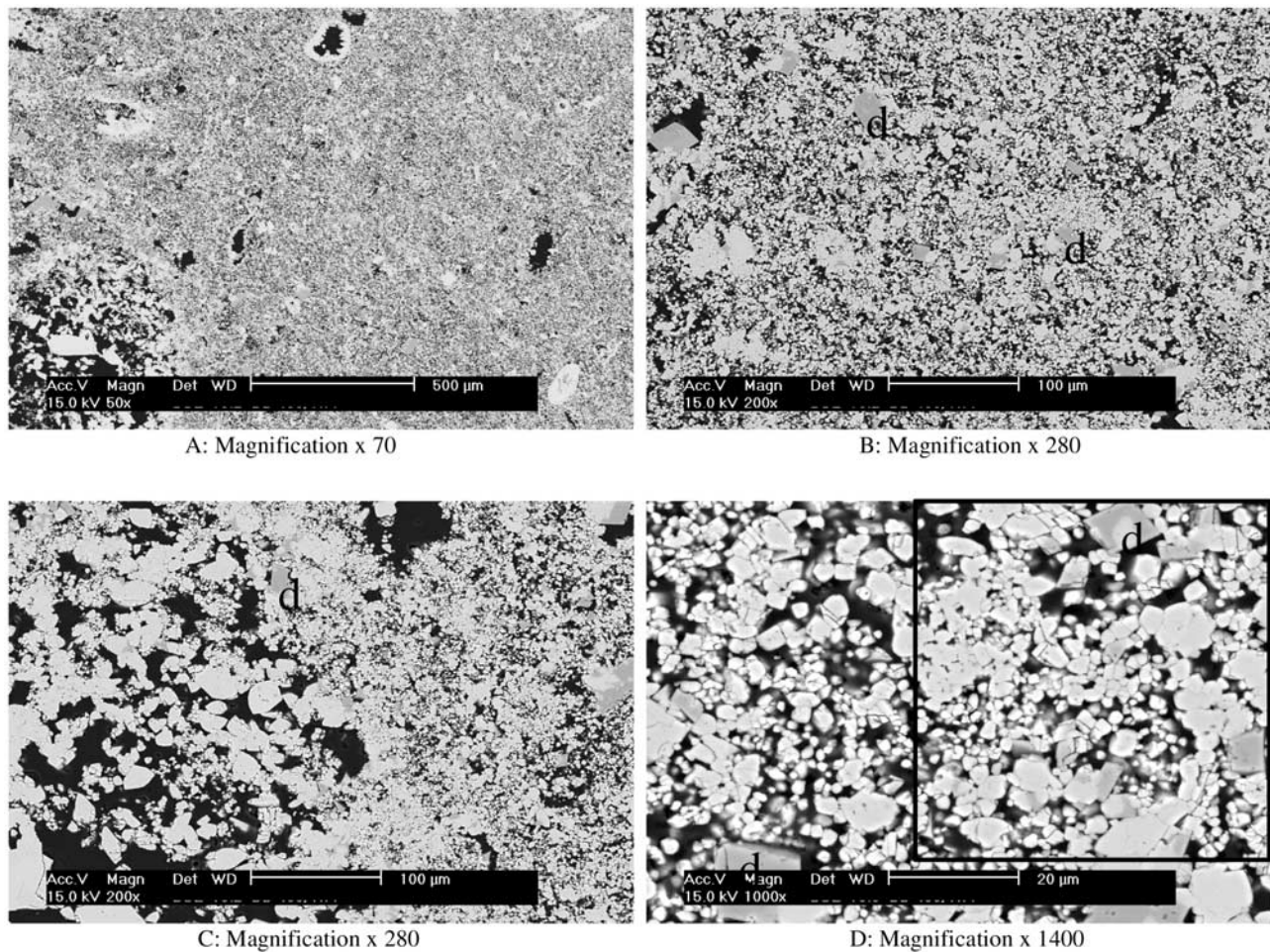
image. Finally we input the extracted network into a pore-scale flow simulator to compute multiphase transport properties. At the end of this section we compare predictions from three networks extracted from different core plugs.

## 2.2. Construction of the Pore Network Model

[18] A cretaceous subsurface carbonate core plug from a giant Middle Eastern field containing light oil (viscosity  $1.4\ \text{mPa}\cdot\text{s}$ ) was chosen for the analysis. A thin section was cut and imaged at various resolutions using scanning electron microscopy, SEM, (Figure 2) for the core plug investigated. Two-dimensional high-resolution SEM images are used here as we cannot obtain 3-D images with the required resolution to image all the pore space for this carbonate sample. The plug chosen contains few vugs and has a relatively homogeneous pore structure.

[19] Figure 2 illustrates the characteristics of the pore system in this compacted and leached peloidal packstone (Figure 2a). Note the presence of some isolated vugs/molds (in solid black) scattered in a predominantly microporous matrix (microporosity is shown in gray). Granular components are not distinguishable any more (Figure 2b), probably because of the combined effect of compaction, pervasive leaching and recrystallization. Dolomite crystals (marked d) are scattered throughout the matrix, postdating calcite. Figure 2c shows a detail taken at the boundary between a dissolution vug and the matrix. This shows that the mesopore within the vug is surrounded by, and connected to, a porous system dominated by micro pores. These are generally well connected at the microscopic scale (Figure 2d).

[20] An image with sufficient resolution to observe the majority of the pore space was selected (Figure 3). The image was thresholded into black and white and then

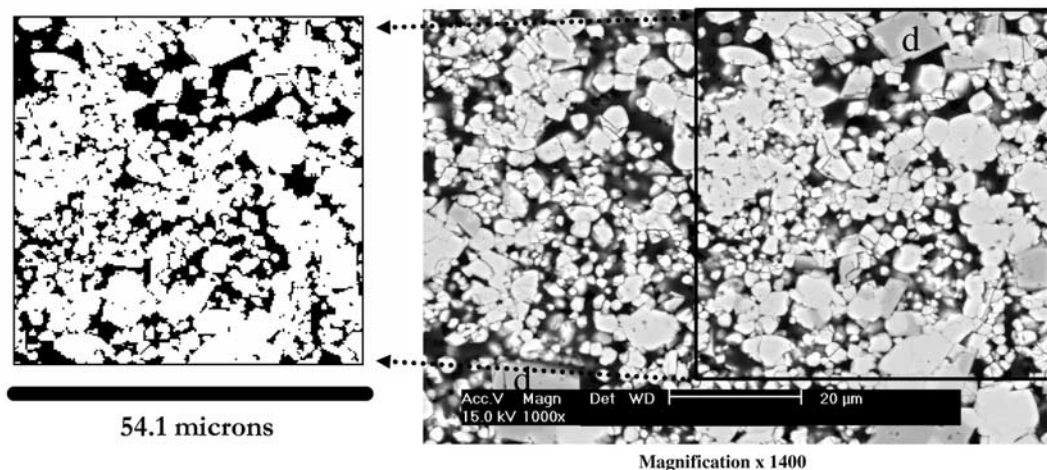


**Figure 2.** (a–d) SEM images at different resolutions taken from a carbonate subsurface core plug. The square in Figure 2d highlights the region used to generate a 3-D image of the pore space. This sample was chosen as it has few vugs and a relatively homogeneous pore structure.

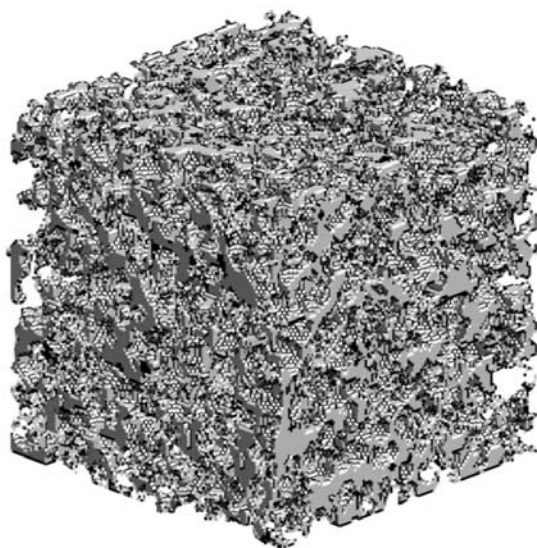
digitized. The image used in this analysis is of size  $54.1 \mu\text{m}$ , taken at a resolution of  $0.27 \mu\text{m}$  ( $200^2$  total voxels). The image at this stage is still two dimensional. Note that for this carbonate, the pore space could not readily be resolved

using micro-CT scanning and therefore no 3-D image is available.

[21] The digitized 2-D image above was input into an algorithm developed by *Okabe and Blunt* [2004] that uses



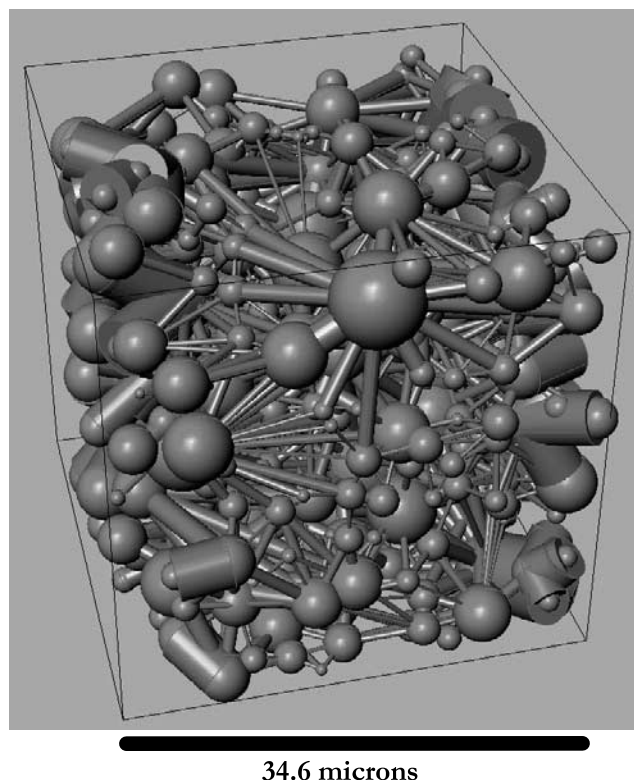
**Figure 3.** Two-dimensional SEM image of the carbonate sample, from which the 3-D network was extracted (black areas denote pore spaces; image was digitized into 200 by 200 voxels with a resolution of  $0.27 \mu\text{m}$ ).



**Figure 4.** Three-dimensional carbonate image obtained using multiple-point statistics ( $128^3$  voxels).

multiple-point statistics to construct a digitized 3-D image through reproducing the same patterns seen in the 2-D image. It is assumed that the image is isotropic. The resultant 3-D image is given in Figure 4.

[22] The 3-D image was then input into the network extraction code, generating a topologically representative network of the carbonate rock. The carbonate network contains 643 pores and 2623 throats. The network is shown



**Figure 5.** The extracted pore network for the carbonate sample obtained from the image shown in Figure 4. The network contains 643 pores and 2623 throats.

**Table 1.** Static Properties of the Carbonate Network Model

| Property                        | Value                 |
|---------------------------------|-----------------------|
| Image size ( $\mu\text{m}$ )    | 34.6                  |
| Image volume ( $\text{mm}^3$ )  | $4.14 \times 10^{-5}$ |
| Image total voxels              | $128^3$               |
| Number of pores                 | 643                   |
| Number of throats               | 2623                  |
| Average connection number       | 7.9                   |
| Number of connection at inlet   | 41                    |
| Number of connections at outlet | 64                    |
| Net porosity (%)                | 29                    |
| Absolute permeability (mD)      | 3.1                   |
| Formation factor                | 7.2                   |

in Figure 5. The network extraction ran in approximately 5 days and reached a maximum processing memory of approximately 16 GB on a Linux cluster. The multiphase flow simulations in contrast took less than a minute to run.

### 2.3. Comparison of Predicted and Measured Static Properties

[23] The network properties are given in Table 1. Table 2 gives a comparison of the absolute permeability predicted by the carbonate network, computed using the lattice Boltzmann method and the experimentally measured value using the core plug that the image comes from: there is a reasonably good agreement between the network-predicted permeability and that measured on a larger core plug.

[24] In Figure 6, the pore and throat radii are plotted against the number of network elements. As expected the throats have much smaller radii than the pores; the pores are well connected and, on average, are connected to eight other pores. In contrast, sandstones, where the pores and throats are of comparable size also typically have a lower coordination number, of around 4 or lower [Øren and Bakke, 2003]. Also note that all the throats have radii less than  $1 \mu\text{m}$ ; these elements could not be imaged directly in 3-D using micro-CT scanning at current resolutions.

### 2.4. Comparison of Predicted and Measured Multiphase Transport Properties

[25] Below we will briefly summarize the experiments conducted to measure the various two-phase flow parameters for this field. We will then compare the network model results with the experimental results for the same core plug.

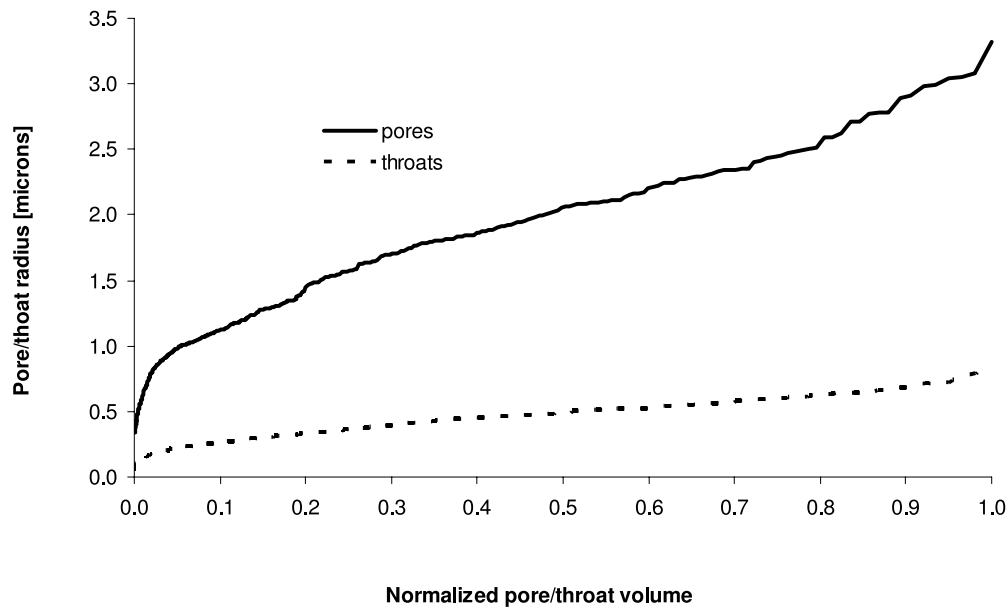
#### 2.4.1. Laboratory Experiments Conducted

[26] For this reservoir, a comprehensive special core analysis (SCAL) program was conducted and many samples were analyzed to capture the large variation in this hetero-

**Table 2.** Comparison of Predicted and Computed Versus Measured Absolute Permeability<sup>a</sup>

|                                        | Predicted<br>Absolute<br>Permeability | Computed<br>Absolute<br>Permeability | Experimental<br>Absolute<br>Permeability |
|----------------------------------------|---------------------------------------|--------------------------------------|------------------------------------------|
| Absolute permeability<br>to brine (mD) | 3.1                                   | 1.9                                  | 3.4                                      |

<sup>a</sup>Predicted permeability is from the network model (this work), computed permeability is from the lattice Boltzmann method, and experimental permeability is from the core plug experiment.

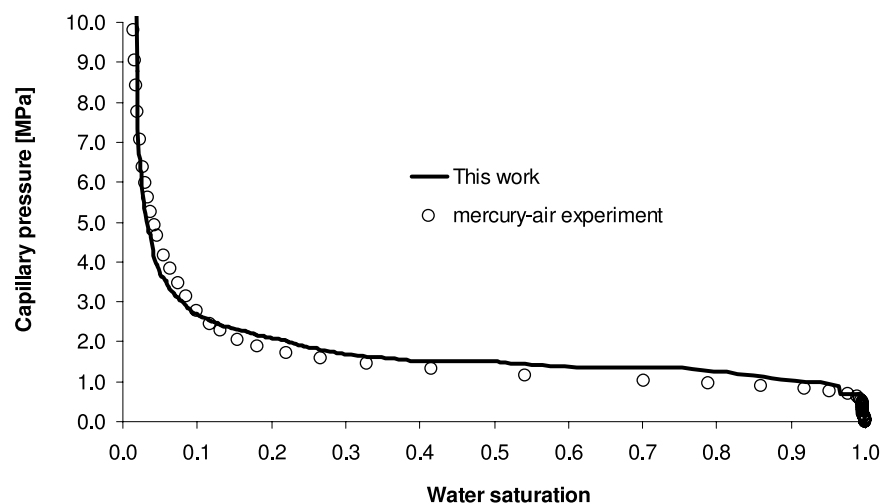


**Figure 6.** Pore and throat radii plotted against network element number.

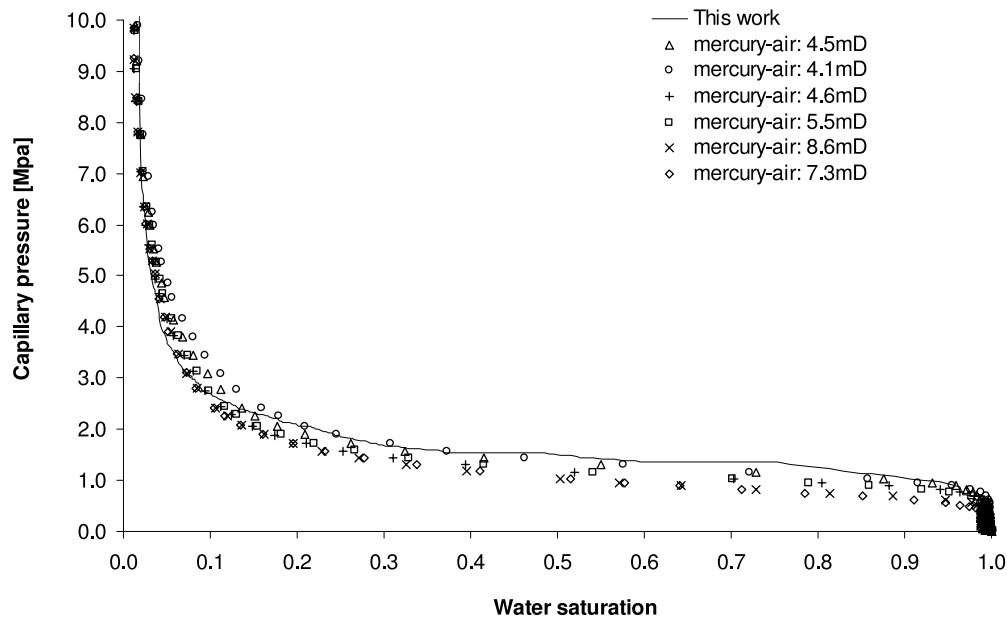
geneous carbonate field. The permeability varies over four orders of magnitude ranging from less than a milliDarcy to more than a Darcy. The porosity of the field ranges between 10 to over 30%; however, most of the initial oil resides in rock types with porosity in a range between 20 and 30%. Core plugs of size 3.78 cm in width and 4.90 cm in length were drilled for various experiments. 250 samples were taken for porosity and permeability measurement. A subset of 80 samples was selected from different permeability and porosity ranges representing key geological facies for the subsequent SCAL program, the details of which have been described by *Masalmeh and Jing* [2004]. The program includes rock characterization, CT scanning, NMR measurements, mercury-air capillary pressure, and analysis of scanning electron microscope thin-section images. The program also includes oil/water primary drainage capillary pressure

measurements, aging the samples in crude oil at irreducible water at reservoir temperatures (120°C) and pressures (27 Mpa) for four weeks to restore wettability and then conducting spontaneous imbibition measurements followed by waterflood and secondary oil injection centrifuge experiments.

[27] 50 samples were selected for mercury-air and water-oil centrifuge capillary pressure measurements. We selected a plug that has a representative value of porosity and permeability for one permeability class: 1–10 mD. The investigation of other permeability classes will be the subject of further work. For mercury injection, mercury is always the nonwetting phase, whereas for the centrifuge experiments, the wettability of the sample represents that of the oil and brine at field conditions. The average saturation in a core plug is recorded at a set of centrifugal speeds and



**Figure 7.** Predictions using the carbonate network compared with laboratory measurements of primary drainage capillary pressure (mercury-air system).



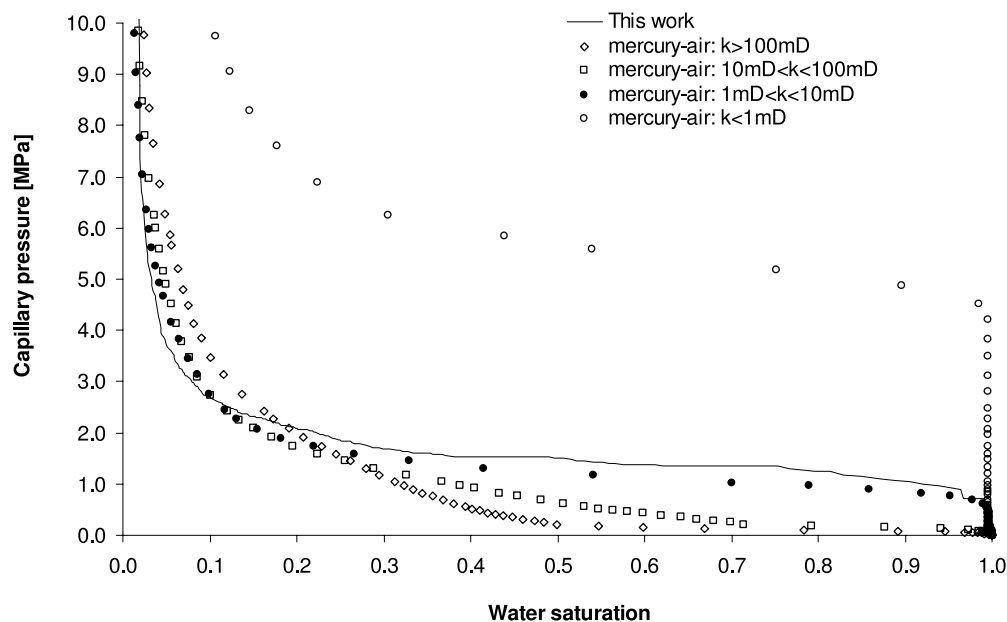
**Figure 8.** Network-predicted versus measured primary drainage capillary pressure: six core plugs with a similar permeability range (mercury-air system).

capillary pressure versus saturation point is calculated at each speed. The capillary pressure can be calculated from the centrifuge raw data using either analytical [Forbes, 1997] or numerical methods [Maas and Schulte, 1997]. Numerical simulations using MoReS, the Shell in-house simulator, were used to model the experimental results and the capillary pressure and displaced-phase relative permeability were found that best fit the measurements [Masalmeh and Jing, 2004]; the centrifuge measurements do not measure the relative permeability of the injected phase.

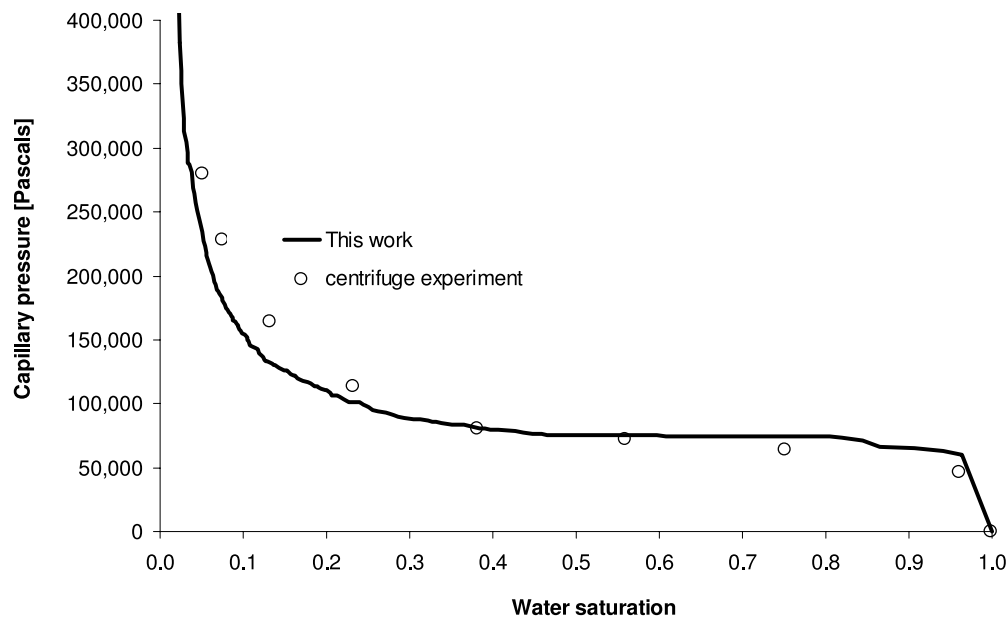
[28] Waterflood capillary pressure was measured for 32 samples. No spontaneous imbibition of water was observed on any of the samples following aging with crude. This indicates that all the carbonate samples had oil-wet characteristics.

#### 2.4.2. Two-Phase Fluid Flow Predictions

[29] The carbonate network generated above is now input into a two-phase, public domain pore-scale simulator [Valvatne and Blunt, 2004] to predict multiphase fluid flow properties. We first predict mercury-air primary drainage



**Figure 9.** Network-predicted versus measured primary drainage capillary pressure: four core plugs from different permeability classes (mercury-air system).



**Figure 10.** Comparison of network-predicted versus measured primary drainage capillary pressure (brine-oil centrifuge experiment).

and compare with the experimental data for the chosen core plug. The core plug we analyzed has a porosity of 29% and an absolute permeability to brine of 3mD. We input a mercury-air system in our carbonate network model with mercury-air interfacial tension of 485 mN/m and a receding contact angle of  $40^\circ$  [Valvatne and Blunt, 2004]. We plot the comparison in Figure 7.

[30] In Figure 8 we show mercury-air primary drainage capillary pressure curves for core plugs of similar permeability from the same field. In Figure 9 we show the curves for core plugs of different permeability classes highlighting the heterogeneity in this field. Note that the capillary pressures for different permeability classes are quite distinct and that our predictions are in the range of behavior seen for the selected permeability with only a slight overestimation of capillary pressure at low saturation.

[31] The fluid properties are then modified to represent the conditions of the displacement experiments. A reservoir brine-oil interfacial tension of 22.5 mN/m, receding contact angle of  $30^\circ$  in primary drainage, brine viscosity of 0.66 mPa s, oil viscosity of 1.4 mPa s, brine density of 1098 kg/m<sup>3</sup> and oil density of 782 kg/m<sup>3</sup> are used. The contact angle was chosen on the basis of the core analysis of Masalmeh and Jing [2004]. We now calculate primary drainage capillary pressure and relative permeability using the brine-oil system.

[32] Relative permeability was only measured on a few core samples. In this section we will compare against one set of measurements for the core plug whose capillary pressure is shown in Figure 7. The capillary pressure and relative permeability predictions compared with experiment for primary drainage are given in Figures 10 and 11.

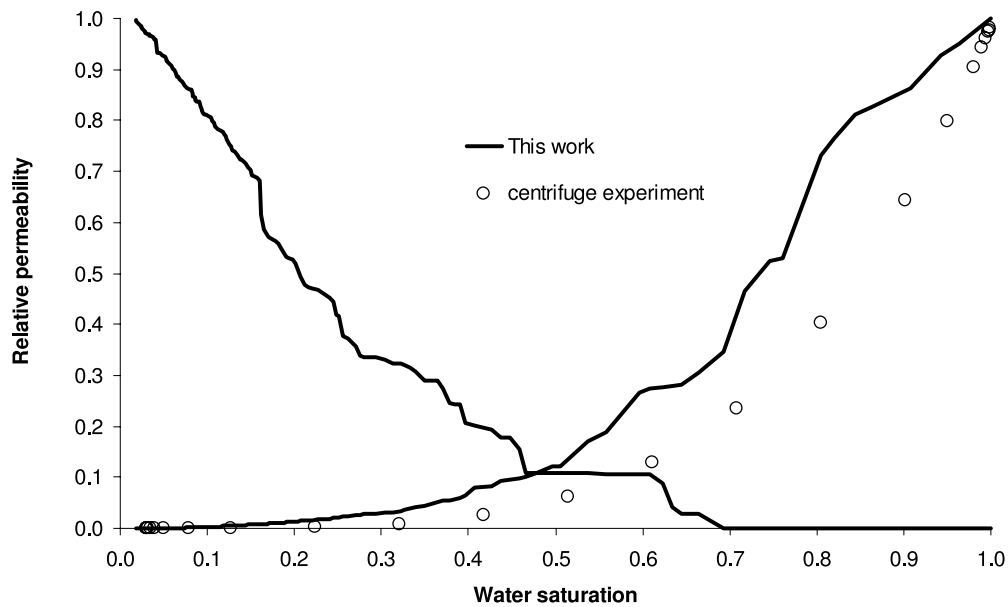
[33] As expected, the network-predicted capillary pressure using a brine-oil system compares well with the centrifuge experiment. Physically, this is an identical displacement process to the mercury-air capillary pressure shown in Figure 7, except for a scaling to account for

interfacial tension and contact angle. Our extracted network captures the pore size distribution and makes good predictions in this case.

[34] The predicted water relative permeability is overestimated compared to the experimental results, although the trend and shape of the curve is correct; the oil relative permeability was not measured. We may overestimate the connectivity of the medium in our network extraction algorithm giving a large coordination number, leading to an overprediction of relative permeability [Al-Kharusi and Blunt, 2007]. Furthermore, we use a very small network that may not be representative of the whole core sample. The good prediction of capillary pressure implies that we have estimated pore and throat sizes correctly, but we would appear to assign too high a flow rate to those larger elements occupied by the nonwetting phase.

[35] Following primary drainage, we inject water in the network. For waterflooding, we modify the contact angles in the network to match the experimentally measured capillary pressure (Figure 12). We distribute contact angles at random to pores and throats in the model. We assume a uniform distribution of contact angle with a range of  $20^\circ$ . We then adjust the mean contact angle to obtain the best match to the capillary pressure; this is a one parameter fit. The waterflood capillary pressure is matched by assuming contact angles representative of a weakly oil-wet system, ranging between  $96^\circ$  and  $116^\circ$ ; see Figure 13. Note that the capillary pressure is entirely negative, showing that there is no spontaneous imbibition of water. This means that the injected water is behaving like a nonwetting phase.

[36] The advancing contact angles used in this work (that is contact angles for water displacing oil) are in line with the value of  $110^\circ$  inferred for this core plug by Masalmeh and Jing [2006]. They developed a mathematical method to calculate contact angles by scaling the primary drainage capillary pressure to derive imbibition capillary pressure.

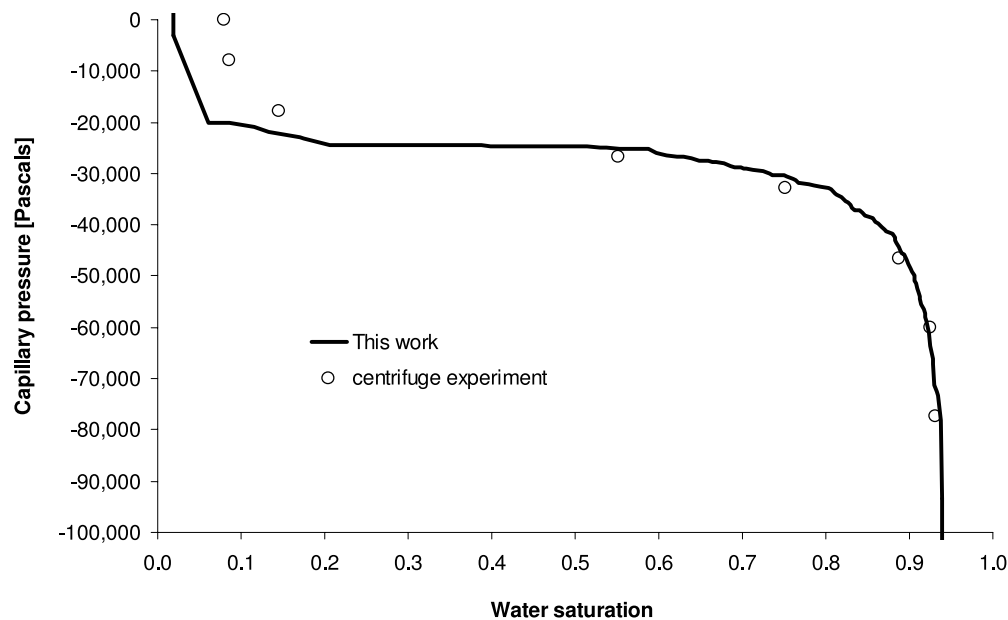


**Figure 11.** Comparison of network-predicted versus measured relative permeability during primary drainage.

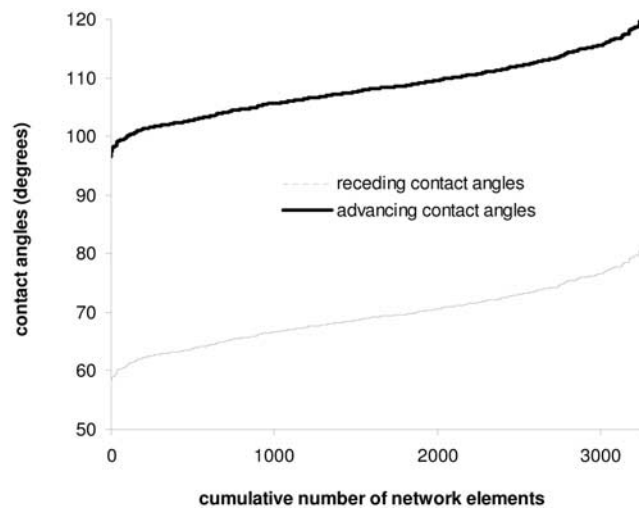
[37] The computed capillary pressure is a good match to the experiments, except at low water saturation. This could be due to the slow displacement of water at low imposed capillary pressure in the experiment, meaning that truly capillary equilibrium conditions have not been reached. Our prediction of the residual oil saturation, around 6%, is excellent. This value is lower than seen in water-wet sandstones, around 30% [Øren and Bakke, 2003; Valvatne and Blunt, 2004], for two reasons. First, because of the oil-wet nature of the rock, oil can form layers in the pore space

during waterflooding sandwiched between water in the corners and in the center [Valvatne and Blunt, 2004]. These layers provide connectivity of the oil down to very low saturation. Second, the high coordination number of the network allows phases to remain connected.

[38] After matching the waterflood capillary pressure we then make a prediction of the water and oil relative permeability as shown in Figure 14. The predicted oil relative permeability is in good agreement with the measured values.



**Figure 12.** Comparison of network-predicted versus measured waterflood capillary pressure (brine-oil centrifuge experiment). The contact angles in the network are adjusted to fit the measurements. A one-parameter fit is used where the contact angles are assigned at random with a range of  $20^\circ$  and the mean is varied. The best match, shown here, is with contact angles between  $96^\circ$  and  $116^\circ$ .



**Figure 13.** The network advancing and receding contact angles plotted against cumulative number of network elements.

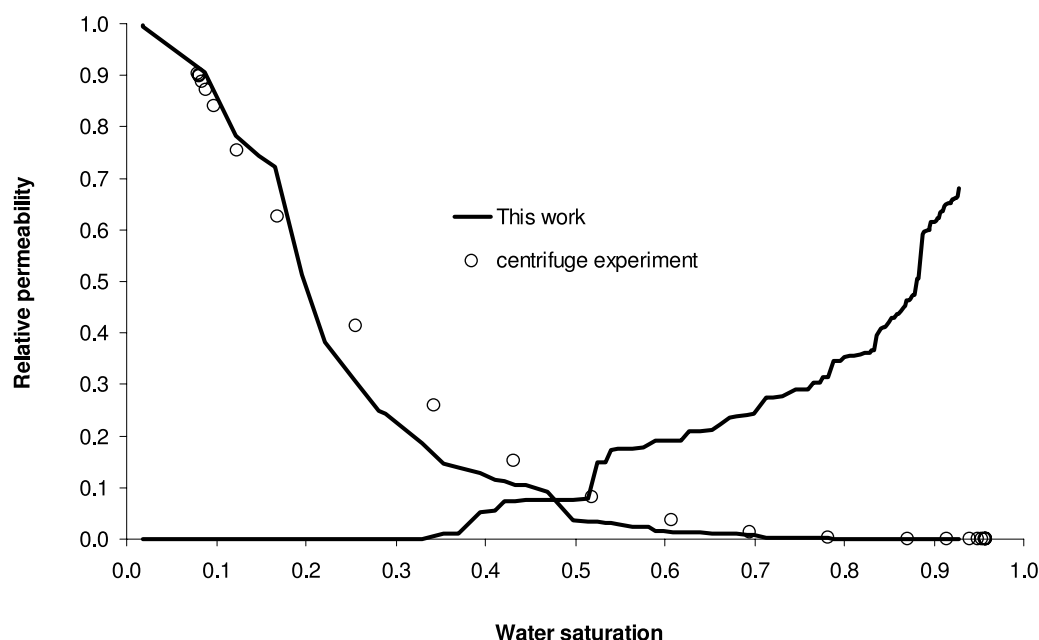
[39] Following waterflooding, we inject oil again. To match the experimental secondary drainage capillary pressure, we allow contact angle hysteresis (Figure 15). The hysteresis is given by a constant linear shift of  $37^\circ$  between the receding (oil displacing water) and advancing contact angles, shown in Figure 13. The receding contact angle estimated by *Masalmeh and Jing* [2006] for this core plug is around  $80^\circ$  which lies at the upper range of our contact angles. Note the significant degree of contact angle hysteresis: while the medium appears to be nonwetting to water during water injection (advancing contact angles greater than  $90^\circ$ ), it is also nonwetting to oil during oil injection (receding contact angles less than  $90^\circ$ ). This degree of

hysteresis is typical for neutrally wet or oil-wet systems [Dullien, 1992].

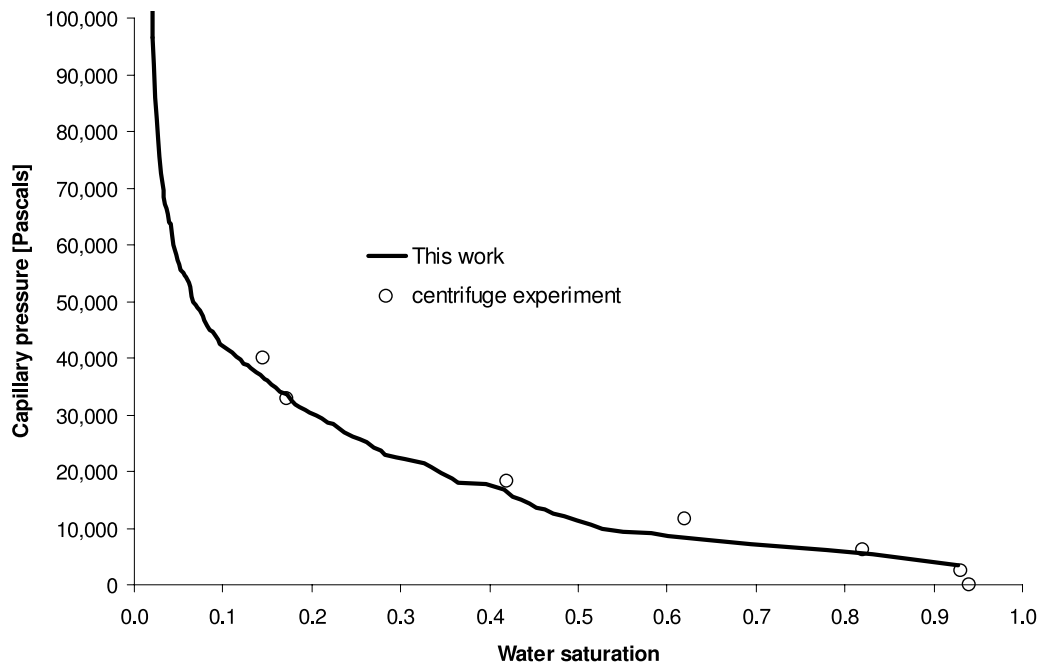
[40] The predicted relative permeability is given in Figure 16. The predicted water relative permeability is again in excellent agreement with the measurements.

[41] Figure 17 shows all the predicted relative permeabilities to highlight the computed hysteresis trends. During primary drainage, there is no trapping of water and, thanks to the high coordination number, both phases are well connected and the relative permeability is high. We would expect the water relative permeability to be higher for waterflooding, where water is the nonwetting phase, than in secondary oil invasion, where it is wetting; this is apparent in Figure 17 for water saturations less than around 0.8. Note that the oil relative permeability is very low beyond a water saturation of 0.5, even though oil continues to flow down to saturations of less than 0.1. The reason for this is that at low oil saturation, the oil is only connected in layers and occupies the centers of very few elements. While these layers do provide continuity of the oil, they have a very low conductance.

[42] The surprising finding is that the oil relative permeability in waterflooding (when oil is the wetting phase) is higher than in secondary oil invasion (when it is non-wetting). During waterflooding, layers of oil are formed sandwiched between water in the center of the pore space and water in the corners; it is these layers that maintain the connectivity of the oil phase down to low oil saturation. During secondary oil invasion, oil now displaces water from the larger pores. This can be done by piston-like displacement, or through the swelling of oil layers. Pores and throats can spontaneously fill with oil from layers as the oil pressure increases. While this leads to an increase in oil saturation, these elements are not well connected; they are only connected to other elements through low conductance layers, and so the relative permeability at a given oil saturation is lower than for both primary drainage (when



**Figure 14.** Comparison of network-predicted versus measured relative permeability during waterflood.



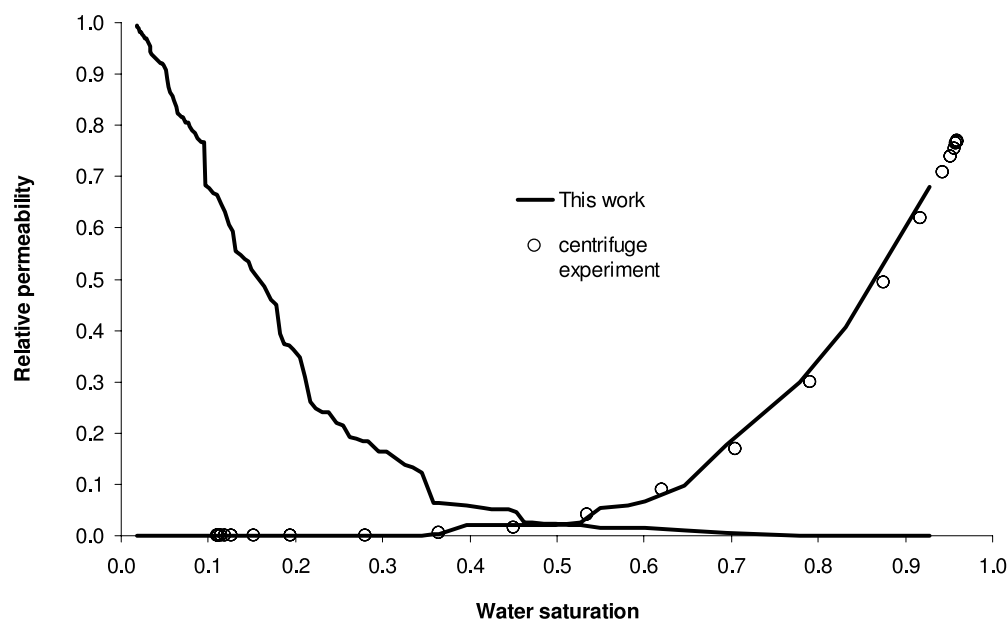
**Figure 15.** Comparison of network-predicted versus measured secondary oil injection capillary pressure (brine-oil centrifuge experiment).

the oil is always connected through the centers of elements) and waterflooding (when the oil is also connected through layers but the saturation for a given conductance is lower). There is no direct experimental evidence for this trend, however, since the centrifuge experiments only measure the relative permeability of the displaced phase.

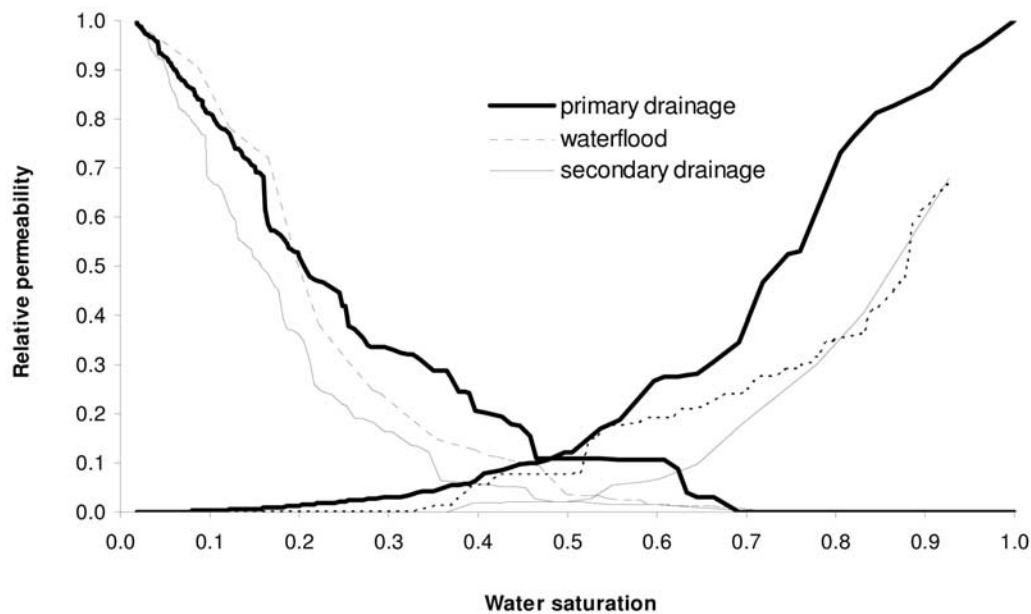
## 2.5. Sensitivity Analysis

[43] All the predictions shown so far have been based on a single network representing a tiny portion of the rock.

Furthermore, we have arbitrarily assigned shapes (or shape factors) to the network. To test the variability in our predictions from extracting networks from images of different plugs, Figure 18 shows the predicted secondary oil invasion relative permeabilities for three networks: plug A shows results from our original network, Figure 16; plug B is a network extracted from a different plug of similar permeability; and plug C is a third network from a different plug of similar permeability that contains vugs. There is significant variation in the predictions: similar variations are



**Figure 16.** Comparison of network-predicted versus measured relative permeability during secondary oil invasion.

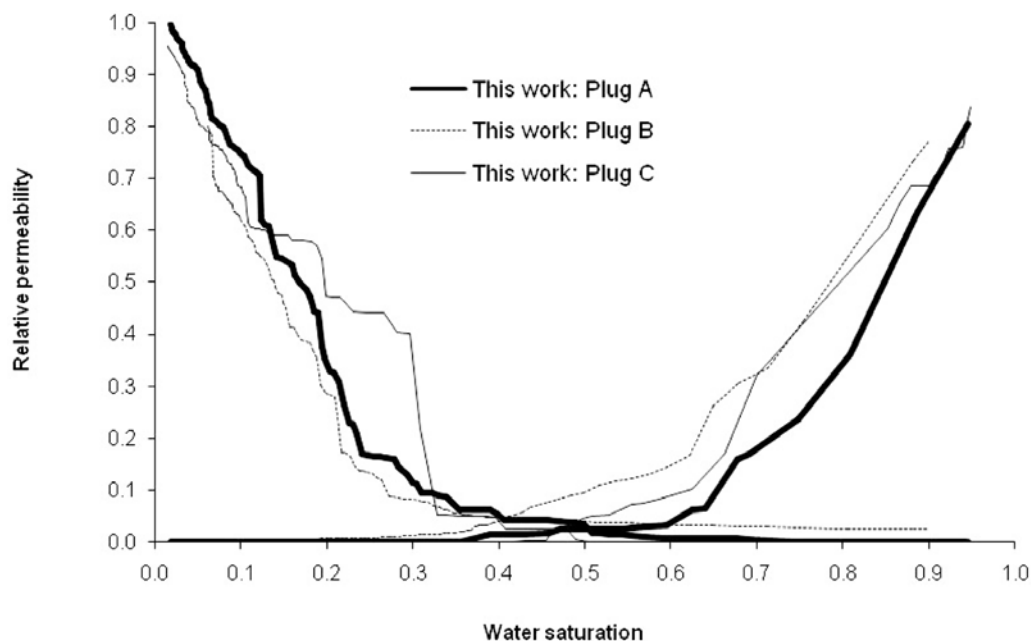


**Figure 17.** Comparison of network-predicted relative permeability data from all three cycles: primary drainage, waterflood, and secondary drainage.

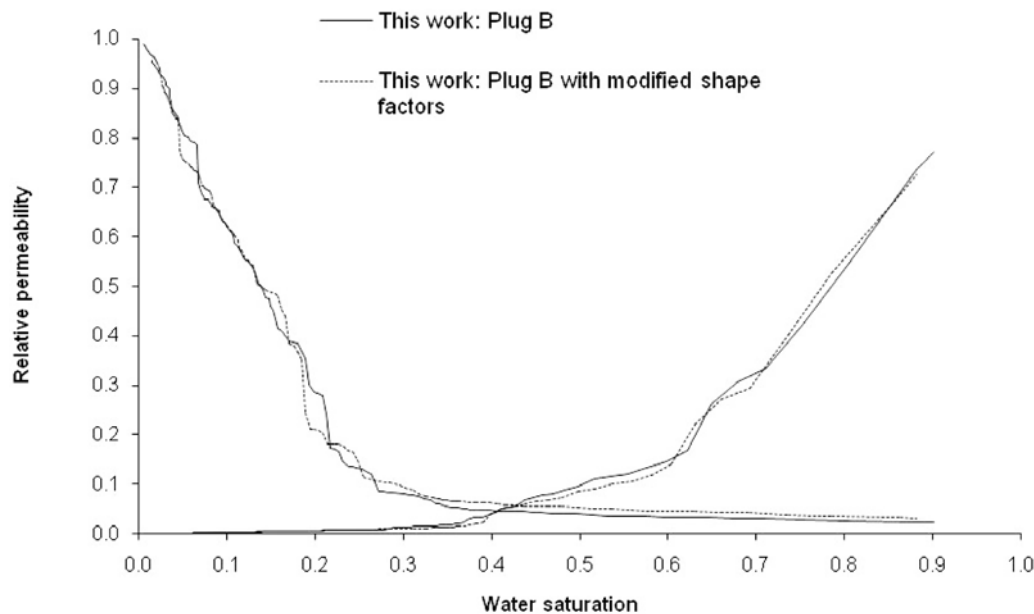
found with other networks and for the other relative permeabilities. In particular the vuggy network leads to relative permeability curves with sharp jumps when large pores become filled. The networks representing plugs B and C now overpredict the water relative permeability for this sample.

[44] To assess the sensitivity of the results to shape factor, Figure 19 compares the predicted secondary oil invasion

relative permeabilities for plug B with a distribution of shape factors and for a fixed value, 0.024, that is half that of an equilateral triangle. 0.024 is approximately the average value of the shape factor distribution applied in all the previous predictions. The results are similar for the two cases, indicating that the exact assignment of shape factor has little impact on the results: the results were insensitive to shape factor for the other networks and for other relative



**Figure 18.** Comparison of predicted relative permeabilities during secondary oil invasion. Plug A refers to the network whose results have been presented previously, plug B uses a network derived from another plug of similar permeability, and plug C has a similar permeability but the pore space contains vugs.



**Figure 19.** Comparison of predicted relative permeabilities for secondary oil invasion for plug B (see Figure 18). The same network topology is used in both cases, but in one a distribution of shape factors for each element is used (as in the previous results), while in the other the shape factor is kept fixed at half the value of an equilateral triangle (0.024).

permeability predictions, as long as we allowed fluid to be retained in the corners of most of the network elements. The conclusion of this section is that there is substantial variability in predictions from different networks, although the results are still consistent with the experimental measurements.

### 3. Discussion, Conclusions, and Recommendations for Future Work

[45] A network to represent the pore space was extracted from a subsurface carbonate core plug. A multistaged workflow was implemented where we started from a 2-D image and finished by predicting relative permeability following the assignment of contact angles to match the measured capillary pressures. We inferred a significant degree of contact angle hysteresis, such that the medium appeared nonwetting to water during water injection and nonwetting to oil during oil injection.

[46] The prediction of the primary drainage capillary pressure was excellent, indicating that our network adequately represented the pore space. With suitable contact angles, we predicted the waterflood residual oil saturation accurately as well as the relative permeability of oil for waterflooding and the relative permeability of water for secondary oil injection. We discovered a surprising hysteresis pattern in relative permeability that was explained by considering the pore-scale configuration of fluids.

[47] While these results are promising, the pore network model may not be representative of the rock investigated when large vugs or other geological variations are present. We saw large variations in predicted relative permeabilities when comparing networks extracted from different core samples with similar permeability. We were limited to constructing networks from images that contained only

$128^3$  voxels, representing a tiny sample of rock that might not be sufficiently large to include key geological features observed in the core plug sample investigated. Vugs and fractures can significantly affect fluid flow at the plug scale and network modeling will fail if these vugs are not properly incorporated into the model. For the systems studied in this work, vugs, although present, were not connected and no consistent impact on relative permeability was apparent.

[48] Further work could focus on generating several networks on the basis of different portions of a core plug, or different plugs, to assess the range of behavior that we might expect. Furthermore, larger networks should be generated, where computer resources permit, allowing a more representative model of the pore space to be generated.

[49] We do not propose a substitute to laboratory measurement; we recommend that pore network modeling should be used to understand the measurements and their associated uncertainties. Pore network modeling could be used to predict transport properties outside the range of parameters studied experimentally. In particular, this workflow could be useful to predict the relative permeability for a reservoir with variable wettability and pore structure across the field where experimental measurements are lacking.

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## RESEARCH ARTICLE

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### Key Points:

- Water retention curve becomes steeper with increasing hydrate saturation
- Relative water and gas permeability is affected by hydrate saturation and morphology
- Brooks-Corey model predicts permeability very well for heterogeneously distributed hydrate system

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## The water retention curve and relative permeability for gas production from hydrate-bearing sediments: pore-network model simulation

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**Abstract** The water retention curve and relative permeability are critical to predict gas and water production from hydrate-bearing sediments. However, values for key parameters that characterize gas and water flows during hydrate dissociation have not been identified due to experimental challenges. This study utilizes the combined techniques of micro-focus X-ray computed tomography (CT) and pore-network model simulation to identify proper values for those key parameters, such as gas entry pressure, residual water saturation, and curve fitting values. Hydrates with various saturation and morphology are realized in the pore-network that was extracted from micron-resolution CT images of sediments recovered from the hydrate deposit at the Mallik site, and then the processes of gas invasion, hydrate dissociation, gas expansion, and gas and water permeability are simulated. Results show that greater hydrate saturation in sediments lead to higher gas entry pressure, higher residual water saturation, and steeper water retention curve. An increase in hydrate saturation decreases gas permeability but has marginal effects on water permeability in sediments with uniformly distributed hydrate. Hydrate morphology has more significant impacts than hydrate saturation on relative permeability. Sediments with heterogeneously distributed hydrate tend to result in lower residual water saturation and higher gas and water permeability. In this sense, the Brooks-Corey model that uses two fitting parameters individually for gas and water permeability properly capture the effect of hydrate saturation and morphology on gas and water flows in hydrate-bearing sediments.

## 1. Introduction

The fundamental expressions for unsaturated soil behavior are anchored in capillarity and water saturation relation  $P_c-S_w$ , typically known as a water retention curve (WRC). The water retention curve essentially captures pore-scale characteristics of the porous media and the gas-fluid-mineral interactions. This curve can be used as an indicator to hydraulic conductivity [Assouline, 2001], soil-water storage [Brady and Weil, 2007], and soil stiffness, strength, and volume changes [Fredlund et al., 1996; Gens and Alonso, 1992; Öberg and Sällfors, 1997; Pedarla et al., 2012]. The relative fluxes of liquid and gaseous phases through porous media are usually captured by relative water and gas permeability  $k_{rw}$  and  $k_{rg}$ . Both the water retention curve and relative permeability are the most critical characteristics to characterize multiphase flow in porous media.

Widely used models of water retention curve and relative permeability [Bear, 1979; Bear and Cheng, 2010; Brooks and Corey, 1964; Corey, 1954; Parker et al., 1987; Stone, 1970; van Genuchten, 1980] contain multiple fitting parameters. The appropriate selection of these parameter values is critical to enhance the prediction accuracy of water and gas production rates for the development of methane gas hydrate deposits [Anderson et al., 2011; Jang and Santamarina, 2011, 2014; Johnson et al., 2011; Kleinberg et al., 2003; Li et al., 2013; Mahabadi and Jang, 2014]. However, this selection has long been a challenge due to the lack of experimental studies that can precisely control and quantify hydrate saturation throughout the tests. Reported direct measurement of relative water permeability in natural sediments is available for Nankai Trough pressure cores after hydrate dissociation [Santamarina et al., 2015]. Recently, percolation theory has been deployed to characterize relative gas and water permeability in hydrate-bearing sediments [Daigle and Rice, 2015;

Ghanbarian et al., 2015]. Yet, none of these studies has been able to capture all the equations and fitting parameters for hydro-thermo-mechanically coupled process analyses in numerical simulators for gas hydrate research [Anderson et al., 2011; Moridis et al., 2008].

This study aims to identify proper parameter values for characterizing the water retention curve and relative permeability in hydrate-bearing sediments using the combined techniques of micro-focus X-ray computed tomography (CT) and pore-network model simulation. The sediment recovered from a hydrate deposit at the Mallik site in Canada is remolded and scanned at in situ stress conditions. A three-dimensional pore network is then extracted from the micron-resolution CT images for pore-network simulation. Hydrate is artificially generated in the pores with various saturations and morphology. The processes of gas invasion, hydrate dissociation, and gas expansion during dynamic gas production are simulated to obtain the proper fitting parameter values for water retention curve and relative permeability of water and gas in hydrate-bearing sediments.

## 2. Fundamentals: Water Retention Curve and Relative Permeability Models

The water retention curve is affected by pore throat size distribution, connectivity and spatial correlation, soil fabric, contact angle, and interfacial tension [Dai and Santamarina, 2013]. Obtaining the water retention curve for sediments with the presence of hydrate faces many experimental challenges, such as long induction time for hydrate formation and poor control of hydrate saturation due to transient hydrate dissolution/dissociation and formation during water and gas flow.

Analytical expressions usually capture the water retention curve using  $P_c$ ,  $S_{rw}$ , and residual water saturation  $S_{rw}$  [Brooks and Corey, 1964; Corey, 1954; Fredlund and Xing, 1994; van Genuchten, 1980]. Most numerical studies on gas hydrates [Anderson et al., 2011; Gamwo and Liu, 2010; Hong and Pooladi-Darvish, 2003, 2005; Moridis and Reagan, 2007; Moridis and Sloan, 2007; Reagan and Moridis, 2008; Reagan et al., 2010; Rutqvist and Moridis, 2007] use the van Genuchten [1980] model:

$$P_c = P_0 \left[ \left( \frac{S_w - S_{rw}}{1 - S_{rw}} \right)^{-\frac{1}{m}} - 1 \right]^{1-m} \quad (1)$$

where  $P_0$  is the gas entry pressure,  $S_{rw}$  is the residual water saturation, and  $m$  is a fitting parameter. Lower  $m$ -value shows steeper  $P_c$ - $S_w$  curve (Figure 1a), typically in sediments with wider pore size distribution.

Additionally, the van Genuchten [1980] model is used to predict the relative water  $k_{rw}$  and gas  $k_{rg}$  permeability [Parker et al., 1987]:

$$k_{rw} = \bar{S}^{0.5} \left[ 1 - \left( 1 - \bar{S}^{1/m} \right)^m \right]^2, \quad \bar{S} = \left( \frac{S_w - S_{rw}}{1 - S_{rw}} \right) \quad (2)$$

$$k_{rg} = (1 - \bar{S})^{0.5} \left[ 1 - \bar{S}^{1/m} \right]^{2m}, \quad \bar{S} = \left( \frac{S_w - S_{rw}}{S_{wmax} - S_{rw}} \right) \quad (3)$$

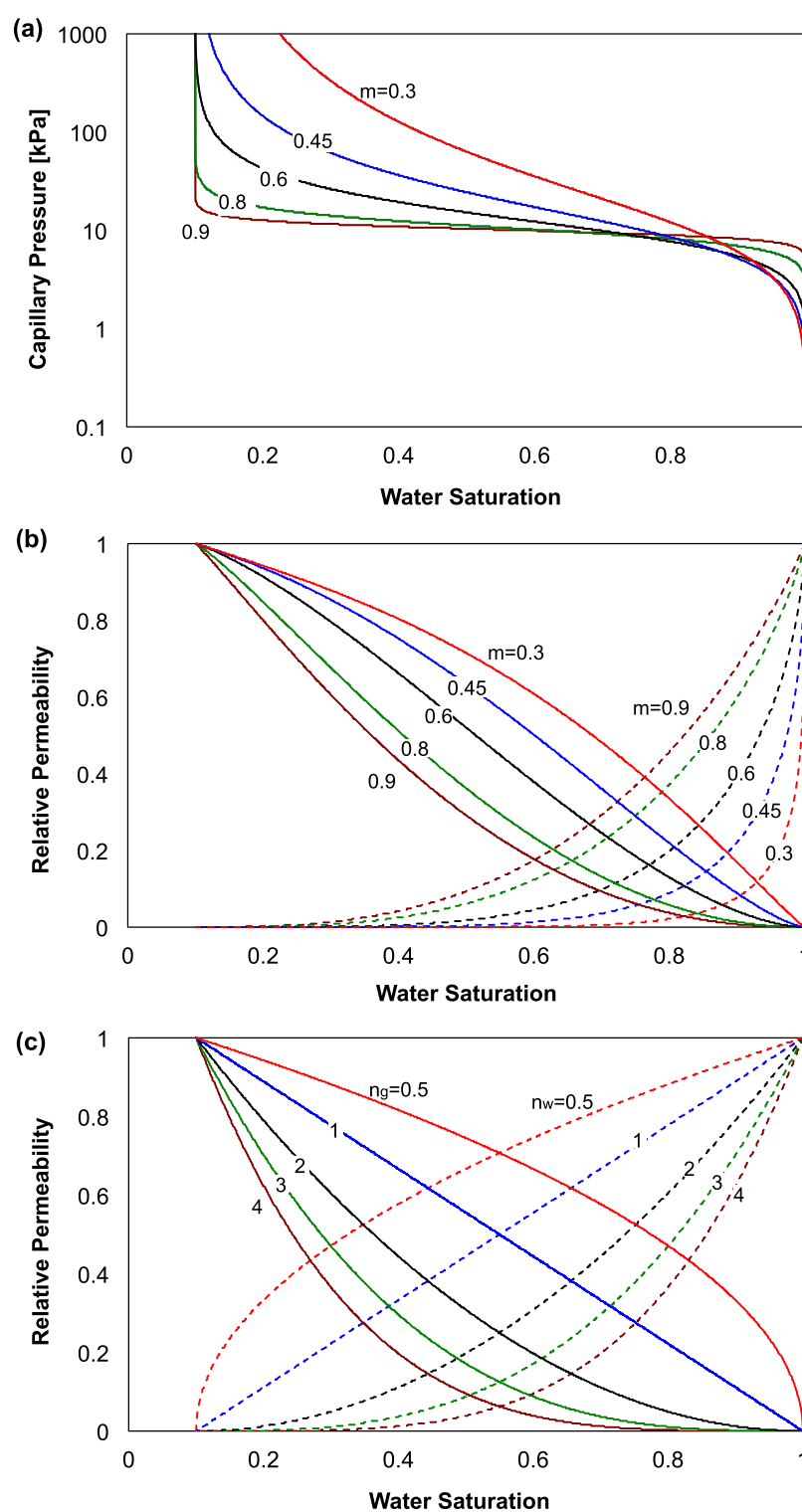
where  $S_{wmax}$  is the water saturation at which gas permeability starts to occur. A lower  $m$ -value depicts lower water permeability but higher gas permeability at a given saturation (Figure 1b).

An earlier relative permeability model [Brooks and Corey, 1964] also frequently used in hydrate simulation studies is:

$$k_{rw} = \left( \frac{S_w - S_{rw}}{1 - S_{rw}} \right)^{n_w} \quad (4)$$

$$k_{rg} = \left( \frac{S_g - S_{rg}}{1 - S_{rw}} \right)^{n_g} \quad (5)$$

where  $S_{rg}$  is the residual gas saturation,  $n_w$  and  $n_g$  are fitting parameters for water and gas permeability. Lower  $n_w$  and  $n_g$  values reflect higher relative water and gas permeability (Figure 1c). Note that two individual fitting parameters,  $n_w$  and  $n_g$ , are used to predict relative water and gas permeability separately in



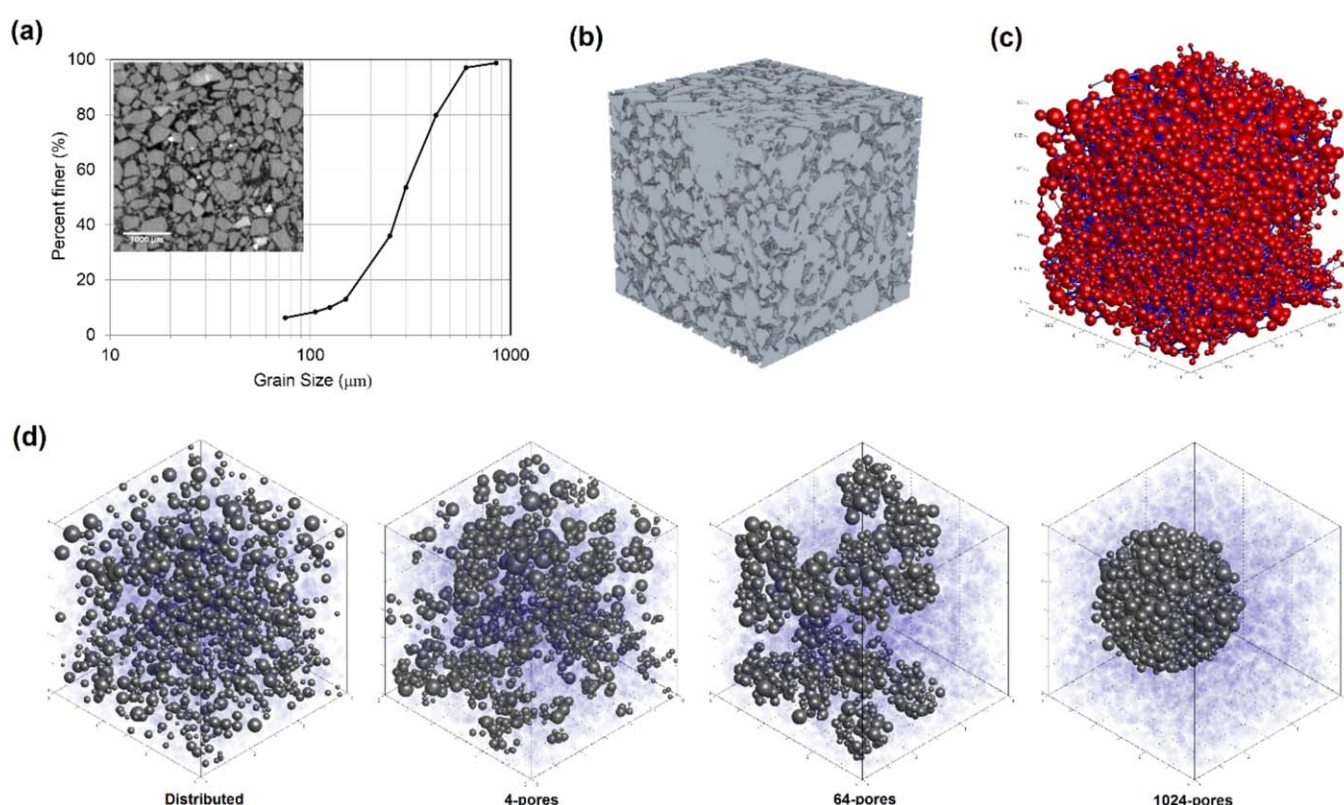
**Figure 1.** Effects of fitting parameters on the water retention curve ( $P_c$ -vs- $S_w$ ) and relative permeability. (a) Water retention curve using the *van Genuchten* model (equation (1)). (b) Relative permeability using the *van Genuchten* model (equations (2) and (3)). As an  $m$ -value increases, relative water permeability increases while relative gas permeability decreases. (c) Relative permeability using the Brooks-Corey model (equations (4) and (5)). As fitting parameters ( $n_g$  and  $n_w$ ) increase, relative water and gas permeability decrease.

Brooks-Corey model, while one common  $m$ -value is used for both water and gas permeability equations in van Genuchten model.

### 3. Methodology

#### 3.1. X-Ray CT Scanning and Pore-Network Extraction

The sediment used for X-ray scanning was recovered from Mallik 5L-38 site at 1091 m depth. A sieve analysis was conducted to obtain the grain size distribution (Figure 2a, refer also to Jenner *et al.* [1999] for the grain size distribution of other specimens at this site). This sediment was oven-dried for 48 hours, and then packed in a rubber sleeve (with an inner diameter of 6.35 mm) housed within a high-pressure vessel made out of beryllium for X-ray scanning. The in-situ effective stress ( $\sim 10$  MPa) was restored during scanning (Note that this is a remolded sample that does not preserve the original pore structure and connectivity of in-situ sediment, but may have a similar porosity due to restored confining stress). The micro-focus scanner has  $\sim 0.68$   $\mu\text{m}$  per pixel resolution with 0.7 mm field of view in all three dimensions. However, we intentionally chose a larger pixel size so the overall scanned volume is large enough to be representative. In this study, the volume of the scanned image is 27  $\text{mm}^3$  (3 mm  $\times$  3 mm  $\times$  3 mm) with 12.5  $\mu\text{m}/\text{pixel}$  resolution. The obtained CT images provide the three-dimensional structure of the scanned sediment, including both the grains and the pore spaces (Figure 2b). Then, a three-dimensional pore network consisting of spherical pores and cylindrical tubes (Figure 2c) is extracted from the CT images using the maximal ball algorithm developed by Silin and Patzek [2006]. The maximal ball algorithm finds spheres inscribed in the pore wall. As a general rule, big spheres located in the center of pore spaces of the sediment image become pores of



**Figure 2.** Pore-network model generation and hydrate distribution in the pore-network model. (a) Grain size distribution of the specimen recovered from the Mallik 5L-38 site. The soil is classified as poorly graded sand with fines based on the Unified Soil Classification System (ASTM D2487). The coefficient of uniformity is  $C_u = D_{60}/D_{10} = 0.33 \text{ mm}/0.13 \text{ mm} = 2.6$  and the coefficient of curvature is  $C_c = D_{30}^2/(D_{10}D_{60}) = (0.23 \text{ mm})^2/(0.13 \text{ mm} \times 0.33 \text{ mm}) = 1.2$  ( $D_x$  is the particle diameter representing that  $X$  percent of the total particles is smaller than  $D$ ). The inset shows a cross section of the CT image of the Mallik sand specimen. (b) Three-dimensional representative pore space (3 mm  $\times$  3 mm  $\times$  3 mm) obtained by  $\mu\text{CT}$  scan. (c) Pore network model extracted from the pore space CT images. The pore network model consists of 5376 pores and 18770 tubes with a mean tube connectivity per pore (coordination number) of  $cn = 6.8$ . Mean pore radius is  $\mu[R_{\text{pore}}] = 60 \text{ } \mu\text{m}$ , mean tube size is  $\mu[R_{\text{tube}}] = 11 \text{ } \mu\text{m}$ , and mean tube length is  $\mu[L_{\text{tube}}] = 41 \text{ } \mu\text{m}$  ( $\max[R_{\text{pore}}] = 185 \text{ } \mu\text{m}$ ,  $\min[R_{\text{pore}}] = 21 \text{ } \mu\text{m}$ ,  $\max[R_{\text{tube}}] = 50 \text{ } \mu\text{m}$ ,  $\min[R_{\text{tube}}] = 1 \text{ } \mu\text{m}$ ,  $\max[L_{\text{tube}}] = 190 \text{ } \mu\text{m}$ ,  $\min[L_{\text{tube}}] = 0.04 \text{ } \mu\text{m}$ ). (d) Various types of hydrate morphology (distributed versus patchy) for a given hydrate saturation  $S_h = 0.2$ . Hydrate forms in randomly distributed individual pores, or in patches of different size consisting of 4 pores, 64 pores, and 1024 pores. Note that hydrate pores are shown in gray and water pores in transparent blue.

the pore-network model and the small spheres inscribed in pore throats are turned into tubes. Further information on inscribing spheres and assigning spheres for pores or tubes is found in *Dong and Blunt* [2009]. The generated pore-network model can be modified to accurately simulate multiphase fluid flows by changing pore and tube size distribution or considering irregular cross-sectional shape of tubes [Valvatne and Blunt, 2004], but in this study, the extracted pore-network model was used without further modification. Detailed information on the extracted pore-network model is provided in the caption of Figure 2.

### 3.2. Hydrate Realization: Saturation and Morphology

Three types of hydrate morphology are found in natural sediments: (1) Pore-filling, (2) lenses/veins, and (3) nodules/chunks [Waite *et al.*, 2009]. The hydrate morphology is determined by the effective stress level of the natural sediments, pore and throat size, and hydrate-water interfacial tension [Clennell *et al.*, 1999; Dai *et al.*, 2012]. In our study, we consider only pore-filling model usually found in coarse-grained sediments.

To achieve target initial hydrate saturation (i.e.,  $S_h=0, 0.2, 0.4$ , and  $0.6$  in this study), hydrates are assigned in randomly chosen pores, not in tubes. Hydrate pore selection disregards their size. And it is assumed that the selected pores are entirely filled with hydrates. The assumption of either 100% or 0% pore occupancy by hydrates reflects Ostwald ripening. In a long-term, Ostwald ripening in coarse-grained sediments promotes patchy hydrate formation, whereby hydrate-filled sediment patches ( $S_h=100\%$ ) are embedded in hydrate-free sediments ( $S_h=0\%$ ). Reported data (e.g., P-wave velocity and hydraulic conductivity at different hydrate saturation) and physical analyses suggest hydrate-bearing sands contain a heterogeneous and patchy hydrate distribution [Dai *et al.*, 2012]. And the laboratory observation performed over two weeks for THF hydrate confirms the formation of hydrate patch [Mahabadi *et al.*, 2016].

Based on the fact that hydrate patch size and morphology affects the physical properties of hydrate-bearing sediments in orders of magnitude [Dai and Santamarina, 2013; Dai *et al.*, 2012; Yun *et al.*, 2005], various hydrate morphology are considered in this study: (1) hydrates in individually distributed pores (Figure 2d) and (2) hydrates in patchy formations of different sizes, occupying 4, 64, or 1024 pores for a given hydrate saturation (Figure 2d). The size of hydrate patch is arbitrarily chosen to explore the effect of hydrate patch size on water retention curve and relative permeability.

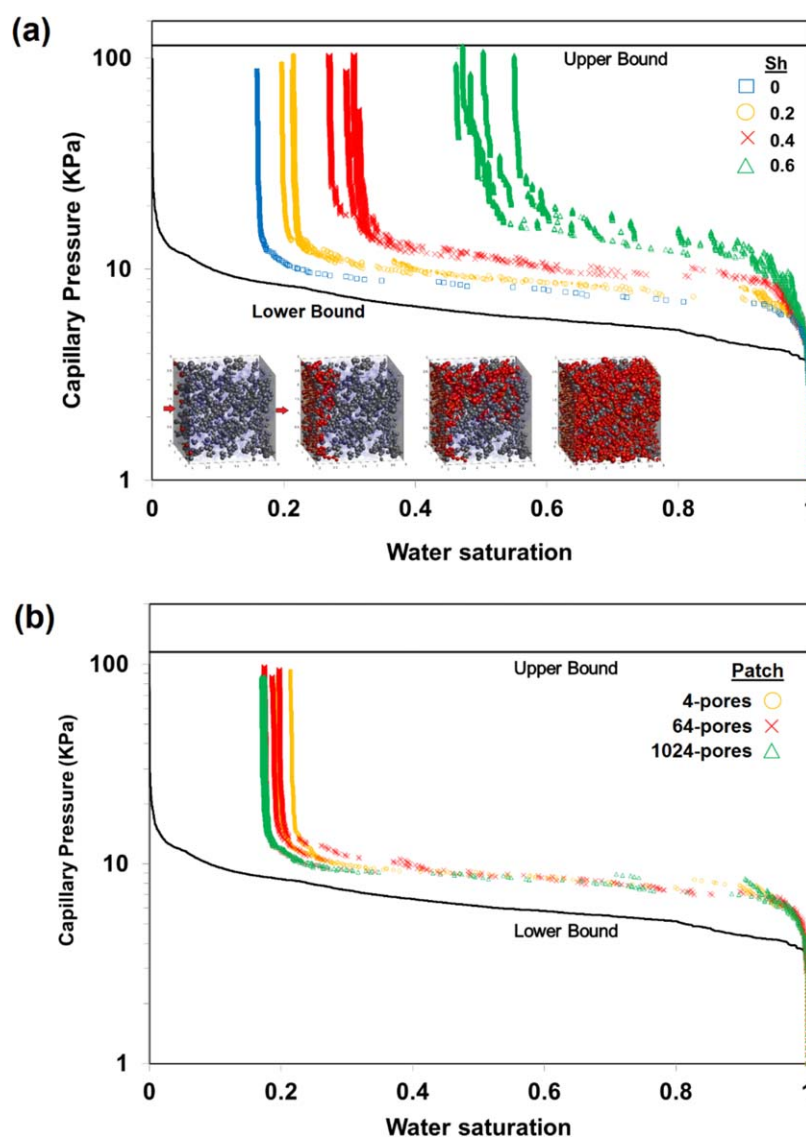
Due to hydrate occupancy in pores, the pore connectivity of pore-network model is reduced, which affects residual water saturation as well.

### 3.3. Water Retention Curves in Hydrate-Bearing Sediments

The water retention curve is obtained by simulating the gas invasion process into water-saturated, hydrate-bearing sediments. Gas invasion is enforced at tubes located on the inlet boundary. Quasi-static gas invasion controlled by capillarity is assumed so the viscosity effect can be disregarded. If the gas pressure is higher than the summation of the capillary and water pressure of the water-filled tube ( $P_g > P_w + P_c$ ), the gas can invade the tube. The capillary pressure at gas-water interface in a tube with radius  $R_{tube}$  is captured by the Laplace equation  $P_c = 2T_s \cos \theta / R_{tube}$ , where in this study the water-gas interfacial tension is  $T_s = 0.072$  N/m; the contact angle is assumed  $\theta = 0^\circ$  to represent a completely water-wet surface (note: refer to *Al-Futaisi and Patzek* [2003] for the effects of contact angle on computed relative permeability). With pressure increases, gas starts invading through the largest tube on the inlet boundary. And as capillary pressure increases, gas invades more water-filled tubes, leading to a decrease in water saturation; thus, the water retention curve is computed until water drainage stops.

### 3.4. Relative Permeability After Hydrate Dissociation

Solid hydrate turns into gas and water during hydrate dissociation. Hydrate dissociation and the volume expansion of the dissociated gas is controlled by gradually lowering the pressure at pore-network inlet and outlet boundaries from  $P=15$  MPa to  $0.1$  MPa at constant  $T=287$  K. The gas volume expansion upon dissociation and during depressurization is computed by the modified Peng-Robinson equation of state [Jang and Santamarina, 2011; Stryjek and Vera, 1986]. The depressurization is kept slow enough so that the endothermic effect during hydrate dissociation is negligible. As soon as the P-T condition comes out of the hydrate phase boundary ( $P=13.9$  MPa and  $T=287$  K), hydrates start to dissociate. The water from dissociated hydrate is drained out of the pore-network (for all cases, water pores are forming percolating path), but the gas from dissociated hydrates remains in the pore space that was initially occupied by hydrates. At the P-T condition of  $P=13.9$  MPa and  $T=287$  K, the volume of the gas dissociated from hydrates is the same as



**Figure 3.** Water retention curves in hydrate-bearing sediments. (a) Effects of hydrate saturation ( $S_h=0, 0.2, 0.4$ , and  $0.6$ ) on computed water retention curves. Upper and lower bounds are obtained by injecting gas into tubes arranged in a series forming a single line from smallest to largest (upper bound) or in parallel (lower bound). Inset figures show gas invasion into the hydrate-bearing pore-network from left to right boundary. Note that gas pores are colored red, hydrate pores gray, and water pores transparent blue. (b) Effects of hydrate morphology (all at constant hydrate saturation  $S_h=0.2$ ) on water retention curves.

the volume of initial hydrates. Therefore, pores occupied by initially hydrates turn into gas pores (for  $S_h=0.2$  case, the saturations for hydrate, gas, and water prior to dissociation are  $S_h=0.2$ ,  $S_g=0$ ,  $S_w=0.8$ , respectively, and become  $S_h=0$ ,  $S_g=0.2$ ,  $S_w=0.8$ , right after dissociation at  $P=13.9$  MPa and  $T=287$  K). Then, further depressurization allows gas to expand into neighboring water pores. Gas expansion continues as long as gas pressure is higher than the summation of capillary pressure and water pressure, and as long as water drainage path is available. At around water saturation  $S_w=0.15\sim 0.2$ , gas expansion stops mainly due to the loss of water drainage path.

The methane solubility in water without hydrate is assumed zero due to relatively small methane solubility in water compared to large methane amount contained in methane hydrates. The gas expansion algorithm and water and gas permeability calculation during each gas expansion step are explained in detail in the authors' previous publication [Jang and Santamarina, 2011; Jang et al., 2011; Mahabadi and Jang, 2014].

## 4. Simulation Results

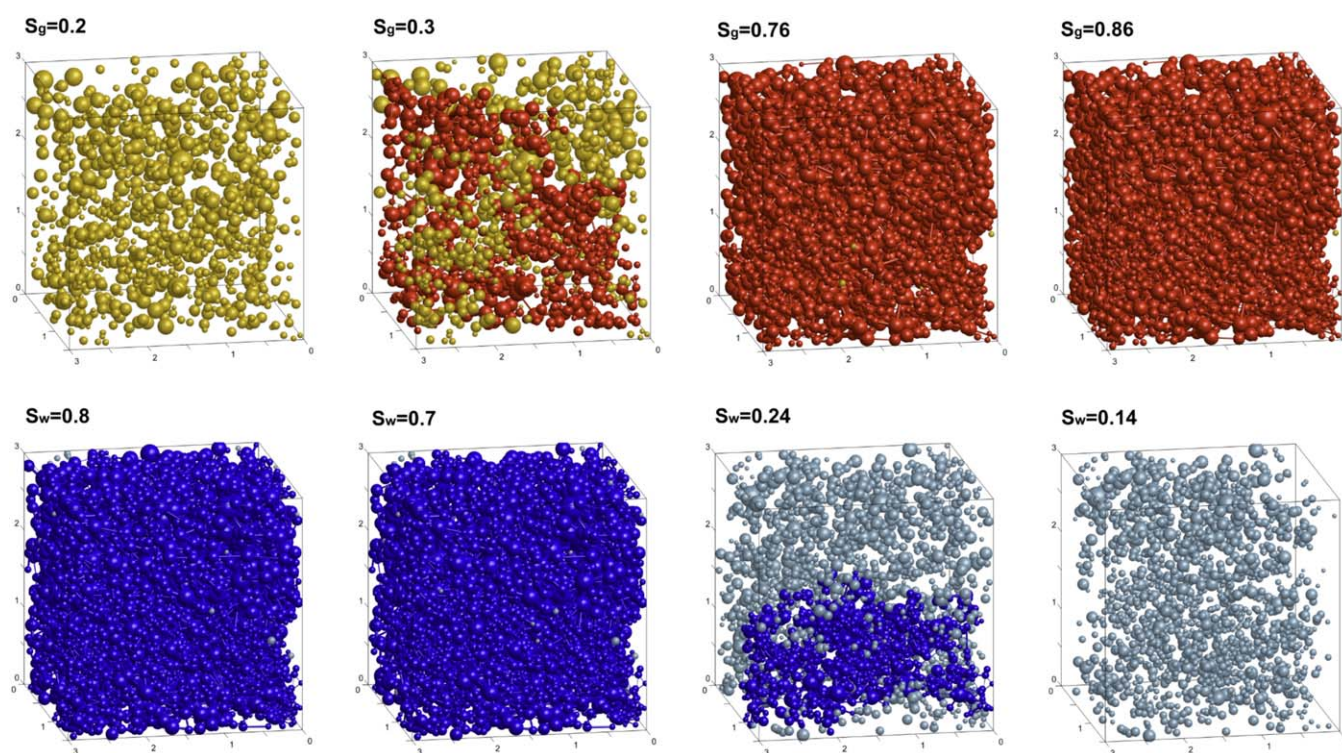
### 4.1. Computed Water Retention Curves

Figure 3a shows the computed retention curves in sediments with various initial hydrate saturations, i.e.,  $S_h=0, 0.2, 0.4$  and  $0.6$ . Hydrates (black spheres Figure 3a, inset) are randomly distributed in individual pores. One simulation run is performed for  $S_h=0$  case and five simulation runs for each of other  $S_h>0$  cases. All cases use the identical pore-network. The process of gas invasion into the hydrate-bearing pore network model is illustrated as red spheres in the Figure 3a inset. As the initial hydrate saturation increases, the gas-entry pressure increases, and the capillary pressure at a given water saturation also increases. The higher initial hydrate saturation condition results in the higher residual water saturation.

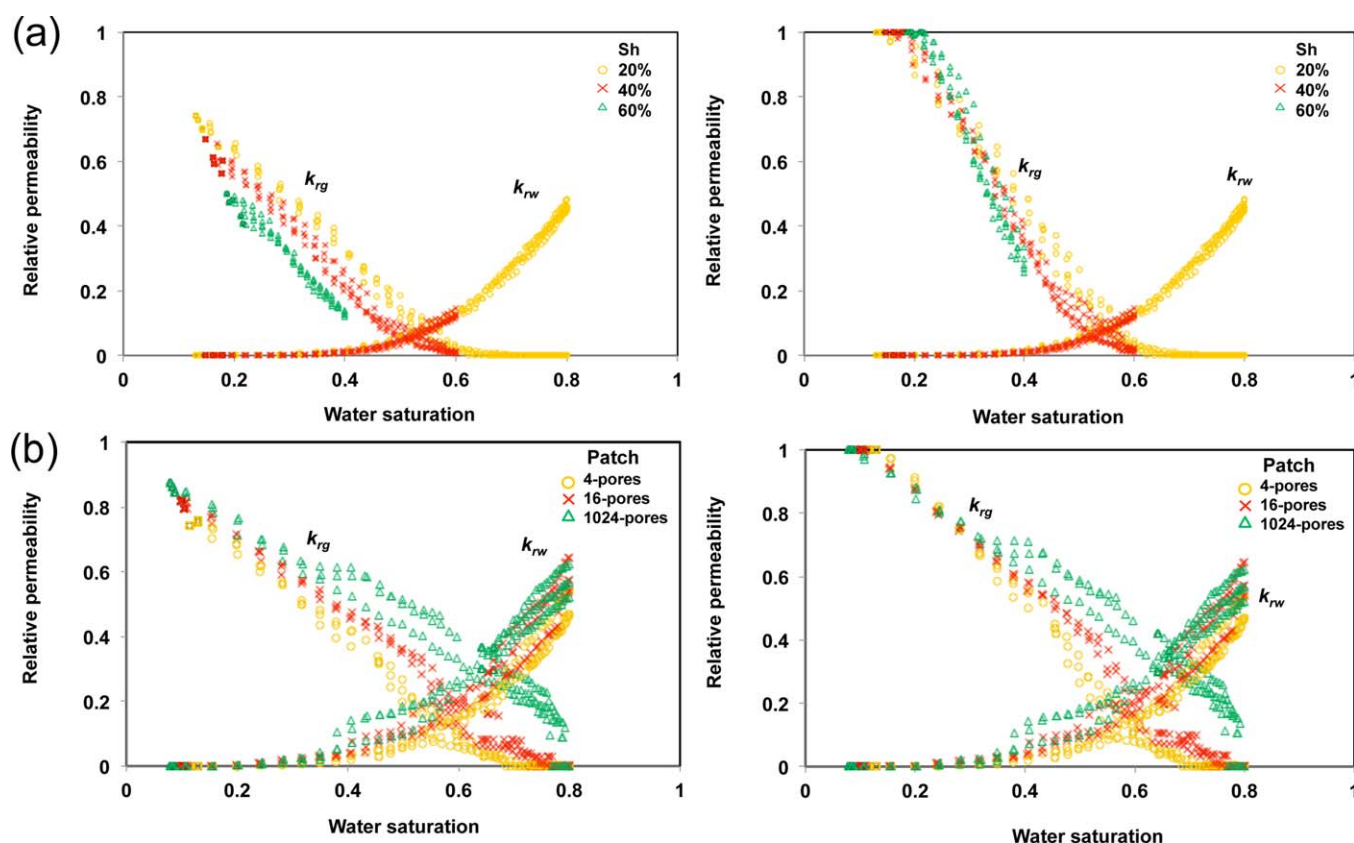
Figure 3b shows the computed water retention curves in sediments with the same initial hydrate saturation ( $S_h=0.2$ ) but different hydrate patch size (i.e., the hydrate patch is composed of 4, 64, and 1024 pores). Three simulation runs are performed for each patch size and the spatial distribution of hydrate patch varied at each simulation. Sediments with larger hydrate patches tend to result in lower gas-entry pressure and also slightly lower residual water saturation (Figure 3b).

### 4.2. Relative Permeability During Gas Expansion

Figure 4 illustrates the gas expansion and water drainage processes during gas expansion in sediments with initial hydrate saturation  $S_h=0.2$  ( $S_h=0$  and  $S_w=0.8$ ). All the hydrates dissociate into gas ( $S_g=0$  and  $S_w=0.8$ , first column in Figure 4). And gas expands with further depressurization (second, third, and fourth columns in Figure 4). With further gas expansion, water continues draining, but gas conductivity does not occur until the gas phase forms a percolation path at gas saturation  $S_g=0.3$  (second column in Figure 4). Later, the water phase gradually loses percolation paths, and the gas phase builds more gas flow paths (third column in Figure 4a). Eventually, gas keeps expanding into neighboring water-filled pores until water drainage stops at  $S_g=0.86$  and  $S_w=0.14=S_{wv}$  (fourth column in Figure 4a).



**Figure 4.** Illustrations of gas and water flow processes during gas expansion in sediment undergoing hydrate dissociation ( $S_h=0.2$ ). Note that gas pores forming conductivity paths are colored red, and isolated gas pores are colored yellow in upper figures. Water pores forming conductivity paths are colored blue, and isolated water pores are colored light blue in lower figures.



**Figure 5.** Relative permeability in hydrate-bearing sediments. (a) Effect of hydrate saturation on relative gas and water permeability for initial hydrate saturations  $S_h = 0.2$  (yellow), 0.4 (red) and 0.6 (green). Five simulation runs are performed for each hydrate saturation condition. (b) Effect of hydrate patch size on relative gas and water permeability. Three simulation runs are performed for each size of hydrate patch: 4 pores (yellow), 64 pores (red) and 1024 pores (green). The hydrate saturation is  $S_h = 0.2$ . Note that in left figures, the water (or gas) conductivity is normalized by the water (or gas) conductivity at 100% water (or gas) saturation, however, the gas conductivity at the residual water saturation is used to normalize gas conductivity in right figures.

The relative permeability of water (or gas) is the conductivity at a given water saturation normalized by the water (or gas) conductivity at 100% water (gas) saturation. The gas conductivity at the residual water saturations can be also used for normalization.

In order to explore the effect of hydrate saturation and patch formation on relative water and gas permeability, the water (or gas) conductivity at a given water saturation is normalized by the water (or gas) conductivity at 100% water (or gas) saturation (Figures 5a and 5b, left). However, in order to fit van Genuchten and Brooks-Corey equations, the gas conductivity at the residual water saturation is used for normalization (Figures 5a and 5b, right).

Results show that relative gas permeability increases, and relative water permeability decreases as gas expands into water-filled pores (Figure 5). As shown in left figure in Figure 5a, sediments with lower initial hydrate saturation result in (1) higher relative gas permeability and (2) lower residual water saturation. This trend of relative gas and water permeability as a function of initial hydrate saturation is consistent with published results [Mahabadi and Jang, 2014]. Higher gas connectivity developed for lower hydrate saturation conditions can explain this trend.

Additionally, with the increase in hydrate patch's size, both water and gas permeability increases dramatically particularly for  $S_w > 0.4$  (Figure 5b, left). When the water saturation is lower than  $S_w = 0.4$ , the effect of hydrate patch size on relative gas and water permeability becomes insignificant. Moreover, larger patch size results in lower residual water saturation.

Gas conductivity values are normalized by the gas conductivity at the residual water saturation for each simulation run in order to fit the van Genuchten equation and Brooks-Corey equation (Figures 5a and 5b, right).

**Table 1.** Fitting Parameters for Water Retention Curves and Relative Permeability.<sup>a</sup>

| Numerical Results            | van Genuchten Model   |                 |                                          |                               |            |                                                                                                                           |                 | Brooks-Corey Model |               |               |
|------------------------------|-----------------------|-----------------|------------------------------------------|-------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------|---------------|---------------|
|                              | Water Retention Curve |                 |                                          | Relative Permeability         |            |                                                                                                                           |                 |                    |               |               |
|                              | $P_0$ [kPa]           | $S_{rw}$        | $m$                                      | $S_{rw}$                      | $S_{wmax}$ | $m_g$                                                                                                                     | $m_w$           | $S_{rw}$           | $n_g$         | $n_w$         |
| 0                            | 8.0                   | 0.16            | 0.91                                     |                               |            |                                                                                                                           |                 |                    |               |               |
| 0.2                          | $8.7 \pm 0.2$         | $0.21 \pm 0.01$ | $0.90 \pm 0.01$                          | $0.14 \pm 0.02$               | 0.80       | $0.87 \pm 0.00$                                                                                                           | $0.88 \pm 0.01$ | $0.14 \pm 0.02$    | $2.6 \pm 0.1$ | $3.1 \pm 0.1$ |
| 0.4                          | $10.2 \pm 0.4$        | $0.29 \pm 0.02$ | $0.87 \pm 0.01$                          | $0.17 \pm 0.01$               | 0.75       | $0.85 \pm 0.01$                                                                                                           | $0.85 \pm 0.01$ | $0.17 \pm 0.01$    | $3.1 \pm 0.1$ | $3.3 \pm 0.1$ |
| 0.6                          | $13.8 \pm 0.5$        | $0.49 \pm 0.04$ | $0.81 \pm 0.01$                          | $0.20 \pm 0.01$               | 0.60       | $0.80 \pm 0.01$                                                                                                           | $0.80 \pm 0.02$ | $0.20 \pm 0.01$    | $3.5 \pm 0.1$ | $3.3 \pm 0.2$ |
| Patch size (at $S_h = 0.2$ ) |                       |                 |                                          |                               |            |                                                                                                                           |                 |                    |               |               |
| 4-pore                       | $8.6 \pm 0.2$         | $0.20 \pm 0.01$ | $0.91 \pm 0.01$                          | $0.12 \pm 0.01$               | 0.80       | $0.70 \pm 0.01$                                                                                                           | $0.89 \pm 0.01$ | $0.12 \pm 0.01$    | $2.1 \pm 0.1$ | $3.1 \pm 0.2$ |
| 64-pore                      | $8.4 \pm 0.2$         | $0.19 \pm 0.01$ | $0.91 \pm 0.01$                          | $0.10 \pm 0.00$               | 0.80       | $0.55 \pm 0.02$                                                                                                           | $0.97 \pm 0.02$ | $0.10 \pm 0.00$    | $1.6 \pm 0.1$ | $2.6 \pm 0.2$ |
| 1024-pore                    | $8.0 \pm 0.2$         | $0.17 \pm 0.00$ | $0.92 \pm 0.01$                          | $0.08 \pm 0.00$               | 0.80       | $0.25 \pm 0.03$                                                                                                           | $1.07 \pm 0.02$ | $0.08 \pm 0.00$    | $1.1 \pm 0.2$ | $2.3 \pm 0.2$ |
| Recommendations              |                       |                 |                                          | Uniformly Distributed Hydrate |            | Note: Impacts of Heterogeneously Distributed/Patchy Hydrate                                                               |                 |                    |               |               |
| van Genuchten model          | WRC                   | $P_0$           | $P_0^{HBS} = P_0^{HFS} (1 + S_h)$        |                               |            | Lower $P_0$                                                                                                               |                 |                    |               |               |
|                              |                       | $S_{rw}$        | $S_{rw}^{HBS} = S_{rw}^{HFS} (1 + 5S_h)$ |                               |            | Lower $S_{rw}$ with increasing patch size                                                                                 |                 |                    |               |               |
|                              |                       | $m$             | $m^{HBS} = m^{HFS} (1 - 0.15S_h)$        |                               |            | Negligible impact when $S_h = 0.2$                                                                                        |                 |                    |               |               |
|                              | Relative Permeability | $S_{rw}$        | $S_{rw}^{HBS} = 0.1 + 0.17S_h$           |                               |            | Lower $S_{rw}$                                                                                                            |                 |                    |               |               |
|                              |                       | $S_{wmax}$      | $S_{wmax}^{HBS} = 1 - 0.7S_h$            |                               |            | No impact                                                                                                                 |                 |                    |               |               |
|                              |                       | $m_g$<br>$m_w$  | $m^{HBS} = m^{HFS} (1 - 0.19S_h)$        |                               |            | Significantly lower $m_g$ , decreasing with increasing patch size<br>Higher $m_w$ , increasing with increasing patch size |                 |                    |               |               |
| B-C model                    | Relative Permeability | $S_{rw}$        | $S_{rw}^{HBS} = 0.1 + 0.17S_h$           |                               |            | Lower $S_{rw} \sim 0.1$                                                                                                   |                 |                    |               |               |
|                              |                       | $n_g$           | $n_g^{HBS} = 2 + 2.6S_h$                 |                               |            | Significantly lower $n_g$ , $n_g^{HBS} = 1 + 2.6S_h$                                                                      |                 |                    |               |               |
|                              |                       | $n_w$           | $n_w^{HBS} \sim 3 - 3.5$                 |                               |            | Lower $n_w$ , $n_w^{HBS} \sim 2 - 3$                                                                                      |                 |                    |               |               |

<sup>a</sup>Note: HBS = hydrate-bearing sediments; HFS = hydrate-free sediments..

## 5. Analyses and Discussions

### 5.1. Pore-Network Model Simulation—Relevance to Experimental Tests

The shape of pore geometry in real sediments is irregular, while the pore-network model composed of cylindrical tubes and spherical pores simplifies the real pore geometry of sediments, which can result in differences between numerical simulation and experimental study. For example, the maximum ball algorithm used for pore-network model extraction finds spheres inscribed onto the pore surface [Dong and Blunt, 2009]. Therefore, the cross-sectional area of tubes tends to be smaller than the actual cross-sectional area of real pore throats. The simulation results obtained by using the Lattice Boltzman Method show that the capillary pressure of the cylindrical tube inscribed in other tubes with polygonal cross-sectional shape is smaller than the capillary pressure of polygonal shape tubes [Mahabadi et al., 2015]. In addition, the process of finding inscribed spheres for various pore throat shapes may result in homogenization of pore throat size distribution of real sediments. Therefore, water retention curves obtained from the pore-network model may show more uniform pore throat size distribution, which increases m-value in the  $P_c$ -vs-saturation curve, as shown in Figure 1a.

### 5.2. Flow in Hydrate-Bearing Sediments—Recommended Parameter Values

The method of least squares is used to fit the equations (1)–(5) to the water retention curves and relative permeability simulation results. When the least square method is applied to the unevenly distributed data points, biased fitting parameters can be obtained. Therefore, a median curve is found for distributed data. Then, evenly distributed data points on the median curve are used for the least squares method to find fitting parameters with a minimum L2-norm. Fitting parameters are calculated at each individual simulation run for each initial hydrate saturation condition and each hydrate morphology case. Then, the averaged fitting parameters of 5 or 3 simulation runs are obtained.

Table 1 summarizes the best fitting values for the van Genuchten model and the Brooks-Corey model based on the simulations. The results of water retention curve simulation suggest that the three parameters in the van Genuchten WRC model (i.e.,  $P_0$ ,  $S_{rw}$ , and  $m$ ) depend on hydrate saturation: increased hydrate saturation in sediments leads to (1) higher gas entry pressure  $P_0$ , (2) higher residual water saturation  $S_{rw}$ , and (3)

steeper  $P_c$ - $S_w$  curve (meaning lower  $m$  value). For patchy hydrate formation, as the size of hydrate patch increases for a given hydrate saturation, the  $m$ -value increases slightly (from  $m=0.91$  to  $0.92$ ), and the gas-entry pressure decreases (Table 1). The gas-entry pressure for the largest hydrate patch case (a patch consisting of 1024 pores) is  $P_0=8.0$  kPa which is as low as the gas-entry pressure of the hydrate-free sediment. It is obvious that sediments with heterogeneous hydrate distribution tend to have lower gas entry pressure than those with uniformly distributed hydrate.

The relative water and gas permeability during gas expansion also depend on both hydrate saturation and hydrate morphology. During hydrate dissociation, gas expands into the neighboring water-filled pores and forms interconnected clusters, which further interconnect to each other to become larger gas clusters and eventually percolate through the porous media. In sediments with distributed hydrate, the gas permeability of the sediment with higher  $S_h$  is lower than that of the sediment with lower  $S_h$  (Figure 5a), due to less interconnected gas clusters in a higher  $S_h$  case than that in a lower  $S_h$  case. Hydrate morphology has even more significant effect on the gas permeability (Figure 5b). At a given saturation, the relative gas permeability increases dramatically as the size of hydrate patch increases, e.g.,  $k_{rg} = \sim 0.5$  in the case of a larger hydrate patch (1024-pore size), but  $k_{rg} = \sim 0.1$  in the case of a smaller hydrate patch (4-pore size) at around  $S_w=0.55$ . However, the variance in gas permeability due to hydrate morphology diminishes toward residual water saturation, i.e., at gas saturation  $S_g > \sim 0.7$  or water saturation  $S_w < \sim 0.3$ .

With regard to water permeability, initial hydrate saturation in sediments with uniformly distributed hydrate has marginal effect. But in sediments with a patch hydrate, larger hydrate patch results in higher relative water permeability because gas expansion from larger hydrate patches has less effect on plugging water paths than does a smaller hydrate patch or distributed hydrates.

More importantly, the simulation results also suggest that using one common fitting parameter  $m$ -value in the van Genuchten model for both gas and water permeability is not appropriate to characterize the water/gas flows during gas production from hydrate deposits. In sediments with uniformly distributed hydrate, the  $m$ -value for both gas and water permeability decreases with increasing hydrate saturation. Under identical hydrate saturation ( $S_h = 0.2$  in this case, Table 1) but various hydrate morphology, however, the  $m$ -values for gas and water permeability show evident discrepancies; and sediments with larger hydrate patches tend to result in lower residual water saturation  $S_{rw}$  and higher gas and water permeability (Table 1). In this sense, the Brooks-Corey model that has two different fitting parameters specifically for gas  $n_g$  and water  $n_w$  permeability appropriately predict the flows in sediments with patchy hydrate. The fitting parameter values for the Brooks-Corey model for sediments with various hydrate saturation and morphology are listed in Table 1.

## 6. Conclusions—Recommendations

Based on the results and analyses in this study, we propose the following recommendations for selecting proper parameters and their values (summarized also in Table 1) when characterizing gas and water flows in sediments during hydrate dissociation:

1. For water retention curves in hydrate-bearing sediments, the air-entry pressure  $P_0$ , residual water saturation  $S_{rw}$ , and  $m$ -value of the van Genuchten model depend on hydrate saturation  $S_h$ . The effect of hydrate patch size on the WRC steepness  $m$  is negligible, particularly at  $S_h=0.2$ .
2. In relative permeability characterization using the van Genuchten model, the residual water saturation  $S_{rw}$  and maximum water saturation  $S_{wmax}$  should be corrected by considering hydrate saturation  $S_h$ . Also, the same  $m$ -value can be used for both gas and water permeability, but only in sediments with uniformly distributed hydrate; hydrate morphology greatly impacts the  $m$ -value for gas permeability.

The Brooks-Corey model characterizes the gas and water permeability using two separate parameters. Hydrate saturation slightly affects the fitting parameter for water permeability; but the gas permeability must be corrected at different hydrate saturation  $S_h$ . Again, hydrate morphology dramatically affects the gas permeability. In particular, a more heterogeneous specimen (with a larger hydrate patch) tends to result in both higher gas and water permeability.

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## RESEARCH ARTICLE

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## Special Section:

Rock Physics of the Upper Crust

## Key Points:

- Mudstone carbonate rock was imaged with X-ray microcomputed tomography, and a calcite dissolution model was numerically implemented
- Microstructural heterogeneity caused preferred fluid pathways that localized dissolution to a subset of the total connected porosity
- The widening of narrow pores caused seemingly porosity-independent permeability evolutionary trend, which transitioned to channelization

## Supporting Information:

- Supporting Information S1

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## Evolution of permeability and microstructure of tight carbonates due to numerical simulation of calcite dissolution

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**Abstract** The current study concerns fundamental controls on fluid flow in tight carbonate rocks undergoing CO<sub>2</sub> injection. Tight carbonates exposed to weak carbonic acid exhibit order of magnitude changes in permeability while maintaining a nearly constant porosity with respect to the porosity of the unreacted sample. This study aims to determine—if not porosity—what are the microstructural changes that control permeability evolution in these rocks? Given the pore-scale nature of chemical reactions, we took a digital rock physics approach. Tight carbonate mudstone was imaged using X-ray microcomputed tomography. We simulated calcite dissolution using a phenomenological numerical model that stands from experimental and microstructural observations under transport-limited reaction conditions. Fluid flow was simulated using the lattice-Boltzmann method, and the pore wall was adaptively eroded at a rate determined by the local surface area and velocity magnitude, which we use in place of solvent flux. We identified preexisting, high-conductivity fluid pathways imprinted in the initial microstructure. Though these pathways comprise a subset of the total connected porosity, they accommodated 80 to 99% of the volumetric flux through the digital sample and localized dissolution. Porosity-permeability evolution exhibited two stages: selective widening of narrow pore throats that comprised preferential pathways and development and widening of channels. We quantitatively monitored attributes of the pore geometry, namely, porosity, specific surface area, tortuosity, and average hydraulic diameter, which we qualitatively linked to permeability. This study gives a pore-scale perspective on the microstructural origins of laboratory permeability-porosity trends of tight carbonates undergoing transport-limited reaction with CO<sub>2</sub>-rich fluid.

## 1. Introduction

Prediction of subsurface permeability from geophysical data is crucially important to a number of applications in the geosciences, namely, hydrocarbon extraction, CO<sub>2</sub> sequestration, and ground contaminant research. For many rock types, including pure sandstones, permeability can be reasonably inferred from estimates of subsurface porosity using an appropriate rock physics model [e.g., Kozeny, 1927; Carman, 1937; Bourbié and Zinszner, 1985]. However, the pore geometries of tight carbonates are so complex and heterogeneous that permeability becomes nonunique, spanning nearly 4 orders of magnitude, with respect to porosity [e.g., Lucia, 1983, 1995; Vanorio *et al.*, 2015]. The lack of a functional relationship between permeability of carbonates and microstructural properties makes modeling fluid flow and interpreting subsurface hydraulic properties in carbonate reservoirs challenging.

Recent experiments [Grombacher *et al.*, 2012; Vanorio *et al.*, 2015] conducted in the transport-limited reaction regime using carbonate mudstones have shown that porosity-permeability changes of comparable magnitude can be obtained through the dissolution of calcite by injection of CO<sub>2</sub>-rich fluid into the pore space. Interestingly, Vanorio *et al.* [2015] found that the permeability of these samples increased between 0.5 and 1 order of magnitude while the porosity remained constant within experimental error. The current study seeks to provide quantitative constraints on the physical, pore-scale parameters that control macroscopic permeability during dissolution of calcite in tight carbonate rocks.

The most commonly used rock physics model that relates permeability to pore-scale parameters is the classic semiempirical Equation Kozeny-Carmen Equation [Kozeny, 1927; Carman, 1937],

$$k = C \frac{\phi^3}{S^2 \tau^2} \quad (1)$$

which relates permeability  $k$ , to porosity  $\phi$ , tortuosity  $\tau$ , specific surface area  $S$ , and an empirical scaling constant  $C$ . For a bundle of tortuous tubes, equation (1) is exact; however, real rock microstructures exhibit complexity and spatial heterogeneity on multiple scales relevant for porous flow. Therefore, equation (1) is valid only under simple geometric transformations of idealized pore geometries or to model compaction of regularly packed granular material. Furthermore,  $S$  and  $\phi$  are highly correlated, as they both functions on the hydraulic diameter integrated along the length of the tube [Lock *et al.*, 2002]. Determining whether a change in  $S$  or  $\phi$  is responsible for an observed change in permeability is intractable without completely constraining the microstructure of the rock. To make matters worse, parameters  $S$  and  $\tau$  cannot be directly measured on whole rocks using conventional rock physics techniques, so they are typically left as ad-hoc parameters to fit experimental data.

In order to account for the complexity of real rock pore geometries, Bourbié *et al.* [1987] developed a rock physics model to absorb the various hydraulic parameters into a single power law function of porosity:

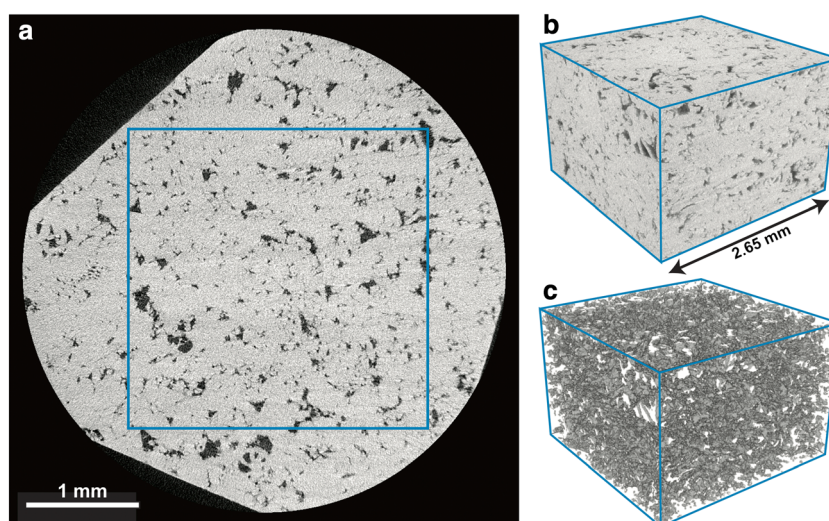
$$k = k_0(\phi/\phi_0)^n \quad (2)$$

Here  $n$  is the so-called porosity exponent that empirically accounts for the morphology and topology of the pore geometry and  $k_0$  is the permeability at some reference porosity  $\phi_0$ . Like equation (1), the porosity exponent takes on integer values for simple transformations. For example,  $n = 2$  for uniform scaling of a uniform pipe lattice geometry about pore centerlines [Turcotte and Schubert, 2014]. In addition, it can be shown through pore network modeling that  $n \approx 3$  for a tetrakaidekahedral grain pack having a random distribution of pore diameters [Zhu and Hirth, 2003]. This packing geometry agrees well with compaction trends of Fontainebleau sandstone [Bourbié and Zinszner, 1985; Bourbié *et al.*, 1987; Doyen, 1988; Gomez *et al.*, 2010] and sintered quartzite [Wark and Watson, 1998]. Higher values of  $n$  have also been observed for rocks near the percolation threshold [Bourbié *et al.*, 1987] and for rocks experiencing chemical alteration, which is discussed at length in reviewer paper Bernabé *et al.* [2003]. In general though,  $n$  is an all-inclusive, empirical quantity that has little to no justification outside the realm of simple pore geometries, e.g., sandstone, or idealized models of complex pore geometries.

In the last two decades, experimental inquiry has sought new ways to improve our understanding of permeability evolution in porous rocks. There are a number of experimental examples of fluid-rock interactions that exhibit changes in permeability virtually independent of porosity. Pertinent to this study, Vanorio *et al.* [2015] observed 0.5 to 1 order of magnitude increase of permeability without a measureable change of porosity—corresponding to a virtually infinite  $n$ —during injection of CO<sub>2</sub>-rich water into stiff carbonate mudstones exhibiting negligible compaction under confining stress. Similar results were obtained by Noirié *et al.* [2004], who observed a 1 order of magnitude increase in permeability within a 0.5% increase in porosity in the early stages of CO<sub>2</sub> injection into carbonate packstones without confining pressure, resulting in an initial porosity exponent of  $n = 150$  that decayed and then stabilized to  $n = 13$ . Noirié *et al.* [2004] attributed their result to the dissolution of micrite constricting hydraulic pathways. Conceptually, this makes sense, as micrite grains correspond to the phase predominantly being dissolved due to high reactive surface area [Vanorio, 2015], but a comprehensive, time-lapse analysis of the parameters describing pore morphology and local flow field is needed to lead the way to predictive models.

In addition to dissolution, precipitation reaction can also give rise to porosity-independent permeability behavior [e.g., McCune *et al.*, 1979; Xu *et al.*, 2004; Adam *et al.*, 2013]. Precipitation can be thought of as the inverse process to dissolution. Instead of removing material that may constrict pore throats, precipitation causes crystals to nucleate on pore walls, decreasing the connectivity and hydraulic radii of the pore space available to fluid flow [Bernabé *et al.*, 2003].

In this study, we seek to understand, if not porosity, what are the specific changes in the rock microstructure that control permeability evolution during reaction with CO<sub>2</sub>-rich fluid? Rock physics experiments do not afford direct access to the pore geometry during experimentation due to the impracticality to compromise between different scales of observation, i.e., plug versus pore length scale. Therefore, in this study, we took a digital rock physics approach that involved numerical simulation of calcite dissolution in real carbonate geometries obtained using X-ray microcomputed tomography ( $\mu$ -CT). The sample



**Figure 1.** Visualization of tight carbonate X-ray  $\mu$ -CT image. (a) Tomographic slice sampled perpendicular to the cylindrical axis of the 4.3 mm rock pillar. The gray scale value is indicative of material: black (pore) and gray (calcite). The blue box denotes the outlines the cropped subvolume. (b) Volume rendering of cropped subvolume. (c) Volume rendering of the segmented pore geometry that was used as input to the LBM and dissolution models.

used in this study is the digital representation of a carbonate mudstone used for a  $\text{CO}_2$ -injection experiment [Grombacher *et al.*, 2012; Vanorio *et al.*, 2015]. To numerically simulate dissolution, we adaptively eroded the pore-grain interface based on local reactive surface area and local velocity magnitude, which we assumed as a proxy for solvent flux. Porosity, specific surface area, tortuosity, and hydraulic radius, were monitored over the course of the numerically simulated dissolution, and changes to those parameters are discussed in the context of their contribution to permeability evolution. Using the results and visualizations of the evolving pore geometry, we argue that porosity-independent behavior in carbonate mudstones can be induced by high dissolution rates in narrow pore throats located along high-conductivity pathways. Over time, dissolution will lead to a channelization by enhancing preexisting pathways of least hydraulic resistance, which is a different mechanism than wormholing. We also compare the results of our dissolution model with permeability-porosity trends of  $\text{CO}_2$  injection experiments.

## 2. Methods

### 2.1. Sample Selection, Preparation, and Imaging

The rock type used in this study is a mudstone carbonate from the Peschici Formation, Italy. The initial mineralogy, texture, porosity, permeability, and pore stiffness were characterized prior to this study [Vanorio *et al.*, 2015]. The sample exhibited a tight micrite matrix with dispersed 5–200  $\mu\text{m}$  diameter pores. They also possessed high pore stiffness, suggesting that the rock did not experience compaction when exposed to reactive fluid. Thus, we did not have to account for dissolution-induced compaction in this numerical study.

To fit within the field of view of the  $\mu$ -CT scanner, a 4.3 mm pillar was cored from the rock. The tight carbonate pillar was imaged at a resolution of 4.42  $\mu\text{m}$  per voxel using a tabletop X-ray  $\mu$ -CT scanner. Figure 1a shows a visualization of a slice taken parallel to the cylindrical axis of the pillar. In the image, the gray scale value of each voxel is indicative of X-ray attenuation. The gray voxels (high gray scale value) represent calcite, while black voxels represent pores.

### 2.2. Data Reduction of $\mu$ -CT Data

The fluid flow code required us to crop the data to a box, so we cropped a  $2.65 \times 2.65 \times 1.81 \text{ mm}^3$  subvolume (Figure 1b) from the gray scale volumetric data set, which was largest percolating subvolume that would fit within the boundaries of the digital core sample.

The gray scale volumetric data could not be used directly as input to the fluid flow software. We first segmented the gray scale subvolume, producing into a binary image consisting of voxels whose value is either 0 (calcite) or 1 (pore). To facilitate the segmentation process, we applied an anisotropic diffusion filter Anisodif3d [Kovesi, 2000], which we overloaded for computation on a graphics processing unit (GPU). The MATLAB-base implementation was based on previous image processing work [Perona and Malik, 1990; Gerig *et al.*, 1992]. The resulting image was segmented using a global threshold (Figure 1c).

### 2.3. Computation of Velocity Field

Laminar flow was numerically simulated in the segmented pore geometry using the Lattice-Boltzmann method (LBM). The essence of single-phase LBM is to model fluid flow by advecting a particle collision function whose parameters are tuned to exactly mimic reproduce physics of laminar flow [Ladd, 1993]. Fluid flow is driven by a pressure boundary condition, which is implemented in the LBM model using a density perturbation function. We used an OpenMP parallelized of the code presented by Keehm [2003]. Prior to running the LBM solver, unconnected porosity was removed from the binary pore geometry to reduce computational size. To ensure that the physics in the pore throats was well resolved, the binary geometry was oversampled without interpolation by a factor of 2, so the computational resolution was 2.21  $\mu\text{m}$  per voxel. The velocity field was obtained once the permeability stabilized to a steady state; i.e., permeability did not change more than 0.0001% for 40 iterations. The flux was obtained by volume averaging the component of the velocity field parallel to the imposed pressure gradient. Given the known pressure boundary condition and dimensions of the model, Darcy's law was used to determine permeability.

We benchmarked the LBM code by solving the classic flow between two infinite, parallel plates problem, which has a closed-form solution for the velocity and volumetric flow rate. In summary, we found good agreement between numerical and analytical results. For a region as few as four voxels across, the maximum error associated with the velocity and volumetric flux was 5% and 6%, respectively. Furthermore, since dissolution tends to widen pores—increasing the number of unit cells that comprise the pore throat—the error decreases as dissolution progresses.

### 2.4. Dissolution Model

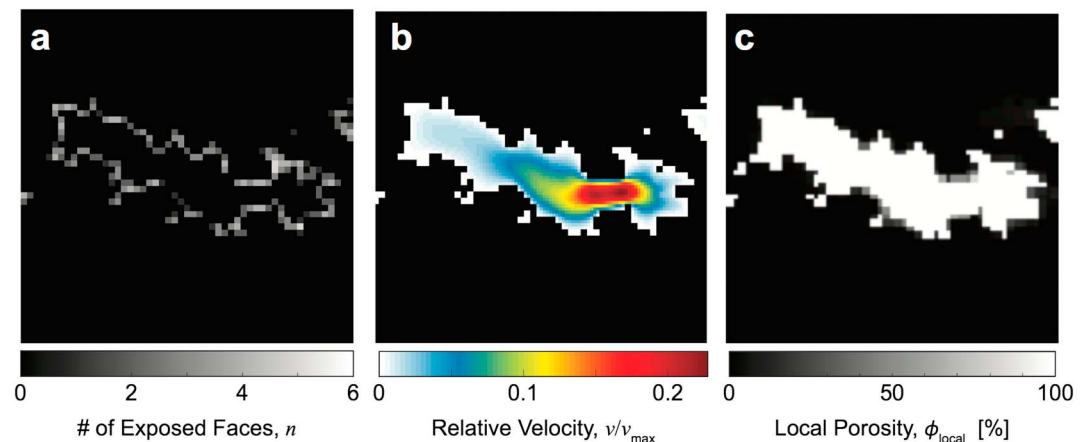
We numerically simulated calcite dissolution using a phenomenological approach. Rather than solving the advection-diffusion equation directly [e.g., Jiang and Tsuji, 2014; Pereira Nunes *et al.*, 2016a, 2016b; Soulaire *et al.*, 2016], our model assumes the local velocity field as a proxy for solvent flux. By doing so, we implicitly ignore diffusion, which we argue is a valid simplification for a transport-limited, high-flow dissolution regime where solvents are advected through the pore geometry faster than they can change the chemistry of the fluid due to reaction with the pore wall. Though our model is not first-principles-based, it builds upon basic physicochemical principles that underpin transport-limited reaction behavior, namely, (1) the reaction in nonpercolating pathways is fully buffered and (2) the reaction is most rapid in regions where solvent flux is high.

Our approach is equivalent to solving the zeroth-order reaction rate equation at each boundary grain voxel  $i$ ,

$$\frac{dm_i}{dt} = -n_i \frac{v_i}{v_{\max}} R_0 \quad (3)$$

where  $m_i$  is mass and  $R_0$  is the maximum reaction rate, which we assumed was  $R_0 = 2.57 \times 10^{-4}$  mol/s, an empirical value given by Plummer *et al.* [1978] for the forward reaction of calcite with weak carbonic acid. We imposed dissolution at high-flux regions by scaling  $R_0$  by the ratio  $v_i/v_{\max}$ , where  $v_i$  is the mean velocity magnitude calculated in a five-voxel neighborhood of  $i$  and  $v_{\max}$  is the maximum velocity in the pore volume. The dependence of reaction rate on surface area was accounted for by parameter  $n_i$ , which scales the reaction rate at each voxel by the number of voxels faces exposed to the virtual reactive fluid.

In order to enable a continuous erosion of the pore wall, we first converted the binary pore geometry into a porosity field. As dissolution proceeded, voxel porosities increased. In Figures 2a and 2b, we show the local surface area and relative velocity magnitude, respectively, and in Figure 2c, we show the local porosity at boundary voxels. Once the porosity reached a value of 1, grain voxels were converted to pore voxels. If the amount of material that was dissolved during a single iteration was more than the volume of the



**Figure 2.** Visualization of the dissolution criteria, given by equation (3), around a single example pore. (a) Number of pore-wall voxel faces exposed to the virtual fluid. (b) Relative velocity magnitude. (c) Local porosity field on the pore-wall after 20 dissolution iterations (0.21 h). Note that lower porosity indicates higher local dissolution rate.

voxel, the remainder was distributed among voxels within the neighborhood defined by a 26-voxel cubic structure element.

The dissolution time step ( $\Delta t$ ) was chosen to ensure numerical stability; no more than one third of a voxel could dissolve at a single time step. To prevent the dissolved pore geometry from wandering from the original pore geometry, the model computed the Euclidean distance map [Kimmel *et al.*, 1996] after every dissolution iteration. If the maximum wander exceeded 1.5 times the voxel length, dissolution ceased and the resulting geometry was used as input to the LBM code to compute the velocity field. Each time the LBM code was initialized, the pore geometry was reevaluated for connectivity, allowing previously unconnected porosity to become connected if it intersected with the updated pore geometry.

### 2.5. Characterization of Hydraulic Parameters

The geometric parameters of interest in this study were connected porosity, tortuosity, specific surface area, and hydraulic diameter. The connected porosity was determined simply by the number of voxels belonging to the connected pore binary volume divided by the total number of voxels in that volume.

Tortuosity was calculated directly from the velocity field outputted from LBM code with the following equation [Duda *et al.*, 2011]:

$$\tau = \frac{\langle u_{\text{mag}} \rangle}{\langle u_{\parallel} \rangle} \quad (4)$$

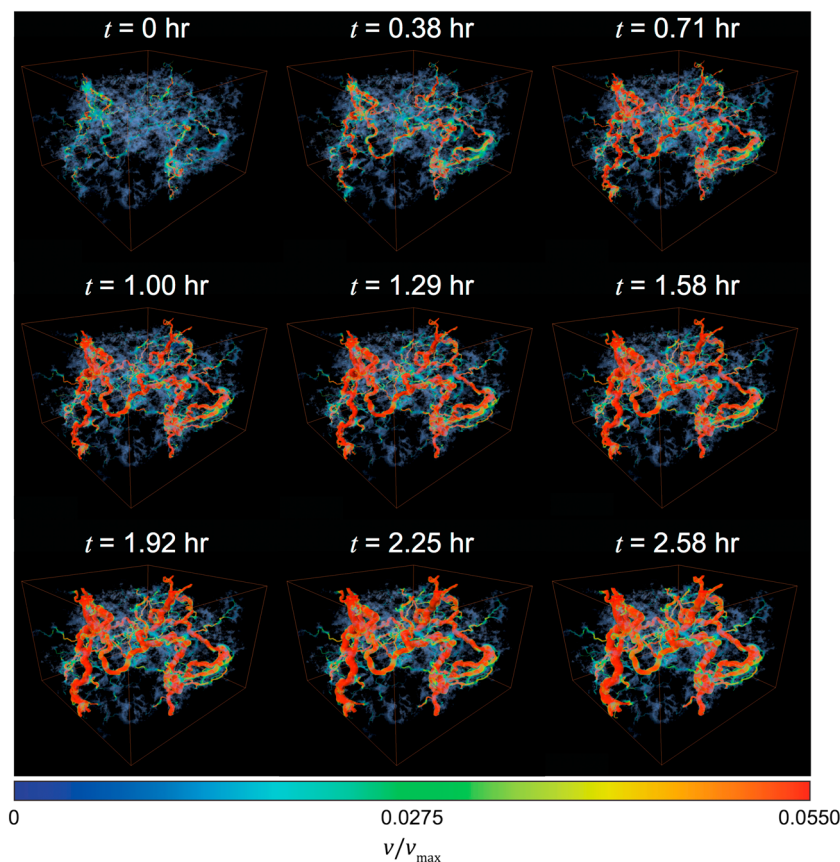
where  $u_{\text{mag}}$  is the local velocity magnitude,  $u_{\parallel}$  is the local velocity parallel to the flow axis, and the notation  $\langle \dots \rangle$  indicates a volume-average over the pore volume. Equation (4) provides an easy way to estimate tortuosity, and as Duda *et al.* [2011] showed, equation (4) is mathematically equivalent to computing tortuosity from average length of streamlines if those streamlines were weighted according to the volume flux integrated along each streamline.

The specific surface area was computed by summing the local surface area of the pore-solid interface and then dividing by the volume of the whole digital rock. The local surface area was obtained by convolving the inverse of the binary pore geometry with a 26-voxel structure element. Then a binary mask composed of pore-wall voxels was applied.

## 3. Results

### 3.1. Velocity Spatial Distribution

In Figure 3, we show volume renderings of the velocity evolution for different times during the dissolution simulation. In the figure, the velocity is normalized by the maximum velocity in the pore geometry. Even in



**Figure 3.** Evolution of velocity during dissolution simulation. Zero-velocity regions are faded to enhance visualization of the proportion of the pore space that conducts fluid flow.

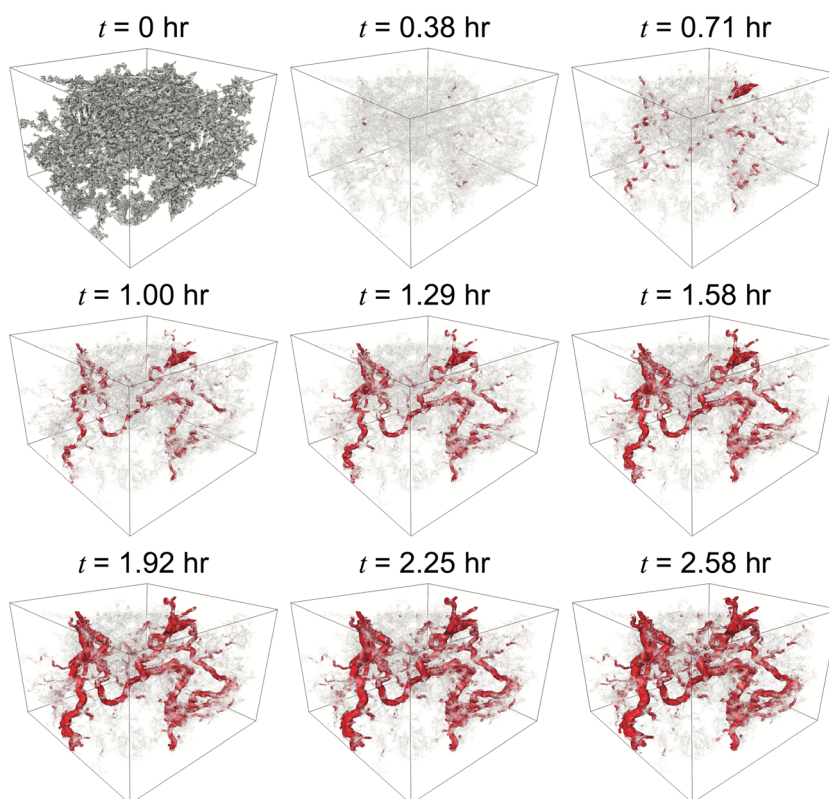
the initial pore geometry ( $t = 0$  h), fluid flow is highly channelized, with only a subset of the total porosity along at least three pathways having a relative fluid velocity greater than 0.01. Over the course of dissolution, the relative velocity magnitude increases only within pores that reside along these high-velocity pathways, while flow in the remaining pores is comparatively stagnant.

### 3.2. Evolution of Pore Geometry

We monitored the pore geometry during simulated dissolution. In Figure 4, we plot the evolution of the pore geometry of the digital carbonate rock due to the numerically induced calcite dissolution. The initial, unreacted connected pore geometry is displayed at  $t = 0$  h (gray) and faded for all subsequent times to enhance the visualization of the porosity created by or made available to flow via dissolution (red).

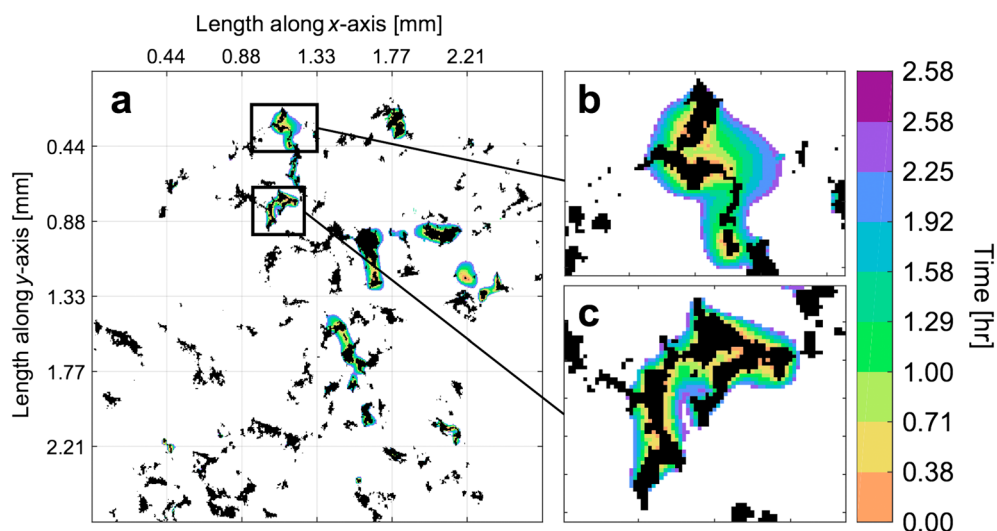
The time-lapse visualization of the pore geometry points to two stages of evolution. During Stage 1 ( $t = 0$ – $0.71$  h), dissolution was sparse, limited to patches dispersed throughout the rock volume. Yet those patches appear to be spatially correlated with the high-velocity regions in Figure 3. During Stage 2 ( $t = 0.71$ – $2.58$  h), those isolated regions grew along the same hydraulic pathways, until they merged, then becoming channels. The newly formed porosity was composed of voxels that had been converted from calcite, rather than by preexisting, unconnected pores that became available to flow through dissolution. Though we allow unconnected porosity to become connected during dissolution, connectivity does not appear to increase significantly throughout dissolution. Instead, dissolution widens preexisting, connected pores along high-velocity pathways.

For a pore-scale view of the evolving pore morphology, we also plot in Figure 5a contours of the connected pore geometry on a two-dimensional cross-section sampled orthogonal to the flow direction. Each contour represents a time snapshot of the binary connected pore geometry. Two example pores are shown in detail in Figures 5b and 5c.

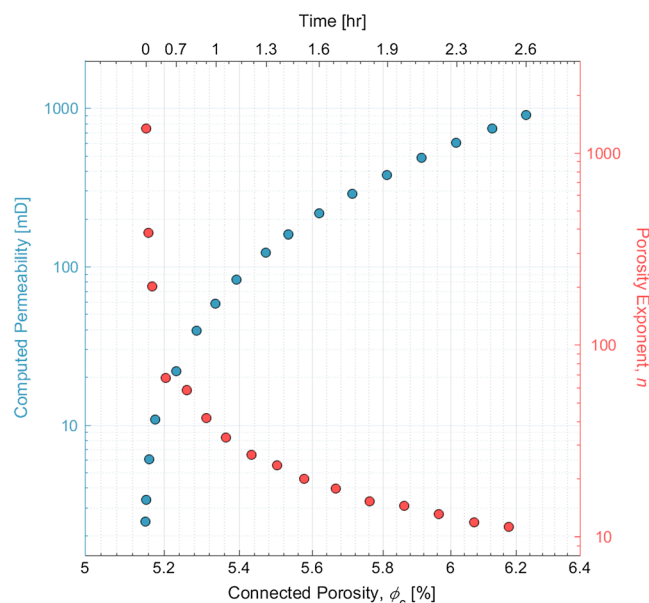


**Figure 4.** Volume renderings depicting the evolution of the connected pore geometry during dissolution simulation. Voxels created or made accessible to flow through dissolution are colored red. The initial pore geometry is shown in gray at  $t = 0$  h and is faded for all subsequent times ease visualization of the created pore space.

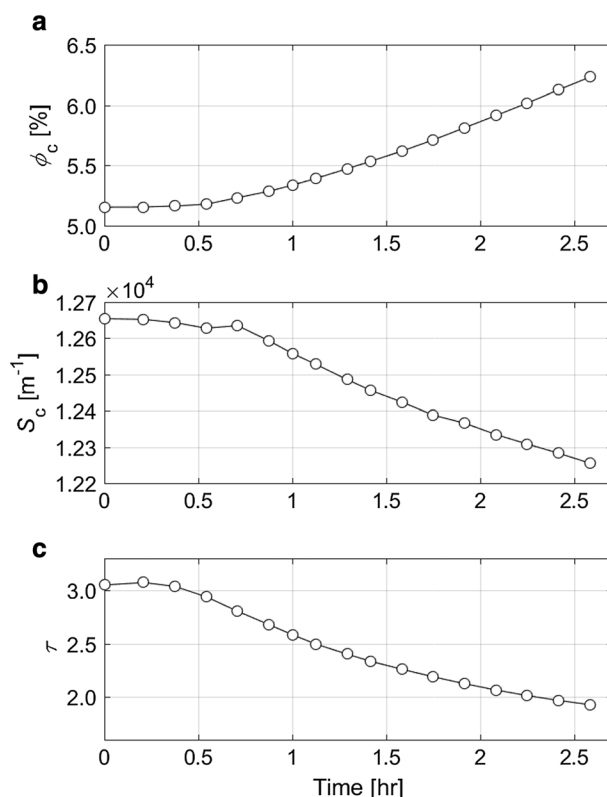
During Stage 1, dissolution was characterized by smoothing of the pore wall; dissolution only removed pores that having the highest local surface area. With the exception of those high-surface area pores, the pore diameter measured on the cross section remained approximately constant at  $\sim 60 \mu\text{m}$ . By  $t = 0.71$  h, dissolution occurred along the entire perimeter of the pore wall. Each time step advanced the position of the pore wall



**Figure 5.** Dissolution overprint. (a) Slice of binary pore geometry overlaid by contours of the dissolution history. Slice is oriented orthogonal to the flow axis. (b and c) Close-ups of two select regions in Figure 5a that depict the evolving pore geometry.



**Figure 6.** Permeability (blue) and porosity exponent (red) plotted as a function of connected porosity (bottom x axis) and time (top x axis). The x and y axes are plotted in log-scale so that the porosity exponent ( $n$ ) is the instantaneous slope of permeability-connected porosity curve.



**Figure 7.** Evolution of connected pore geometry hydraulic attributes. (a) Porosity of the connected pore geometry,  $\phi_c$ , versus time. (b) Specific surface area of connected pore geometry,  $S_c$ , versus time. (c) Tortuosity,  $\tau$ , computed directly from the velocity field using equation (4). results of calcite dissolution simulation.

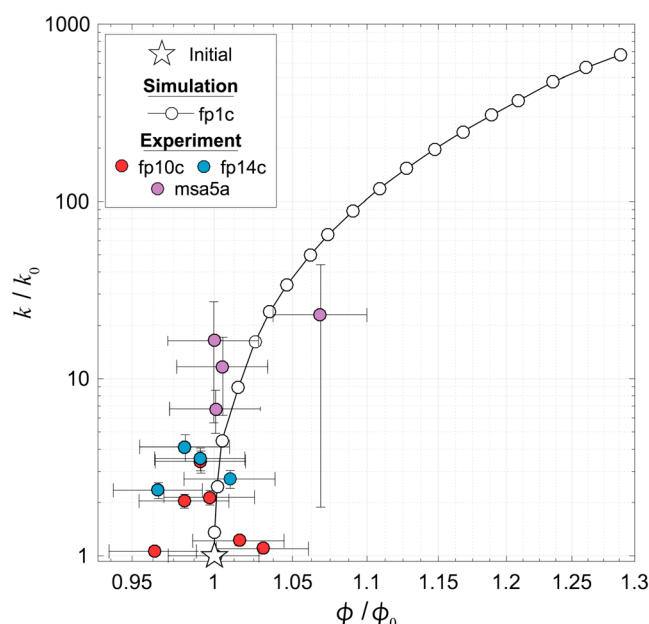
further into the calcite matrix and tended to round the pore. Although, the pores retained by of their original shape and had not advanced so far by  $t = 2.58$  h to have become tubular. By the end of the dissolution simulation, at  $t = 2.58$ , the diameter of pores were approximated  $100 \mu\text{m}$ .

### 3.3. Permeability and Attributes of the Pore Geometry

We plot the permeability versus connected porosity trend in Figure 6 (blue circles). Permeability increased from 2 mD to 906 mD from the initial pore geometry to the final, dissolved pore geometry at  $t = 2.58$  h. The increase of connected porosity due to dissolution over the same time interval was small by comparison, only  $\sim 1\%$  of the initial value at  $t = 0$  h.

We also give the porosity exponent  $n$  (red circles), which quantifies the rate of change of  $\log_{10}(k)$  with respect to  $\log_{10}(\phi_c)$ . The evolution of  $n$  indicates two stages of the pore geometry evolution. During Stage 1 ( $t = 0-0.71$  h),  $n$  decreased from 1334 to 67, the largest increase of permeability with respect to connected porosity during the simulation. During Stage 2 ( $t = 0.71-2.58$  h),  $n$  decreased from 58 to 6. There is some evidence of a transitional stage between  $t = 0.71-1.29$  h where  $n$  behaves nonlinearly in log-log space. Afterward,  $n$  decreases more or less linearly until the end of the dissolution simulation. It is unclear whether the porosity exponent would have converged on a constant value if the simulated dissolution had continued, which would have indicated uniform dissolution along the length of the channels.

In addition to permeability, we also track attributes of the pore geometry relevant to fluid flow. Our choice of parameters, namely, connected porosity  $\phi_c$  (Figure 7a), specific surface area  $S_c$  (Figure 7b), and tortuosity  $\tau$  (Figure 7c), was motivated by Kozeny-Carman formalism



**Figure 8.** Permeability-porosity comparison of dissolution simulation results and CO<sub>2</sub>-injection trends conducted on natural carbonate mudstones [Vanorio *et al.*, 2015]. Initial porosities ( $\phi$ ) and permeabilities ( $k$ ) of both experimental and digital samples were different, so porosities and permeabilities were normalized by the porosity ( $\phi_0$ ) and permeability ( $k_0$ ) of the initial, unreacted sample. The star marker represents the porosity and permeability of the initial normalized porosity and permeability. The colored markers represent experimental results. The error bars on experimental data represent  $2\sigma$  uncertainty of the normalized porosities and permeabilities, which was calculated by propagating the error in the measured porosity ( $\sigma_\phi = 1\%$ ) and permeability ( $\sigma_k = 2\%$ ). The empty markers represent the simulation results from this study.

ized porosity ( $\phi/\phi_0$ ) and permeability ( $k/k_0$ ) values. The error bars on experimental values in Figure 8 represent the two-sigma uncertainty values. The normalized porosity and permeability values for the simulations results are also plotted.

We want to emphasize here that direct comparison of rock physics data is not the primary goal of this research as it poses two difficulties. First, the volume use for digital rock physics simulation in this study is order 1000 times smaller than that use for experiments. We do not presume that our choice of sub-volume is representative of the samples. Given the high degree of heterogeneity present in carbonate samples, there is no guarantee that the same microstructures are swept by the reactive fluid in experiments and simulations. Second, compaction—which we explicitly ignore in our simulations—may hamper permeability evolution by closing compliant pores during the dissolution process [Vanorio *et al.*, 2011]. Though carbonate mudstones are relatively stiff, compared with carbonate packstones, they do experience some ( $\sim 0.2\%$ ) compaction subject to confining pressure during CO<sub>2</sub> injection experiments [Vanorio, 2015].

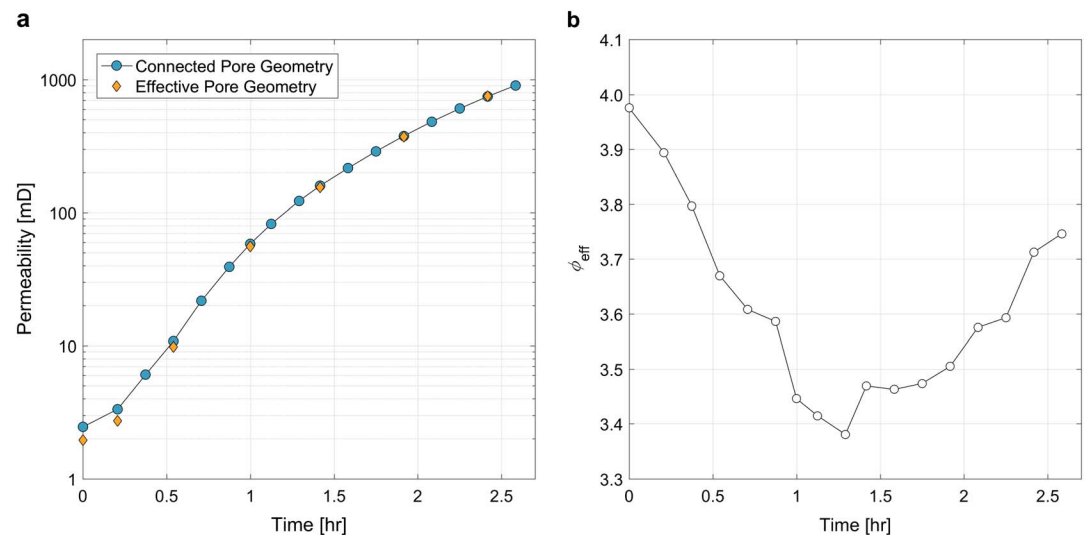
Nevertheless, our simulation does seem to agree with experiments in terms of porosity-permeability relationship during Stage 1. The experimental data, with the exception of three data points, fall within uncertainty of simulation results, so we infer that porosity-independent permeability behavior of experiments is primarily due to dissolution; precipitations of CaCO<sub>3</sub> in the pore space and compaction are likely secondary processes. Furthermore, the fact that experimental data overlap only Stage 1 of our simulation results suggests that channels did not form in the experimental samples, which is corroborated by microstructural analysis of the injected samples [Vanorio *et al.*, 2015]. However, we cannot say at this point that the absence of channel formation in injection experiments is due to insufficient duration of the experiment or to hampering by compaction.

(equation (1)). Despite the permeability increase during Stage 1,  $\phi_c$ ,  $S_c$ , and  $\tau$  were constant. During Stage 2, porosity increased while specific surface area and tortuosity decreased, consistent with equation (2).

## 4. Discussion

### 4.1. Comparison With Experimental Data

Here we compare our simulation results with CO<sub>2</sub> injection experiments. We plot in Figure 8 our simulation and experimental results for three carbonate mudstones [Vanorio *et al.*, 2015]. For experimental data, porosity was measured using He porosimetry. Initial porosities and permeabilities of the three samples ranged 4.3–10.6% and 8–222 mD, respectively. To compare our simulation results with experimental data, we normalized the experimental and simulated porosity and permeability data by those of their respective unreacted samples. Uncertainty of the measured porosity and permeability values for the injection experiments were 1% and 2%, respectively, which we propagated to the normal-



**Figure 9.** Evolution of the attributes of the effective pore geometry. (a) Permeability of connected and effective pore geometry versus time. (b) Effective porosity,  $\phi_{eff}$ , versus time.

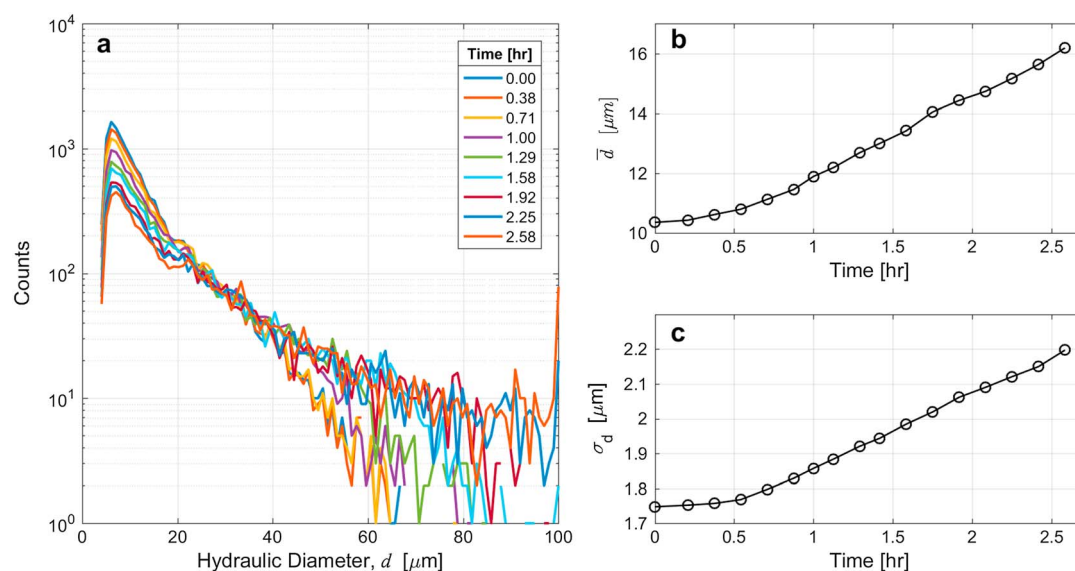
#### 4.2. From Preferred Pathways to Channels

The spatial distribution of velocity in the carbonate mudstone pore geometry is spatially heterogeneous. As is evident by the velocity magnitude visualization at  $t = 0$  h in Figure 3, flow is highly channelized, with a fraction of porosity accommodating a disproportionate share of the fluid flux. These so-called preferential fluid pathways have also been observed in modeling studies concerning fluid flow in random pore media [Martys and Garboczi, 1992; David, 1993; Yang et al., 1993; Pyrak-Nolte and Montemagno, 1994] as well as in real rock geometries [Liu et al., 2005; Karpyn et al., 2007; Smith et al., 2013; Miller et al., 2015; Menke et al., 2016]. They are best interpreted as pathways of least hydraulic resistance that result from heterogeneity imposed by the complex depositional and diagenetic history of the rock.

The coupling between velocity and dissolution rate constitutes a positive feedback that localizes dissolution and high flow rates to preferential fluid pathways. Thus, the heterogeneity of the initial rock microstructure acts as an initial condition to the dissolution reaction and evolution of the pore geometry. At the onset of dissolution, pores having the highest velocity surface-to-volume ratio residing along these preferential fluid pathways dissolve first. Throughout Stage 1, these regions grow until dissolution occurs along the entire length of the preferential fluid pathways, forming channels. By Stage 2, the permeability of the rock is controlled these channels, which spatially correlate with the locations of the preferential pathways present in the initial, unreacted pore geometry.

#### 4.3. Geometric Attributes and Porosity-Permeability Behavior

Despite the 1 order of magnitude increase of permeability during Stage 1, Kozeny-Carman parameters, namely, connected porosity, specific surface area, and tortuosity, in Figure 7 are constant. One interpretation of this seemingly unintuitive result is that a significant proportion of pores containing comparatively stagnant flow, yet contribute to the total porosity and surface area. These pores can be thought of as a background signal, and the porosity created through dissolution is comparatively small. In our digital pore geometry, stagnant pores are those that do not reside along preferred fluid pathways. In order to isolate late the changes to permeability to those pores that actually accommodate fluid flow, we make a distinction between effective and ineffective pores. To this end, we labeled individual pores by applying a morphological watershed algorithm on the connected pore geometries and used the velocity field to calculate the flux into each pore. We then removed ineffective pores, defined as pores whose flux was less than a threshold value, which we denote as a tunable parameter  $p$ . We find that a value of  $p = 0.05$ , or 5% of the average pore flux, satisfactorily reproduces the permeability trend within 0.1–20% of the connected pore geometry permeability (Figure 9). More details on the methodology for partitioning pores into effective and ineffective porosity can be found in



**Figure 10.** Evolution of the hydraulic diameter as measured using a skeletonization of the effective pore geometry. (a) Hydraulic diameter histogram of the effective pore geometry. (b) Time evolution of the geometric mean hydraulic diameter. (c) Time evolution of the geometric standard deviation hydraulic diameter.

the Online Supplement [Beucher and Meyer, 1992; Meyer, 1994; Kimmel et al., 1996; Bernabé et al., 2003; Porikli and Tuzel, 2006; Klasing et al., 2009; Holzer et al., 2012].

Interestingly, dissolution gives rise to some unintuitive porosity behavior, given in Figure 8b. Conventional thinking, driven by the Kozeny-Carman formalism, is that permeability scales with the cube of porosity. However, the effective porosity first exhibits a decrease from  $\phi_e = 4.0$ –3.4% during the period  $t = 0$ –1.29 h, followed by an increase from  $t = 1.29$ –2.58 h. We argue that this behavior is a result of localized dissolution at preferential fluid pathways. That is, as dissolution widens preferential fluid pathways and decreases the tortuosity of the pathways, they accommodate a larger fluid flux, so much that less conductive pores are dropped from the fraction that comprises the effective pore geometry. The dropped porosity outweighs the porosity produced through dissolution of narrow pores, so the total effective porosity decreases during Stage 1. As the narrow pores then continue to widen, the velocity distribution along each preferential path becomes more uniform, which, in turn, induces dissolution over a larger surface area. By  $t = 1.0$  h, dissolution occurs over the entire length of every percolation pathway, resulting in a net porosity gain for all subsequent times.

#### 4.4. Hydraulic Diameter

Despite their widespread use in rock physics, the singularly interesting porosity behavior suggests that it is not a good indicator of permeability evolution during dissolution. More fundamentally than porosity or specific surface area, permeability depends on the mean hydraulic diameter integrated along flow paths [Lock et al., 2002]. In order to ascertain the time evolution of the hydraulic diameter distribution, we skeletonized the effective pore geometry using a medial-axis thinning algorithm [Lee et al., 1994; Kerschnitzki et al., 2013] and then retrieved the hydraulic diameter at each point along the skeleton by interpolating the Euclidean distance map. Because the distance map given the distance to the nearest calcite voxel for every pore voxel, the hydraulic diameter is the circumscribed in the cross section of the pore. More information about obtaining the hydraulic diameter through skeletonization is listed in the supporting information [Lee et al., 1994; Zhu et al., 2011; Kerschnitzki et al., 2013; Miller et al., 2014].

We plot the hydraulic diameter distributions in Figure 10a along with the time evolution of the geometric mean and standard deviation (Figures 10b and 10c). Over the course of dissolution, there is an inversion of the pore diameter about  $\sim 25 \mu\text{m}$ , which we interpret as conversion narrow pore throats into wider pores, and is consistent with experimental observations [Shulakova et al., 2016]. Despite the stagnancy of the porosity and specific surface area of the connected pore geometry (Figures 7a and 7b) and unintuitive

effective porosity behavior (Figure 9b), the hydraulic diameter best mimics—at least qualitatively—the permeability evolution. Furthermore, the mean hydraulic diameter in Stage 1 exhibits a slower rate of change than Stage 2, which temporally correlates with the transition from patchy dissolution to channelization illustrated in Figure 4. Therefore, we propose that the evolution of the permeability and pore morphology can be conceptualized as first selective widening of high-velocity, narrow pore throats, followed by widening along the entire length of newly formed channels.

As a conceptual example, consider flow through a series-connected pore configuration. Though the volumetric flux in each of the pores is the same due to mass conservation, velocity and the dissolution rate are highest in the narrowest pores, causing those pores to widen faster than other pores in the network. As the narrow pores widen, the velocity magnitude distribution within each preferred fluid pathway—not on the whole pore geometry—becomes progressively homogeneous, which in turn induces dissolution over a larger surface area, eventually inducing dilation over the length of the entire channel.

The two-stage porosity-permeability trend supports this conceptual dissolution behavior model. For the series pore configuration given above, permeability is proportional to the harmonic mean hydraulic diameter and thus is strongly weighted by the narrowest pore in the configuration. Also, the narrowest pores also contribute the least to the total pore volume, which are largely outweighed in the calculation of the geometric mean, explaining both the minute porosity increase and slow rate of change of the mean hydraulic diameter during Stage 1. Therefore, widening of the narrowest pores maximally impacts permeability while minimally impacting porosity, a possible explanation for porosity-independent permeability behavior. During Stage 2, all pores along the preferred fluid pathways, which make up a nonnegligible portion of the porosity and number of pores, widen.

Though a thorough analysis is outside the scope of this study, given the widespread use of remote sensing techniques to detect changes in the subsurface, we briefly discuss the elastic properties of our digital rock. Vanorio *et al.* [2015] and Vanorio [2015] measured the primary wave velocity ( $V_p$ ) for carbonate mudstones as a function of injected pore volumes and confining pressure. They showed that the  $V_p$  decreases during dissolution by  $\sim 0.1$  km/s and is insensitive to within measurement error of the measured porosity. Since changes in  $V_p$  are sensitive to changes in the bulk porosity, we do not expect the small modifications to pore throats observed in our numerical simulations to measurably affect the wave speeds.

#### 4.5. Channelization Versus Wormhole Formation

Though the morphology of channels in this study bare a strong resemblance to so-called wormholes [e.g., Fredd and Fogler, 1999; McDuff *et al.*, 2010; Smith *et al.*, 2013, 2014], we emphasize here that they form via different processes and under different flow conditions. Given the same initial chemistry of the reactive fluid, wormhole formation depends both on the flow rate and the initial microstructure of the rock. In a low-flow regime, solvents react with calcite faster than they can be advected through the pore space. As a result, the reaction front literally burrows a channel into the rock at a rate and direction that depends on the distribution of lithology, porosity, and reactive surface area within the sample [Smith *et al.*, 2014]. In contrast, in a high-flow regime, solvents are advected into the pore space faster than they can dropout by reacting with the  $\text{CaCO}_3$  pore wall. As was pointed out in a recent study [Menke *et al.*, 2016] and we confirm with simulations, the channels formed in a high-flow regime is not due to wormholing but is related to preexisting heterogeneity of the pore geometry.

The distinction between wormholing and channelization is subtle but important in the context of permeability-porosity evolution. Permeability of a rock is controlled by the smallest pore throat along percolation pathways. However, wormholing, prior to breakthrough, modifies only the pores at the inlet of the sample. As a result, the porosity exponent should be nearly zero during wormholing prior to breakthrough. In contrast, the modifications to the pore space we observe in our simulations, especially at the onset of dissolution, are so small yet impactful that permeability increases over an order of magnitude without a measureable change of porosity.

Wormholing and channelization may also have different impacts on the mechanical properties of rocks. The pore spaces produced by wormholing in a low-flow regime are so large that they may cause the rock to compact. Conversely, injection experiments [Vanorio *et al.*, 2015] show that the rock can remain stiff even during some channelized dissolution. The contrasting elastic behavior between wormholing in low-flow regime and

channelization in high-flow regime has importance implications for remotely monitoring rock physics properties in reservoir during enhanced hydrocarbon recovery or CO<sub>2</sub> sequestration methods.

## 5. Conclusion

Numerical simulations of calcite dissolution were conducted on digital carbonate mudstones obtained from  $\mu$ -CT imaging to elucidate changes in microstructure that could explain the porosity-independent permeability behavior observed in CO<sub>2</sub> injection experiments of the same rock type. The goal of the model was to produce a nonuniform transformation of the pore geometry similar to dissolution in a transport-limited regime, compute permeability, and relate changes in permeability to specific attributes of the dynamic pore geometry. As a first approach, we developed a simple, phenomenological dissolution model that coupled the dissolution rate of the pore wall to the neighborhood velocity field obtained through LBM. The digital pore geometry was evolved through time by iterating between simulation of dissolution and fluid flow. At each LBM computation, we calculated permeability from the velocity field, which we linked to changing attributes of the pore geometry.

We found that preexisting heterogeneity in the rock microstructure gave rise to preferred fluid pathways, which accommodated a disproportionate share of the flow rate. Since dissolution rate was coupled to the local velocity field in our model, dissolution only occurred along these pathways. The permeability-porosity evolution of the digital pore geometry appeared to have two stages. Stage 1 was characterized by the dissolution of the narrowest pore throats of these preferential pathways. Though small compared to the total pore volume, widening of narrow pores at a faster rate than others on the same fluid pathways induced an order of magnitude increase of permeability while maintaining a virtually constant porosity. During Stage 2, dissolution occurred along the entire length of the percolation pathways, resulting in a much larger increase of porosity with each subsequent dissolution iteration. Stage 1 simulation results compare favorably with CO<sub>2</sub> injection experiments conducted on tight carbonate mudstones.

We also found that the pore geometry could be partitioned into two porosity classes: effective and ineffective porosities. We found that the same permeability—within a maximum error of 20%—could be obtained by removing pores with comparatively stagnant flow rates. Despite the increase of permeability through the entire dissolution simulation, we found that the effective porosity exhibited unintuitive behavior, sometimes decreasing while permeability increased. Therefore, we concluded that porosity is a poor indicator of permeability for tight carbonates undergoing dissolution from CO<sub>2</sub> injection. Instead, we found that the hydraulic diameter, which increases throughout the dissolution simulation, is a better indicator of the permeability behavior.

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# Water Resources Research

## RESEARCH ARTICLE

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### Key Points:

- Fluid and electric flows are numerically simulated at the pore scale in rocks undergoing precipitation and dissolution
- Simulations show that rock permeability  $k$  can be fairly well estimated with the electrical tortuosity and an appropriate length scale
- The uncertainty of the  $k$  prediction is mainly from the electrical tortuosity, which can underestimate the true hydraulic tortuosity

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## Permeability Prediction in Rocks Experiencing Mineral Precipitation and Dissolution: A Numerical Study

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**Abstract** In this study, we focus on the electrical tortuosity-based permeability model  $k = r_{\text{eff}}^2/8F$  ( $r_{\text{eff}}$  is an effective pore size, and  $F$  is the formation factor) and analyze its applicability to rocks experiencing mineral precipitation and dissolution. Two limiting cases of advection-dominated water-rock reactions are simulated, that is, the reaction-limited and transport-limited cases. At the pore scale, the two precipitation/dissolution patterns are simulated with a geometrical model and a phenomenological model. The fluid and electric flows in the rocks are simulated by directly solving the linear Stokes equation and Laplace equation on the representative elementary volume of the samples. The numerical results show that evolutions of  $k$  and  $F$  differ significantly in the two limiting cases. In general, the reaction-limited precipitation/dissolution would result in a smooth variation of  $k$  and  $F$ , which can be roughly modeled with a power function of porosity  $\phi$  with a constant exponent. In contrast, the transport-limited precipitation/dissolution mostly occurs near the pore throats where the fluid velocity is high. This induces a sharp change in  $k$  and  $F$  despite a minor variation in  $\phi$ . The commonly used power laws with constant exponents are not able to describe such variations. The results also reveal that the electrical tortuosity-based permeability prediction generally works well for rocks experiencing precipitation/dissolution if  $r_{\text{eff}}$  can be appropriately estimated, for example, with the electrical field normalized pore size  $\Lambda$ . The associated prediction errors are mainly due to the use of electrical tortuosity, which might be considerably larger than the true hydraulic tortuosity.

## 1. Introduction

In hydrology and many water-related disciplines, the distribution of permeability  $k$  in the subsurface is crucial for conducting a flow analysis, for example, to assess the aquifer performance under different water usage scenarios (Paniconi & Putti, 2015), to predict the organic solute plume front in contaminated sites (e.g., Rizzo & de Barros, 2017), and to evaluate different waterflooding strategies to recovery additional oils from reservoirs (e.g., Friesen et al., 2017). The permeability of a rock is primarily controlled by its texture (e.g., porosity, particle/pore size, sorting, and surface area) and mineralogy, and it can vary over several orders of magnitude (e.g., Brace, 1980) owing to the heterogeneous nature of the rock (Honarpour et al., 1995). In the last several decades, there has been a continuous effort to explore the links between  $k$  and the rock's texture (and/or mineralogy; e.g., Bryant & Blunt, 1992; Garboczi, 1990; Kwon et al., 2004; Mualem, 1976; Xu & Yu, 2008). These studies have significantly improved our capability of predicting a rock's permeability to understand the fluid flow phenomenon in the subsurface.

From a geological viewpoint, the permeability of a rock is not stationary but continuously evolves during diagenesis. This is because diagenetic processes, for example, compaction, mineral precipitation/dissolution, and weathering, can significantly alter a rock's texture and mineralogy, resulting in the change in  $k$  (Cardell et al., 2008; Noiriel et al., 2004; Popp et al., 2001). To better understand the fluid flow phenomenon in rocks, it is therefore necessary to analyze how diageneses affect a rock's microstructure and thus its permeability. Such analysis is an area of interest for studying many geological systems and processes, such as crustal hydrothermal systems (Steefel & Lasaga, 1994), fluid flow in Earth's mantle (Kelemen et al., 1995), and saline water circulating in carbonates (Whitaker & Smart, 1997). In particular, these analyses are critical for carbonate rocks, of which the permeability-porosity relations do not follow the trend observed in sandstones (e.g., Ehrenberg & Nadeau, 2005) due to more severe precipitation and (or) dissolution. From a practical perspective, the knowledge of diagenetic influences on  $k$ , especially

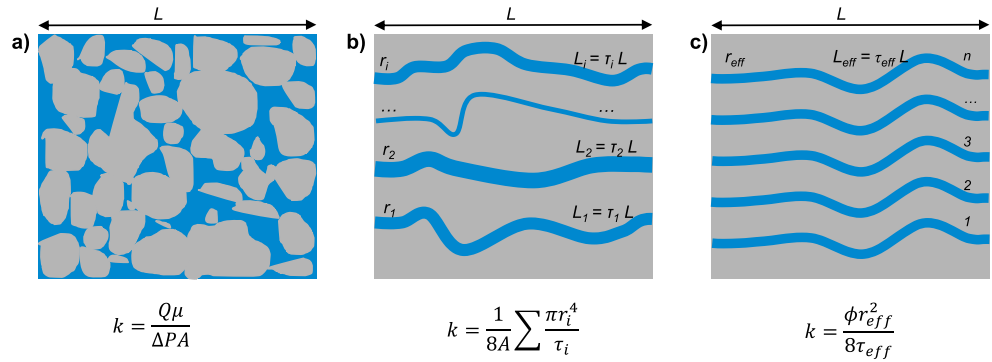
those related to mineral precipitation and dissolution, could benefit many engineering projects such as geological CO<sub>2</sub> sequestration (Luquot & Gouze, 2009), bioremediation and soil improvement with microbially induced calcite precipitation (Achal et al., 2011; Montoya & DeJong, 2015), and oil recovery by injecting acid in reservoirs (Emberley et al., 2004).

To date, there have been many studies on the microstructural and permeability variations of rocks during mineral precipitation and/or dissolution (e.g., see a recent study in Luhmann et al., 2017, and the references therein). These studies can be grouped into two categories: laboratory experiments and pore-scale numerical simulations. Most laboratory experiments are conducted under controlled flow and environmental conditions (e.g., Garcia-Rios et al., 2015), and the samples are characterized in detail with advanced pore-scale imaging techniques (Noiriel, 2015). These experiments provide valuable data for us to understand different precipitation/dissolution patterns in rocks and their controlling factors. As an alternative, a pore-scale numerical simulation has also been widely used in studying the effect of precipitation/dissolution on rock properties (e.g., Miller et al., 2017; Molins et al., 2017; Qin et al., 2016; Yoon et al., 2012). One advantage of numerical simulations is that many microscopic parameters of rocks, which are difficult to determine in experiments, can be directly extracted from the digital representation of the microstructure of the samples (Blunt et al., 2013). Another advantage is that different precipitation/dissolution patterns (e.g., uniform dissolution versus wormholing; Golfier et al., 2002) can be relatively easily controlled in the simulation (e.g., Jiang & Tsuji, 2014). Currently, the main focus of pore-scale simulations is still on developing microscale precipitation/dissolution models that can reproduce the variations in a rock's microstructure and bulk properties as observed in laboratory experiments (e.g., Miller et al., 2017; Nunes et al., 2016).

One of the most important goals in studying rocks experiencing precipitation and/or dissolution is to develop a predictive  $k$  model. Currently, a large body of the relevant research centers on empirical porosity-permeability models such as the power law model and Kozeny-Carman equation (e.g., Bernabé et al., 2003; Zhu et al., 2007). The general conclusion from these studies is that the model parameter(s) varies considerably during the diagenetic processes. Therefore, the permeability predictions from these empirical relations with fixed model parameters are not satisfactory. Recently, attempts have been made to incorporate diagenetic processes in the Kozeny-Carman equation (e.g., Ghezzehei, 2012), and these process-based studies have provided valuable, pore-scale insights into the diagenetic controls on rock's permeability.

At laboratory scales, the permeability of rocks can also be estimated with the aid of geophysical data such as electrical resistivity (Revil & Cathles, 1999; Slater & Lesmes, 2002), induced polarization (Binley et al., 2005; Revil et al., 2015), and nuclear magnetic resonance (NMR) relaxation (Daigle & Dugan, 2011; Kenyon et al., 1988). The underlying principle of these applications is the use of geophysical parameters (i.e., formation factor, imaginary conductivity, induced polarization relaxation frequency, and NMR  $T_2$  relaxation time) to estimate the hydraulic tortuosity (Duda et al., 2011) or the effective pore radius. By doing so, the permeability of the rock can then be estimated from a simple tortuous bundle of capillary tubes model (e.g., Cai et al., 2014) without knowing the detailed rock texture (see detailed discussions in section 2). Due to the noninvasive nature and applicability in field tests, predicting permeability from geophysical data could potentially be a useful means to determine the spatial hydraulic heterogeneity in field scales (e.g., Maurya et al., 2018) if the lab-derived petrophysical relations can be properly upscaled to field scales. Despite the existence of many geophysical-based  $k$  prediction models (see Binley et al., 2015), it is, however, unclear whether they still hold for rocks experiencing different patterns of precipitation or dissolution.

This study aims to understand whether the commonly used, electrical tortuosity-based permeability model,  $k = r_{\text{eff}}^2/8F$  ( $r_{\text{eff}}$  is an effective pore size, and  $F$  is the formation factor; e.g., see Avellaneda & Torquato, 1991), still works for rocks in different mineral precipitation/dissolution scenarios using a pore-scale, numerical approach. In section 2, the theoretical background of the permeability model is reviewed. Section 3 describes the details of the pore-scale methods used to simulate precipitation/dissolution, electric current flow, and fluid flow in rocks. The results, including permeability and formation factor, are presented in section 4. In section 5, permeability estimated from  $k = r_{\text{eff}}^2/8F$  is compared with the hydrodynamic calculations and the origins of the prediction errors are also discussed.



**Figure 1.** Schematics demonstrating different models of a porous rock (pores are indicated in blue, and rock matrix is indicated in gray): (a) a real porous medium, (b) a simplified model consisting of tortuous circular tubes with different size  $r_i$  and length  $L_i$ , and (c) a simplified model having many tortuous circular tubes with same size  $r_{\text{eff}}$  and length  $L_{\text{eff}}$ . The length of the rock sample is  $L$ , and the related  $k$  expression is also shown in each figure. In Figure 1a, only the connected pores are considered for simplifications. The porosities in Figures 1b and 1c are identical, and they may be less than the actual porosity shown in Figure 1a.

## 2. Permeability Prediction—Theoretical Background

The permeability  $k$  of a porous medium (with a dimensional unit of  $[\text{length}]^2$ ) can be determined experimentally by measuring the flow rate  $Q$  induced by a pressure difference  $\Delta P$  across the sample with Darcy's law (e.g., Lu & Likos, 2004), that is,

$$k = \frac{Q\mu}{\Delta PA}, \quad (1)$$

where  $\mu$  is the viscosity of the fluid and  $A$  is the cross-section area of the sample. According to dimensional analysis, the permeability  $k$  is a product of a squared length scale, that is,  $l^2$ , and a dimensionless and scale-invariant quantity  $H$  (Bernabé et al., 2010):

$$k = Hl^2. \quad (2)$$

Based on equation (2), Bernabé et al. (2010) pointed out that the modeling of  $k$  essentially consists of two steps: (1) the determination of the relevant length scale  $l$  and (2) the selecting of the scale-invariant, dimensionless parameter  $H$ . This viewpoint can be used to understand most, if not all, of the existing permeability prediction models including those incorporating geophysical measurements as will be discussed in details later.

The pore space of a real porous rock usually has a complex structure as shown in Figure 1a, and it is extremely difficult to mathematically describe its geometry (Torquato, 2002). In order to theoretically model the fluid flow, the complex pore space is usually represented by a simplified, equivalent pore space. Here “equivalent” means the permeability of porous media with these two pore spaces is the same. Generally, only connected pores of the rock are considered in the simplification because isolated pores usually do not contribute to the fluid flow. One commonly used model is a bundle of tortuous circular capillary tubes with a dispersed radius  $r_i$  and length  $L_i$  as shown in Figure 1b. In this simplified model, the tubes are in parallel and the fluid flow in each tube can be described by the Hagen-Poiseuille equation. The analytical form of  $k$  for the model in Figure 1b can then be derived as

$$k = \frac{1}{8A} \sum \frac{\pi r_i^4}{\tau_i} \quad (3)$$

where  $\tau_i = \frac{L_i}{L}$  is the tortuosity of tube  $i$  ( $L$  being the length of the sample).

Many permeability models, especially for unsaturated porous media, are based on equation (3), and popular examples include those proposed in Burdine (1953), Mualem (1976), Fredlund et al. (1994), and Kosugi (1996). Note that this type of model needs the pore size  $r_i$  distribution as an input. In practice, the pore size distribution of soils is usually estimated from the water retention curve or particle size distribution, and for rocks, it is usually measured with mercury injection capillary pressure curves. Also, since the tortuosity  $\tau_i$

distribution is not available, the average tortuosity  $\langle \tau_i \rangle$  is commonly used instead of the distribution. In practice,  $\langle \tau_i \rangle$  can be assumed as unity if a bundle of straight capillary tubes model is used (e.g., Kosugi, 1996) or a nonunity constant (e.g.,  $\langle \tau_i \rangle = \sqrt{2}$  for sphere packing in Carman, 1937). More commonly,  $\langle \tau_i \rangle$  is assumed as a function of porosity  $\phi$  (for saturated cases) or water content  $\theta$  (for unsaturated cases); the parameters in these functions are sometimes constrained by texture data (Schuh & Cline, 1990) or geophysical data (e.g., resistivity data; Niu et al., 2015).

The complexity of the pore space in Figure 1b can be further reduced by assuming that  $r_i$  and  $\tau_i$  are constant (i.e., effective radius  $r_{\text{eff}}$  and effective tortuosity  $\tau_{\text{eff}}$  as shown in Figure 1c) and keeping the (connected) porosity unchanged. Similarly, the permeability  $k$  can be determined analytically by applying the Hagen-Poiseuille equation and Darcy's law, expressed as

$$k = \frac{\phi r_{\text{eff}}^2}{8\tau_{\text{eff}}}. \quad (4)$$

In practice, the effective tortuosity  $\tau_{\text{eff}}$  is often estimated from resistivity measurements. Ignoring the surface conduction (Revil & Glover, 1998), the electrical conductivity  $\sigma_{\text{eff}}$  of the sample in Figure 1c is expressed as

$$\sigma_{\text{eff}} = \frac{n\pi r^2}{A\tau_{\text{eff}}} \sigma_f = \frac{\phi}{\tau_{\text{eff}}} \sigma_f, \quad (5)$$

in which  $n$  is the total number of tubes and  $\sigma_f$  is the fluid conductivity. Applying the definition of formation factor  $F = \sigma_f / \sigma_{\text{eff}}$  (Archie, 1942), we can have  $\tau_{\text{eff}} = F\phi$  for the model in Figure 1c. Thus, equation (4) can be rewritten as

$$k = \frac{r_{\text{eff}}^2}{8F}. \quad (6)$$

The above equation is the basis of using resistivity measurements to predict  $k$ . The underlying assumption is that the electrical tortuosity ( $F\phi$ ) is the same as the geometric/hydraulic tortuosity  $\tau_{\text{eff}}$ . In this paper, we term equation (6) as the electrical tortuosity-based permeability model. Compared to the dimensional analysis in equation (2), the term  $1/(8F)$  is then the invariant parameter  $H$ .

Now, the next step is to estimate the effective pore size  $r_{\text{eff}}$  (i.e., the relevant length scale  $l$  in equation (2)). The most common way is to use the hydraulic radius  $r_h$  defined as

$$r_h = \frac{2}{S_{\text{por}}} = \frac{2V}{S}, \quad (7)$$

where  $V$  and  $S$  are respectively the volume and surface area of the pore space and  $S_{\text{por}}$  is the surface area-to-pore volume ratio. Both experimental and theoretical studies have indicated strong links between  $S_{\text{por}}$  and some geophysical responses, such as quadrature conductivity  $\sigma''$  (Binley et al., 2005; Börner, 1992; Weller & Slater, 2015) and NMR  $T_2$  relaxation time (Borgia et al., 1996). This is the reason that  $\sigma''$  and NMR  $T_2$  time have been frequently used to aid the  $k$  prediction in porous media (Daigle & Dugan, 2011; Revil & Florsch, 2010; Weller et al., 2015). Note that other factors may also affect the relations between geophysical responses and  $S_{\text{por}}$ . For instance, the change of salinity and frequency could significantly affect the value of  $\sigma''$  (Weller et al., 2011); the existence of inhomogeneous magnetic field and bulk relaxation could also pose difficulties in interpreting the NMR  $T_2$  relaxation time (Behroozmand et al., 2015).

The dynamic pore size  $\Lambda$  proposed in Johnson et al. (1986) has also been used to estimate  $r_{\text{eff}}$  in  $k$  predictions. For porous media, this parameter describes the effects of the internal surface on many processes such as electrical surface conduction and high-frequency viscous damping of acoustic waves (Johnson et al., 1986). The parameter  $\Lambda$  is defined as

$$\Lambda = \frac{2 \int |\mathbf{E}|^2 dV}{\int |\mathbf{E}|^2 dS}, \quad (8)$$

where  $\mathbf{E}$  is the local electrical field and  $dV$  and  $dS$  denote the integrations are over the pore space and grain-fluid interface, respectively. The  $\Lambda$  parameter is a dynamic length that is determined from the solution of the Laplace equation, and it cannot be measured by a simple geometrical analysis (Kostek et al., 1992). In practice,  $\Lambda$  of a sample can be estimated through experiments, for example, by relating  $\Lambda$  to the characteristic

length scale  $l_c$  (Banavar & Johnson, 1987), which corresponds to the threshold pressure  $p_c$  at which the mercury injection curve starts rapidly rising (Katz & Thompson, 1986).

In pore-scale imaging and modeling, some other pore size parameters are also frequently used, including the pore node/body size  $r_d$  and pore throat size  $r_{th}$  (Lindquist et al., 2000). With the recent advancements in imaging techniques (Wildenschild & Sheppard, 2013), extracting these geometrical parameters from microscopic images becomes readily available, and they have been commonly used to reconstruct the rock structure and to predict the permeability (Blunt et al., 2013).

In summary, equation (6) will be tested in this study with the effective pore size  $r_{eff}$  approximated by four pore size parameters,  $r_h$ ,  $\Lambda$ ,  $r_d$ , and  $r_{th}$ . Then, we have the following four electrical tortuosity-based permeability models to analyze:

$$k = \frac{r_h^2}{8F}, \quad (9a)$$

$$k = \frac{\Lambda^2}{8F}, \quad (9b)$$

$$k = \frac{r_d^2}{8F}, \quad (9c)$$

$$k = \frac{r_{th}^2}{8F}. \quad (9d)$$

### 3. Numerical Methods and Samples

In this section, we introduce the pore-scale, numerical methods used to simulate mineral precipitation/dissolution, fluid flow, and electric current flow in porous media. We also explain briefly how the pore size parameters ( $r_h$ ,  $\Lambda$ ,  $r_d$ , and  $r_{th}$ ) are determined from digital representations of the rock microstructure. These numerical methods are applied on two rock samples, and the properties of the samples are described lastly. Note that all the simulations in this study are performed on three-dimensional samples.

#### 3.1. Precipitation/Dissolution Patterns and Their Modeling

When reactive fluids flow through rocks, due to fluid-rock disequilibrium, chemical reactions such as mineral precipitation/dissolution could happen at the solid-liquid interface (e.g., Morse et al., 2007). This pore-scale chemical reaction could alter a rock's microstructure and thus change the rock's bulk property such as permeability. It is generally believed that the reaction rate is quite heterogeneous within the rock and a number of factors could affect the precipitation/dissolution pattern, including local fluid chemistry, mineralogy, and transport properties (e.g., Molins et al., 2017). Thus, a direct simulation of the precipitation/dissolution at the pore scale (e.g., Nunes et al., 2016) is computationally costly and requires the solution of the advection-diffusion-reaction equation.

Fortunately, the main regimes of precipitation and dissolution have been identified based on experimental and numerical studies (e.g., Golfier et al., 2002; Kang et al., 2003; Nunes et al., 2016), and they can be characterized by the combination of Péclet and Damköhler numbers, that is,  $Pe$  and  $Da$ . The  $Pe$  number compares the rate of advection to the rate of diffusion (e.g., see Kang et al., 2003), and it can be defined as  $Pe = uL_c/D_m$  where  $u$  and  $L_c$  respectively are characteristic Darcy velocity and length of the system and  $D_m$  is the diffusion coefficient. The  $Da$  number is defined as the ratio of advective time over reaction time (e.g., Nunes et al., 2016), that is,  $Da = k_r/u$ , where  $k_r$  is the local reaction rate. In this study, we use simplified models to mimic the precipitation and dissolution at the pore scale, aiming to capture the main macroscopic features of the samples as observed in laboratory experiments. In the simulation, the fluid transport is assumed as advection dominated, that is,  $Pe > 1$ . Under this transport condition, two limiting cases will be considered (Nunes et al., 2016): (1) the transport-limited case, in which the reaction at the solid-liquid interface is limited by the diffusion of reactants to and from solid surfaces, that is,  $PeDa < 1$ ; and (2) the

reaction-limited case, in which the reaction is limited by the reaction rate at the solid-liquid interface (i.e.,  $PeDa > 1$ ). In carbonate dissolution, the transport-limited case is typically related to the “wormholing,” while the reaction-limited case generally corresponds to the so-called uniform dissolution phenomena (Golfier et al., 2002).

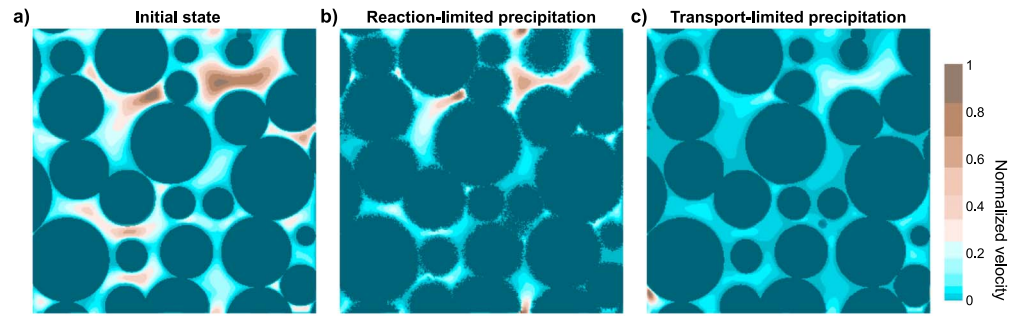
As observed in laboratory experiments, the reaction-limited precipitation/dissolution pattern (defined at the sample scale) is characterized by a roughly uniform growth/removal of minerals at the solid-liquid interface (e.g., Noiriél et al., 2016). Here we use a geometry-based model to simulate the reaction-limited precipitation and dissolution. This model is adopted from the random growth model used in Noiriél et al. (2016) to simulate crystal growth on aragonite grains. The mineral crystals are added (in precipitation) or removed (in dissolution) through a series of discrete steps, and the following summarizes the procedures to realize it in our simulations. First, the voxels at the solid-liquid interface are identified, and this can be done by tracing the phase connectivity to identify a phase change (e.g., see Zhan et al., 2010). In precipitation, these voxels are within the pore space; in dissolution, they are in solid phases. Second, we convert the binary microstructure into a porosity field. By doing this, a continuous growth/erosion of the pore wall can be realized later (Miller et al., 2017). During a time step  $\Delta t$ , the porosity of the interface voxels is updated by a porosity change of  $\Delta\theta < 1$  (i.e., the porosity of the voxel is decreased by  $\Delta\theta$  in precipitation or increased by  $\Delta\theta$  in dissolution), and the locations of the updated voxels are randomly selected following a uniform distribution with a possibility of  $p$ . These two steps are repeated to mimic the heterogeneous mineral precipitation/dissolution at the pore scale. In this process, if the porosity of an interface voxel reaches 0 (in precipitation) or 1 (in dissolution), the phase of the voxels is then converted. The above steps are repeated on the discretized rocks until the target porosity is reached.

Different from the reaction-limited process, the transported-limited (advection-dominated) precipitation and dissolution are even more heterogeneous at the pore scale (e.g., Ott & Oedai, 2015) because the reaction is constrained by the availability of reactants transported to or from the interface, which are related to the local flow conditions. In this study, we use the strategy proposed in Miller et al. (2017) to simulate this limiting case. The method assumes the local fluid velocity field is a proxy of the solvent flux, and it is equivalent to solving the zeroth-order rate equation at the solid-liquid interface (Miller et al., 2017). Overall, the numerical realization is similar to that of the reaction-limited case except that the amount of local porosity change  $\Delta\theta$  during  $\Delta t$  is not a constant but scaled by  $v_m/v_{\max}$ , where  $v_m$  is the local average fluid velocity (e.g., five-voxel neighborhood) and  $v_{\max}$  is the maximum fluid velocity in the pore space. In the simulation, the fluid velocity field of the sample is determined by solving the linear Stokes equation (see section 3.2). As the pore space is continuously altered by mineral precipitation/dissolution, the fluid velocity field in the pore space in our simulation is also updated in the processes.

### 3.2. Fluid and Electric Current Flow Simulations

To determine the fluid velocity field in the pore space of the sample, the linear Stokes equation is numerically solved. In the calculation, we use the Stokes solver developed at the National Institute of Standards and Technology (Garboczi, 1998). After applying a potential gradient across the sample, the code solves the spatial distribution of the pressure and fluid velocity using a staggered-grid finite difference scheme. The permeability  $k$  of the sample can then be determined by applying Darcy's law based on the averaged fluid velocity (e.g., Zhan et al., 2010). The Stokes solver is written in FORTRAN, and the preprocessing code is written in C. For a sample with  $300^3$  voxels, the calculation usually takes about 1 to 3 hr on a personal computer depending on the complexity of the model.

The electric flow in the sample is simulated using the finite difference code DC3d developed in the National Institute of Standards and Technology (Garboczi, 1998). In the simulation, the pore fluid conductivity  $\sigma_f$  is kept unchanged during the mineral precipitation/dissolution. The solid phase is assumed to have a uniform material property, and its electrical conductivity is zero. The conduction contributed from the electrical double layer at the solid-liquid interface (i.e., surface conduction; Revil & Glover, 1998) is ignored, and thus, no extra conductivity is added to either fluid or solid phase in the calculation (Niu & Zhang, 2017). The assumption of zero surface conductivity corresponds to rocks at the high-salinity limit where the conduction from bulk fluid is much higher than the surface conduction. It is noted the (intrinsic) formation factor  $F$  is an intrinsic rock property, which is independent of the fluid and solid-liquid interface properties (e.g., see equation 7a in Schwartz et al., 1989). The code applies a uniform electric field across the sample and then solves



**Figure 2.** Microstructure and fluid velocity field of Sample 1: (a) initial condition ( $\phi = 30.2\%$ ), (b) after reaction-limited precipitation ( $\phi = 20.1\%$ ), and (c) after transport-limited precipitation ( $\phi = 28.2\%$ ). The solid phase is indicated in dark blue. The fluid velocity is normalized by the maximum flow velocity in the pore space. The resolution of the images (size of each voxel) is  $\sim 4.3 \mu\text{m}$ , and the sample volume is  $300^3$  voxels.

the spatial electric potential and current density in the sample with the conjugate gradient method. Based on averaged current density, Ohm's law can be used to calculate the effective electrical conductivity of the sample  $\sigma_{\text{eff}}$  (e.g., Niu & Zhang, 2018). Since  $\sigma_{\text{eff}}$  is solely from the ionic conduction in the bulk solution, the formation factor  $F$  of the sample can be simply determined as  $F = \sigma_t / \sigma_{\text{eff}}$ . The code is also written in FORTRAN, and the calculation time usually is shorter than that of the Stokes solver.

### 3.3. Calculation of Pore Size Parameters

As defined in equation (7), the hydraulic radius  $r_h$  is related to the surface area  $S$  of the solid-liquid interface and the volume  $V$  of the pore space. In the simulation, the solid-liquid interface of the sample is determined by tracing the phase connectivity (e.g., see Zhan et al., 2010). For each voxel, its six neighbors are checked to identify the phase change. If a voxel is isolated with zero connectivity, we simply eliminate it. After the interface is identified,  $V$  and  $S$  can then be easily determined, and then  $r_h$  can be calculated. The determination of  $\Lambda$  is similar to that of  $r_h$ , but the weighting of each voxel is not unity but dependent on the electrical field  $|\mathbf{E}|^2$  according to equation (8). The electrical field  $\mathbf{E}$  is the gradient of the electrical potential field, which is the solution of the Laplace equation.

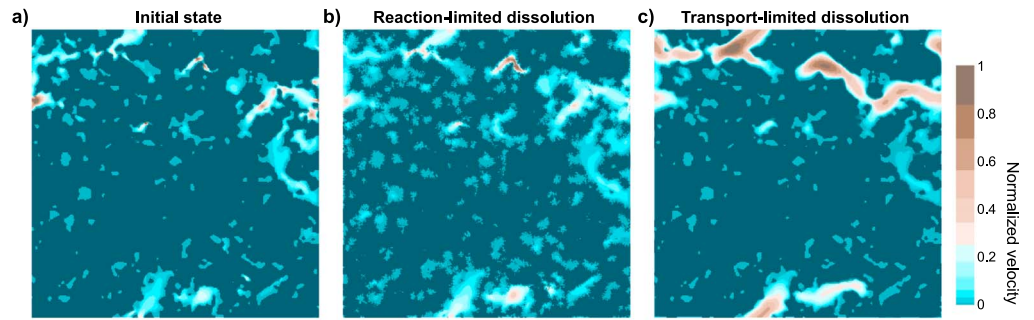
As discussed, the mean pore node size  $r_d$  and pore throat size  $r_{th}$  are only related to the geometry of the pore space, and they can be determined using the concept of maximum balls (e.g., Al-Kharusi & Blunt, 2007) or by medial axis analysis of the pore space (e.g., Lindquist et al., 1996). In this study, we use the modified maximal ball algorithm developed by Dong and Blunt (2009) to determine the distribution of the pore node size and pore throat size. This two-step algorithm searches the nearest solid to define a void ball, and it defines pore nodes and pore throats by affiliating the maximal balls into family trees according to their size and rank (Dong & Blunt, 2009). The results are then fitted with a normal distribution to determine the mean value  $r_d$  and  $r_{th}$ .

### 3.4. Samples

Two samples are used in this study: a synthetic loosely packed ooids sample (Sample 1) and a carbonate mudstone (Sample 2). The ooids sample (Figure 2a) with the initial porosity  $\phi = 30.2\%$  is obtained using the discrete element method by compressing sparsely distributed spherical particles (e.g., Niu & Zhang, 2018). The carbonate mudstone sample has a porosity of  $\sim 13\%$  (from helium measurement) and is from the Wellington Formation, KS; its microstructural image in Figure 3a is from micro-computed tomography scan data after segmentation. In this study, Sample 1 (Figure 2a) is used for precipitation simulations and Sample 2 (Figure 3a) is used for dissolution simulations.

## 4. Simulation Results of Fluid and Electric Current Flows

The limiting scenarios of precipitation and dissolution are numerically simulated using the methods mentioned above for Samples 1 and 2, respectively. Their permeability  $k$  and formation factor  $F$  are also calculated during the precipitation and dissolution, and the results are shown in Figures 4 and 5. To facilitate



**Figure 3.** Microstructure and fluid velocity field of Sample 2: (a) initial state ( $\phi = 13.8\%$ ), (b) after reaction-limited dissolution ( $\phi = 28.8\%$ ), and (c) after transport-limited dissolution ( $\phi = 17.1\%$ ). The solid phase is indicated in dark blue. The fluid velocity is normalized by the maximum flow velocity in the pore space. The resolution of the images (size of each voxel) is  $\sim 4.3 \mu\text{m}$ , and the sample volume is  $300^3$  voxels.

the discussion, we also plot the empirical power law of  $k$  (e.g., Zhu et al., 2007) and Archie's law (Archie, 1942) with varying model parameters in the figures. The two models are expressed as

$$\frac{k}{k_0} = \left( \frac{\phi}{\phi_0} \right)^n \quad (10)$$

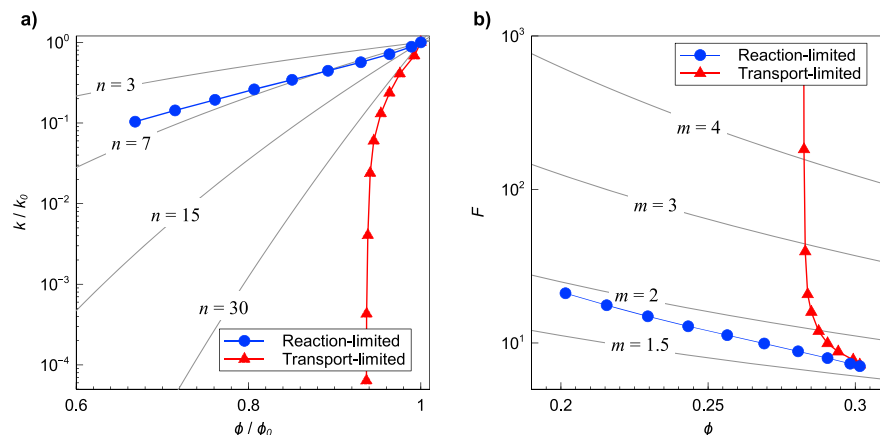
and

$$F = \phi^{-m}, \quad (11)$$

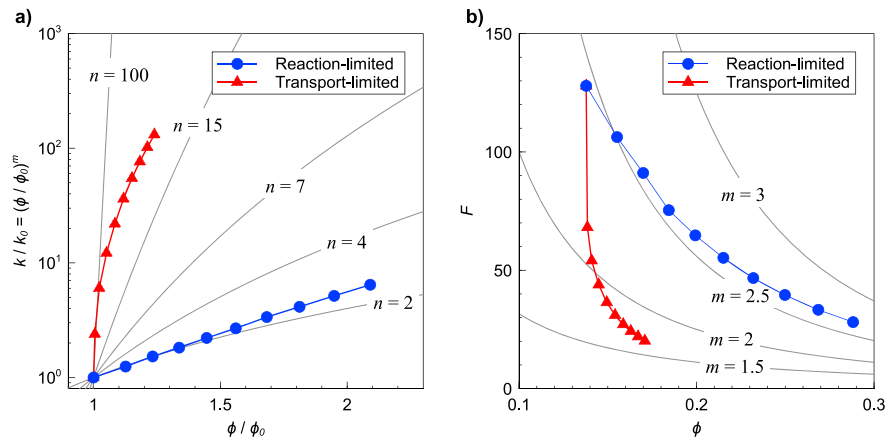
where  $k_0$  is the permeability of the sample at the initial porosity  $\phi_0$ , the exponent  $n$  is an empirical constant, and Archie's porosity exponent  $m$  is usually called cementation factor. The  $k$ - $\phi$  and  $F$ - $\phi$  relations of rocks have been extensively studied with laboratory experiments (e.g., Bernard et al., 2007; Noiriel et al., 2004). In the following, we discuss whether our simulated  $k$  and  $F$  show similar trends as those in published studies and analyze their associated microstructural variations.

#### 4.1. Precipitation (Sample 1)

In simulating the reaction-limited precipitation, the porosity change ( $\Delta\theta$ ) of a voxel during a time step  $\Delta t$  is set as 0.5, and only 5% of the interface voxels are allowed to change phase; a total of 100 time steps are simulated in the calculation, and this induces a porosity reduction of  $\sim 10\%$ , from  $\sim 30.2\%$  to  $\sim 20.1\%$  (Figure 4b). In simulating the transport-limited precipitation, the maximum  $\Delta\theta$  value is 0.3, and 200 time steps are considered in the simulation; the resulted porosity decrease is insignificant, only  $\sim 2\%$  as shown in Figure 4b.



**Figure 4.** Simulation results of Sample 1: (a)  $k$  evolution and (b)  $F$  evolution during reaction-limited and transport-limited precipitation. The parameters  $n$  and  $m$  are the exponents in  $k/k_0 = (\phi/\phi_0)^n$  and  $F = \phi^{-m}$  (i.e., equations (10) and (11)).



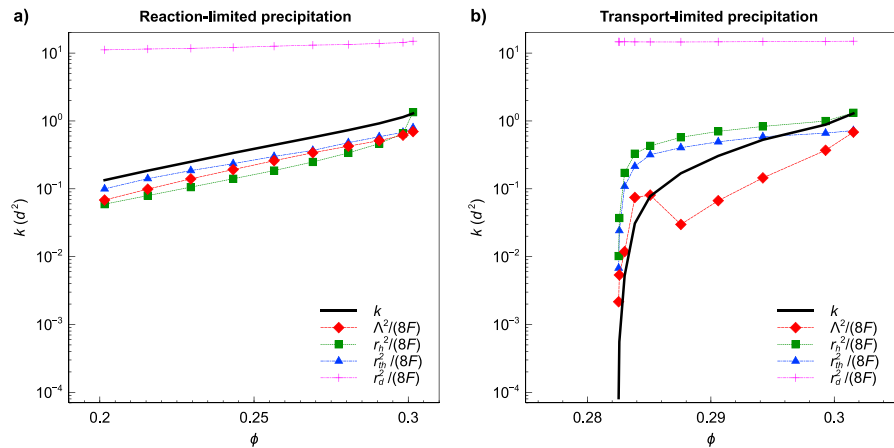
**Figure 5.** Simulation results of Sample 2: (a) permeability  $k$  evolution and (b)  $F$  evolution during reaction-limited and transport-limited dissolution. The parameters  $n$  and  $m$  are the exponents in  $k/k_0 = (\phi/\phi_0)^n$  and  $F = \phi^{-m}$  (i.e., equations (10) and (11)).

The simulated  $k$ - $\phi$  and  $F$ - $\phi$  relations are quite different for the two precipitation. For reaction-limited case (blue symbols in Figure 4), both exponents ( $n$  and  $m$ ) are almost constant without a large variation. For example, in the entire porosity range,  $n$  is close to  $\sim 5$ , while  $m$  is roughly between 1.5 and 2. The microstructure of the sample after the reaction-limited precipitation is shown in Figure 2b, and it appears that, at the sample scale, the precipitated minerals uniformly distribute on the grain surface. These numerical features are consistent with those observed in high-porosity Fontainebleau sandstones (Bourbie & Zinszner, 1985) where the uniform silica overgrowth induces an insignificant variation in  $n$  and  $m$  (Gomez et al., 2010; Revil et al., 2014). It seems that the uniformly precipitated minerals did not significantly change the electric/fluid flow patterns in the pore space (e.g., see Figures 2a and 2b) and that the reaction-limited precipitation may be characterized by a constant porosity exponent  $n$  (or  $m$ ).

For transport-limited cases in Figure 4 (red symbols), both exponents varied considerably despite the minor porosity reduction. For instance,  $n$  quickly exceeds  $\sim 30$  after a porosity decrease of  $\sim 2\%$ , and  $m$  lastly reaches  $\sim 5.1$  when  $\phi = 28\%$ . The increased  $m$  and  $n$  in transport-limited precipitation are associated with the fact that the precipitated minerals accumulate mostly near the pore throats where fluid velocity is relatively high (see Figures 2a and 2c). These numerical results agree with the experimental observations in many published studies (e.g., see Noiriel et al., 2004; Leroy et al., 2017). Thus, it is clear that the transport-limited precipitation has a profound influence on the fluid/electric flow, and it may be characterized by a sharply increased  $n$  (or  $m$ ).

#### 4.2. Dissolution (Sample 2)

The same parameters (e.g.,  $\Delta t$ ,  $\Delta \theta$ , and time step) were used to simulate mineral dissolution in Sample 2, and the calculated  $k$  and  $F$  are shown in Figure 5. For reaction-limited dissolution (blue symbols), both  $k/k_0$  and  $F$  are power functions of porosity  $\phi$  with constant exponents, for example,  $n \sim 2$  in Figure 5a and  $m \sim 2.5$  in Figure 5b. In transport-limited cases (red symbols), however, the associated exponents ( $n$  and  $m$ ) vary largely. For instance, the porosity exponent  $n$  decreases from  $\sim 100$  at  $\phi = 13.8\%$  to  $\sim 20$  at  $\phi = 17.1\%$ . This huge decrease in  $n$  has been experimentally observed in limestones where a  $\sim 2\%$  porosity increase causes  $n$  to change from  $\sim 150$  to  $\sim 15$  (Noiriel et al., 2004). The microstructural variations of the sample (Figure 3) are also quite similar to those of Sample 1 (Figure 2). While in reaction-limited dissolution, the dissolved minerals are almost uniformly distributed on all the solid-liquid interface (Figure 3b), the transport-limited dissolution mainly occurs in some preferred channels (Figure 3c) where the fluid velocity is high (Figure 3a). For both dissolution, the porosity exponent  $n$  and  $m$  can be used to characterize the microstructural variations in rocks.



**Figure 6.** The predicted permeability of Sample 1 during (a) reaction-limited precipitation and (b) transport-limited precipitation. The unit of permeability  $k$  is in  $d^2$ , where  $d$  is the voxel size.

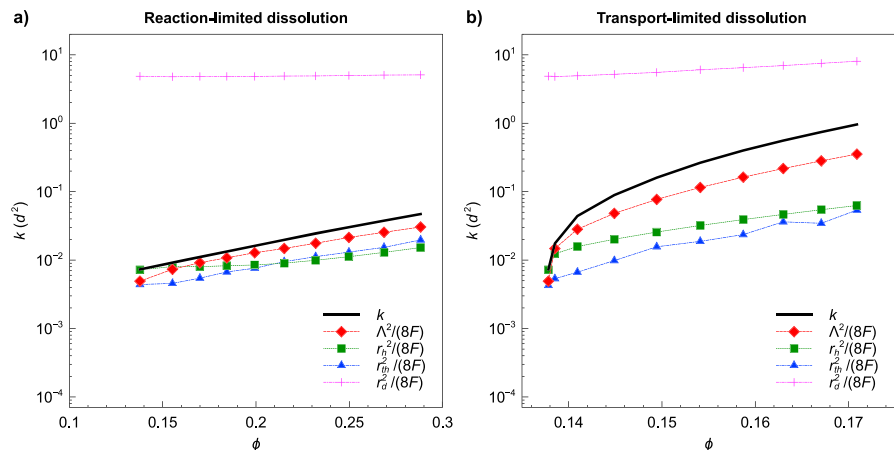
## 5. Permeability Predictions

In this section, we compare the  $k$  predictions to their true values for the two samples undergoing different patterns of precipitation/dissolution. We also discuss the origin of the prediction errors in each scenario. Note that our focus here is the sample's permeability in response to pore-scale precipitation/dissolution, and the results need to be upscaled to field scales for hydrogeological applications. Currently, the upscaling of lab-based, hydrogeophysical relations still remain as an unsolved problem in hydrogeology.

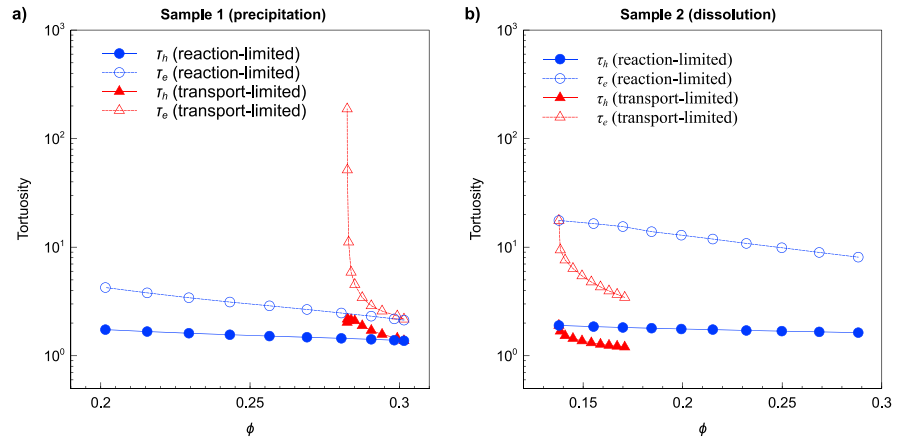
### 5.1. Permeability Predictions

The permeability predicated with (9a) is shown in Figures 6 and 7 for the two samples under different types of precipitation/dissolution. Four pore size parameters were used in the predictions, and they are  $\Lambda$ ,  $r_h$ ,  $r_{th}$ , and  $r_d$ . The permeability determined from hydrodynamic calculations, that is, solving the linear Stokes equation, is also shown for comparison. In Figures 6 and 7, one major finding is that the prediction from the pore body size  $r_d$  significantly overestimates the permeability,  $\sim 2$  orders, for all the cases. This is understandable because it is the pore throat rather than pore body that controls the fluid flow in porous media.

For reaction-limited cases (Figures 6a and 7a), parameters  $\Lambda$ ,  $r_h$ , and  $r_{th}$  give very good permeability estimates, and in general, the discrepancies are less than a half order. In contrast, in transport-limited cases (Figures 6b and 7b), all the predictions deviate noticeably from the true permeability. Among them, the pore size  $\Lambda$  still gives a fair  $k$  estimation, and the associated discrepancy is roughly  $\sim 0.5$  order (e.g., Figure 7b).



**Figure 7.** The predicted permeability of Sample 2 during (a) reaction-limited dissolution and (b) transport-limited dissolution. The unit of permeability  $k$  is in  $d^2$ , where  $d$  is the voxel size.



**Figure 8.** Evolutions of electrical tortuosity  $\tau_e$  and hydraulic tortuosity  $\tau_h$  during (a) precipitation (Sample 1) and (b) dissolution (Sample 2).

This indicates that the applicability of the electrical tortuosity-based permeability ((9a)) largely depends on the patterns of the mineral precipitation and dissolution. It seems the electrical tortuosity-based  $k$  prediction works better for rocks experiencing reaction-limited precipitation and dissolution where the mineral growth and removal are roughly uniformly distributed on the solid-liquid interface. Among these pore size parameters, the electrical field normalized pore size  $\Lambda$  works fairly well for both reaction- and transport-limited cases.

## 5.2. Discussion on the Origin of $k$ Prediction Errors

According to equations (4) and (6), the uncertainty of  $k$  predictions may come from two sources: (1) the length scale  $r_{\text{eff}}$  estimation and (2) the tortuosity  $\tau_{\text{eff}}$  estimation. In equation (6), while different pore size parameters ( $\Lambda$ ,  $r_h$ ,  $r_{\text{th}}$ , and  $r_d$ ) are used to estimate  $r_{\text{eff}}$ , the tortuosity is estimated by the electrical tortuosity  $\tau_e = F\phi$ . In the following, we compare the electrical tortuosity  $\tau_e$  to the true tortuosity, that is, the hydraulic tortuosity  $\tau_h$ , which can be determined from the fluid velocity field, expressed as (Duda et al., 2011)

$$\tau_h = \frac{\langle v_{\text{mag}} \rangle}{\langle v_{\parallel} \rangle}, \quad (12)$$

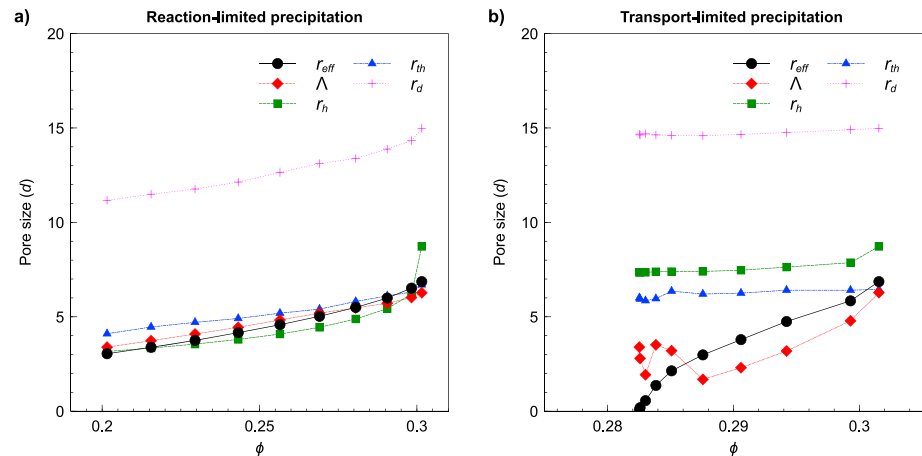
where  $\langle v_{\text{mag}} \rangle$  is the pore volume-averaged fluid velocity magnitude and  $\langle v_{\parallel} \rangle$  is the pore volume-averaged fluid velocity parallel to the applied pressure gradient.

The tortuosity of Samples 1 and 2 is presented in Figures 8a and 8b, respectively. It is shown that for both samples, the calculated electrical tortuosity  $\tau_e$  (empty symbols) is larger than the hydraulic tortuosity  $\tau_h$  (solid symbols) in all the scenarios. In Figure 8, the ratio  $\tau_e/\tau_h$  is roughly  $\sim 3$  for Sample 1 and  $\sim 10$  for Sample 2. They correspond to an underestimation of  $k$  of  $\sim 0.3$  and  $\sim 1$  order. These magnitudes are in accordance with the discrepancies between predicted and true permeability as observed in Figures 6 and 7. This implies that using  $F\phi$  to represent  $\tau_{\text{eff}}$  could underestimate the permeability significantly. Moreover, in reaction-limited cases (blue symbols), both hydraulic and electrical tortuosity changes smoothly. In contrast, the tortuosity in transport-limited cases (red symbols) varies considerably despite the minor porosity change. These distinct responses may be associated with the different microstructural variations as discussed before (e.g., a relatively uniform change of the solid-liquid interface in Figures 2b and 3b and an increased mineral growth/removal near the main flow paths in Figures 2c and 3c).

Now we analyze the accuracy of using  $\Lambda$ ,  $r_h$ ,  $r_{\text{th}}$ , and  $r_d$  to estimate the effective pore size  $r_{\text{eff}}$  in calculating  $k$  (equation (4)). We calculate the true  $r_{\text{eff}}$  value from the true permeability  $k$  and hydraulic tortuosity  $\tau_h$ , expressed as

$$r_{\text{eff}} = \sqrt{\frac{8k\tau_h}{\phi}}. \quad (13)$$

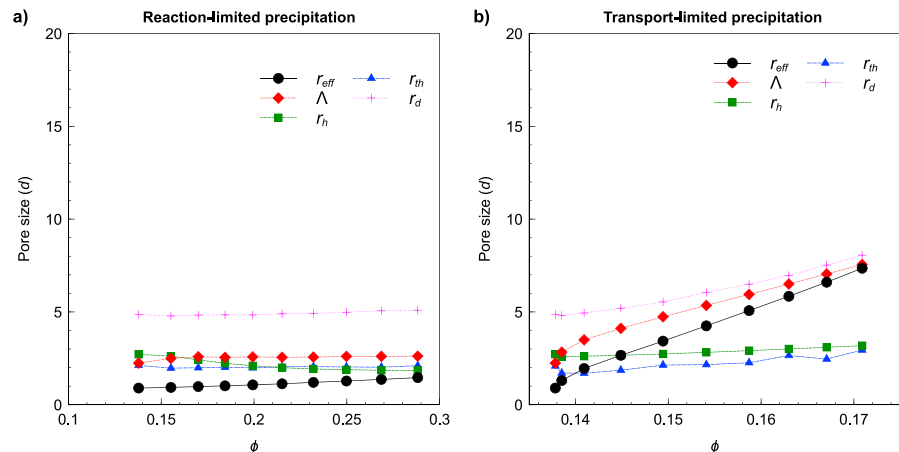
The evolutions of these pore size parameters are shown in Figures 9 and 10 for Sample 1 (precipitation) and Sample 2 (dissolution), respectively. In Figures 9a and 10a, it appears that  $\Lambda$ ,  $r_h$ , and  $r_{\text{th}}$  are quite close to  $r_{\text{eff}}$ ,



**Figure 9.** Evolutions of different pore size parameters of Sample 1 during (a) reaction-limited precipitation and (b) transport-limited precipitation.

indicating that the length scale controlling fluid flow can be safely estimated by any of them for reaction-limited precipitation/dissolution. This also confirms that the errors of  $k$  predictions in Figures 6a and 7a are mainly from the tortuosity estimation. In transport-limited cases, pore size parameters such as  $r_h$  and  $r_{th}$  cannot capture the significant variations of the effective pore size constraining the fluid flow, and thus, they have partially contributed to the discrepancies of  $k$  observed in Figures 6b and 7b. If compared with the tortuosity estimation (i.e.,  $F\phi$ ), the contributions of effective pore size estimations are very minor. It is also clear from Figures 9 and 10 that among these four pore size parameters, only  $\Lambda$  can be reliably used to estimate the length scale characterizing the fluid flow in porous media undergoing different types of precipitation and dissolution.

As shown in Figure 9, the calculated electrical tortuosity is always higher than the hydraulic tortuosity. This is because the shear plane (within which the fluid is not moving) is always away from the solid-fluid boundary. Thus, the fluid does not need to move exactly along the rough solid-fluid boundary. In contrast, ions can move within the shear plane. This means the ions near the solid-liquid interface have to move along the rough boundary surface, resulting in a longer flow path than the fluid. Therefore, the electrical tortuosity is higher than the hydraulic tortuosity as shown in Figure 9. The results imply that we have to be very cautious when using electrical tortuosity to approximate hydraulic tortuosity, especially when the internal surface of the rock is very rough.



**Figure 10.** Evolutions of different pore size parameters of Sample 2 during (a) reaction-limited dissolution and (b) transport-limited dissolution.

## 6. Conclusions

The pore-scale numerical simulations have shown that the permeability-porosity ( $k$ - $\phi$ ) and formation factor-porosity ( $F$ - $\phi$ ) relations of rocks are largely influenced by the patterns of the mineral precipitation and dissolution. In reaction-limited, advection-dominated cases,  $k$  and  $F$  are roughly power functions of porosity  $\phi$  with the exponents ( $n$  and  $m$ ) nearly constant, and in general, the precipitated or dissolved minerals uniformly distribute at the solid-fluid interface. This microstructural alternation is responsible for the porosity change but did not significantly change the characteristics of the fluid and electric current flows in the pore space. In transport-limited, advection-dominated cases, the precipitated and dissolved minerals are mostly near the pore throats where fluid velocity is relatively high. This heterogeneous microstructural alternation could significantly affect the fluid and electric current flows in the pore space, resulting in a huge change in  $k$  and  $F$  despite a minor variation in  $\phi$ . The commonly used power laws with constant porosity exponents are not able to describe such changes. Compared to reaction-limited cases, transport-limited precipitation/dissolution has a more profound influence on fluid/electric flow and it may be characterized by a sharply changed  $n$  or  $m$ .

In general, the electrical tortuosity-based permeability model  $k = r_{\text{eff}}^2/8F$  works well for rocks experiencing precipitation and dissolution if the effective pore size  $r_{\text{eff}}$  is estimated with the electrical field normalized pore size  $\Lambda$ . The error of the prediction is mainly from the electrically estimated tortuosity, that is,  $F\phi$ , which is consistently larger than the true (hydraulic) tortuosity. In reaction-limited cases, the hydraulic radius  $r_h$  and mean pore throat size  $r_{th}$  can also be safely used to estimate  $r_{\text{eff}}$ . In all the cases, the pore body size  $r_d$  significantly overestimates the effective pore size, and thus, it should be avoided in permeability predictions of rocks experiencing precipitation and dissolution.

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